



中壢特高頻雷達

太空測計及操作 (I)

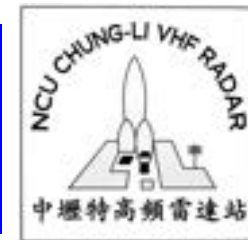
朱延祥、蘇清論

中華民國107年10月8日





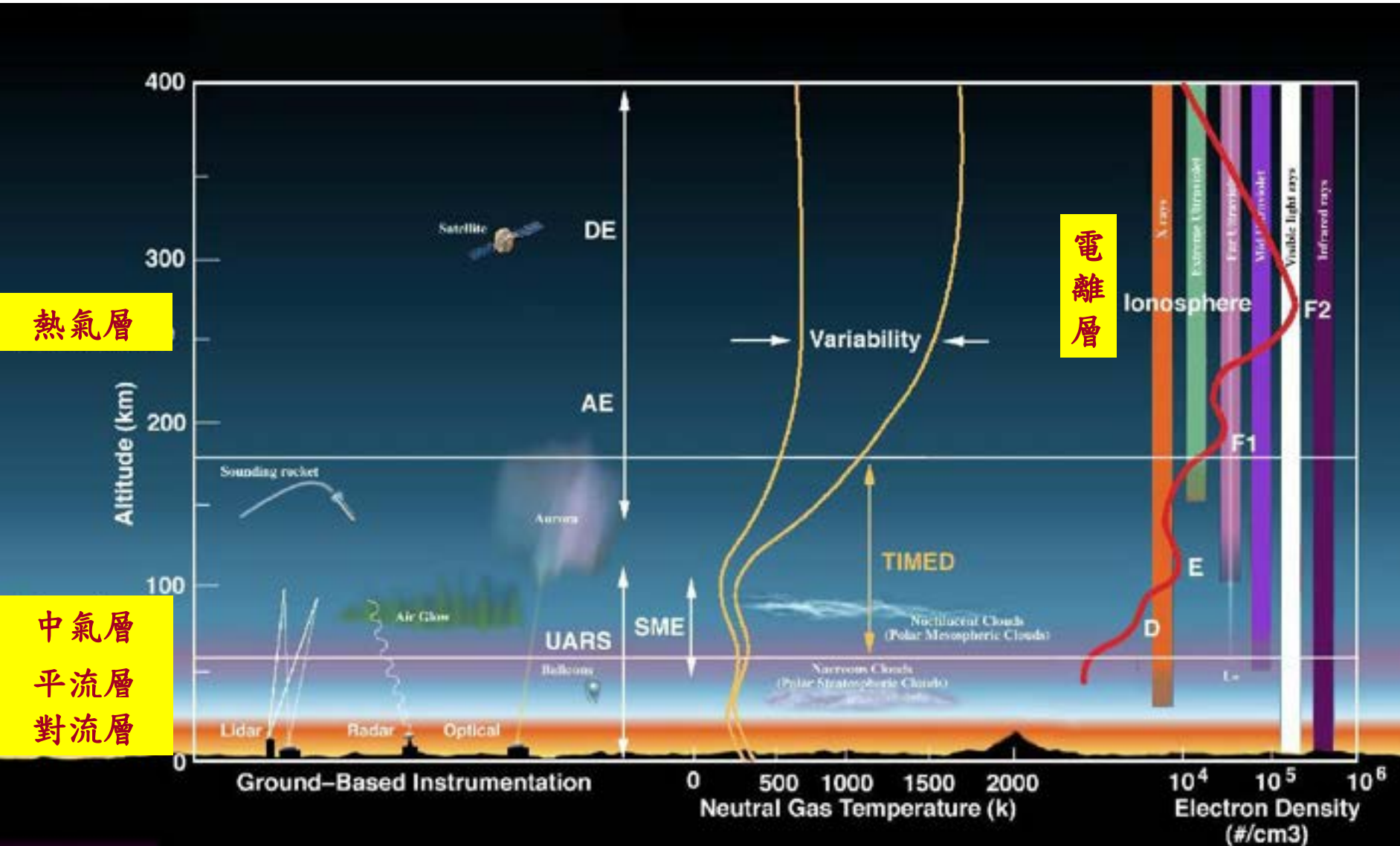
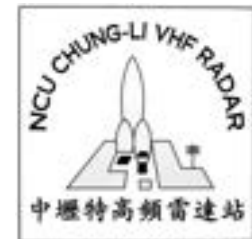
OUTLINE



- 中壢特高頻雷達簡介
- 天線場型模擬與實驗參數
- 中壢特高頻雷達參與實驗簡介



地球大氣與太空環境結構





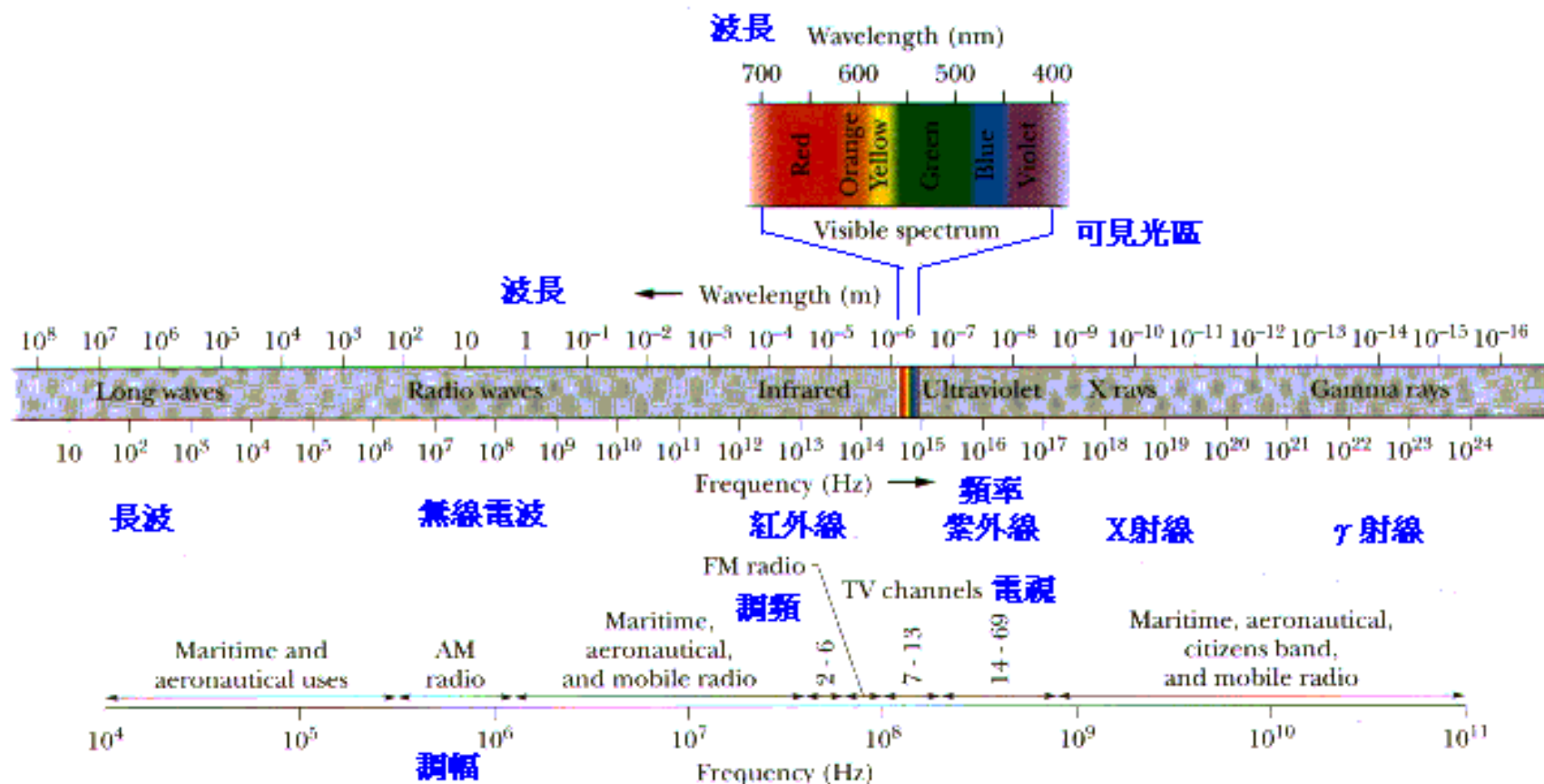
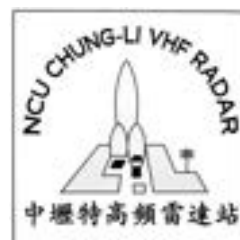
國際電信聯合會之 無線電頻率劃分表



頻帶號碼(N)	頻帶命名	頻率範圍	公制之波長劃分
4	特低頻 (VLF)	3至 30千赫(KHz)	萬公尺波
5	低頻 (LF)	30至 300千赫	千公尺波
6	中頻 (MF)	300至3000千赫	百公尺波
7	高頻 (HF)	3至 30兆赫(MHz)	十公尺波
8	特高頻 (VHF)	30至 300兆赫	公尺波
9	超高頻 (UHF)	300至3000兆赫	十分之一公尺波
10	極高頻 (SHF)	3至 30秊赫(GHz)	百分之一公尺波
11	至高頻 (EHF)	30至 300秊赫	千分之一公尺波
12	—	300至3000秊赫	萬分之一公尺波

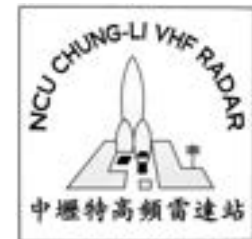


電磁波頻譜





中壢特高頻雷達站位置圖





雷達站簡史



民國70年初 為配合國際中層大氣探測計畫，劉兆漢校長建議
在我國設立特高頻雷達

民國71年9月 中大向國科會提出「極高頻雷達之建造及對中層
大氣之觀測研究」之大型研究計畫，委託美國
TYCHO公司負責建造

民國73年 雷達站開始施工，整理天線場地

民國74年5月31日 首次接收到大氣回波，完成中壢特高頻雷達
站的設置

民國74年6月3日 雷達揭幕式，正式啟用，主要性能為中低層
大氣之觀測研究。



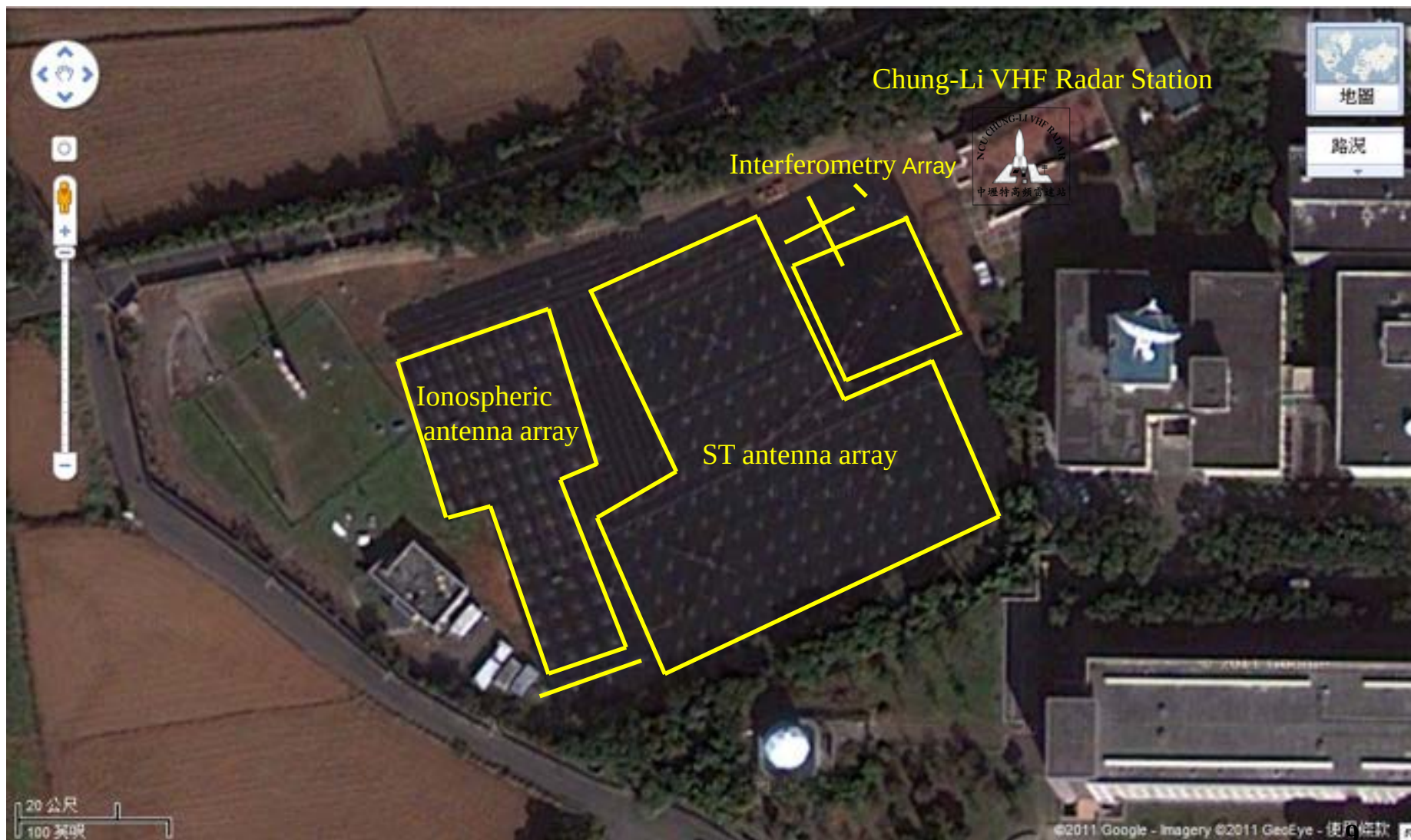
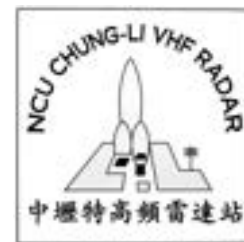
中壢特高頻雷達 性能提升與研究技術發展史



- 民國79年5月 中壢特高頻雷達具有頻率域雷達干涉法的觀測功能。
- 民國81年4月 中壢特高頻雷達增建完成流星定位天線系統，提升中壢雷達具有流星與閃電的觀測能力。
- 民國82年6月 中壢特高頻雷達擴增完成電離層天線陣列，提升中壢雷達具有電離層電子密度不規則體的觀測與空間域雷達干涉法定位能力。
- 民國91年8月 30MHz雙態雷達加入中壢特高頻雷達系統陣列，提升中壢雷達具有雙態的觀測能力。94年5月正式成功接收訊號
- 民國93年8月 啟動52MHz雷達更新計畫，94年12月初步完成3座20 kw發射機/接收機、資料擷取與控制系統更新。95年7月再增加2座20 kw發射機與三個接收機。98年再增加20kw發射機一座。
- 民國101年10月 完成更新十字型不等間距干涉天線系統，同年12月發展完成閃電定位技術。
- 民國104年4月 建構測海雷達天線系統，擴增中壢特高頻雷達觀測海洋能力。

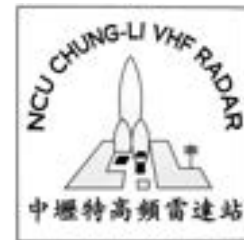


中壢VHF雷達站全貌



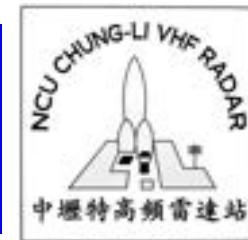


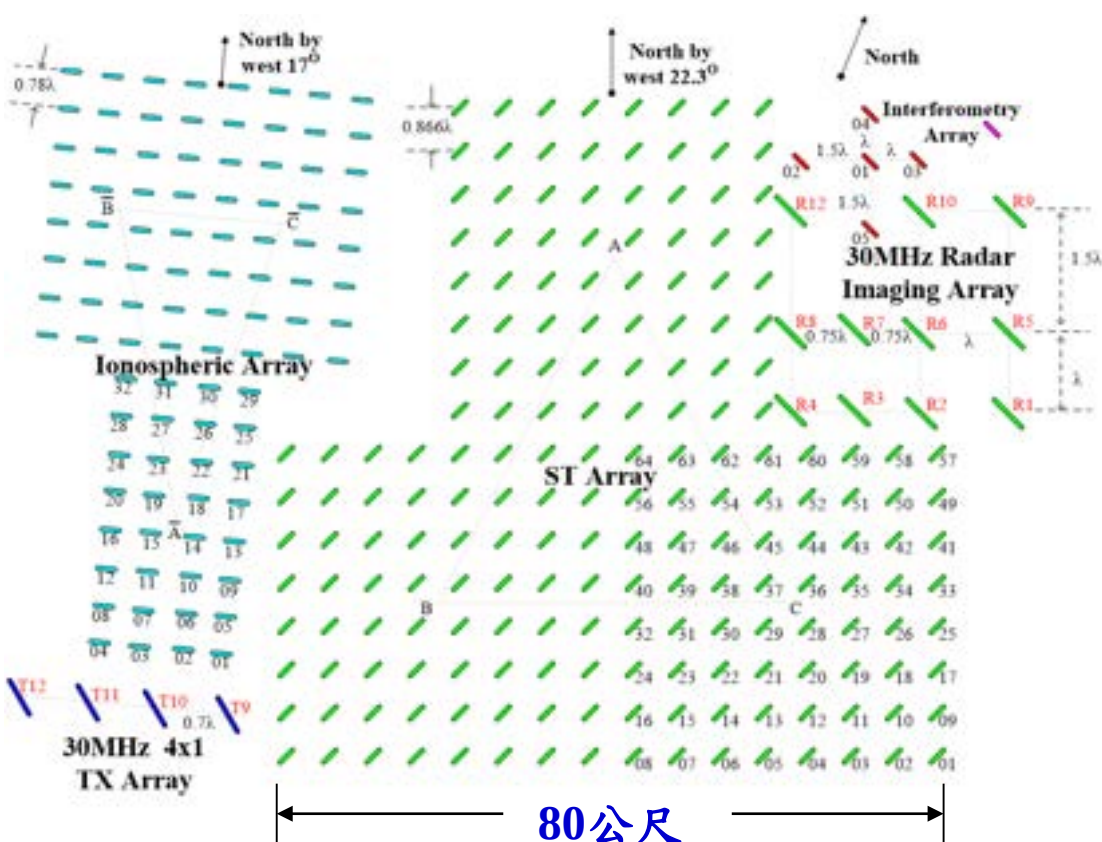
中壢VHF雷達站





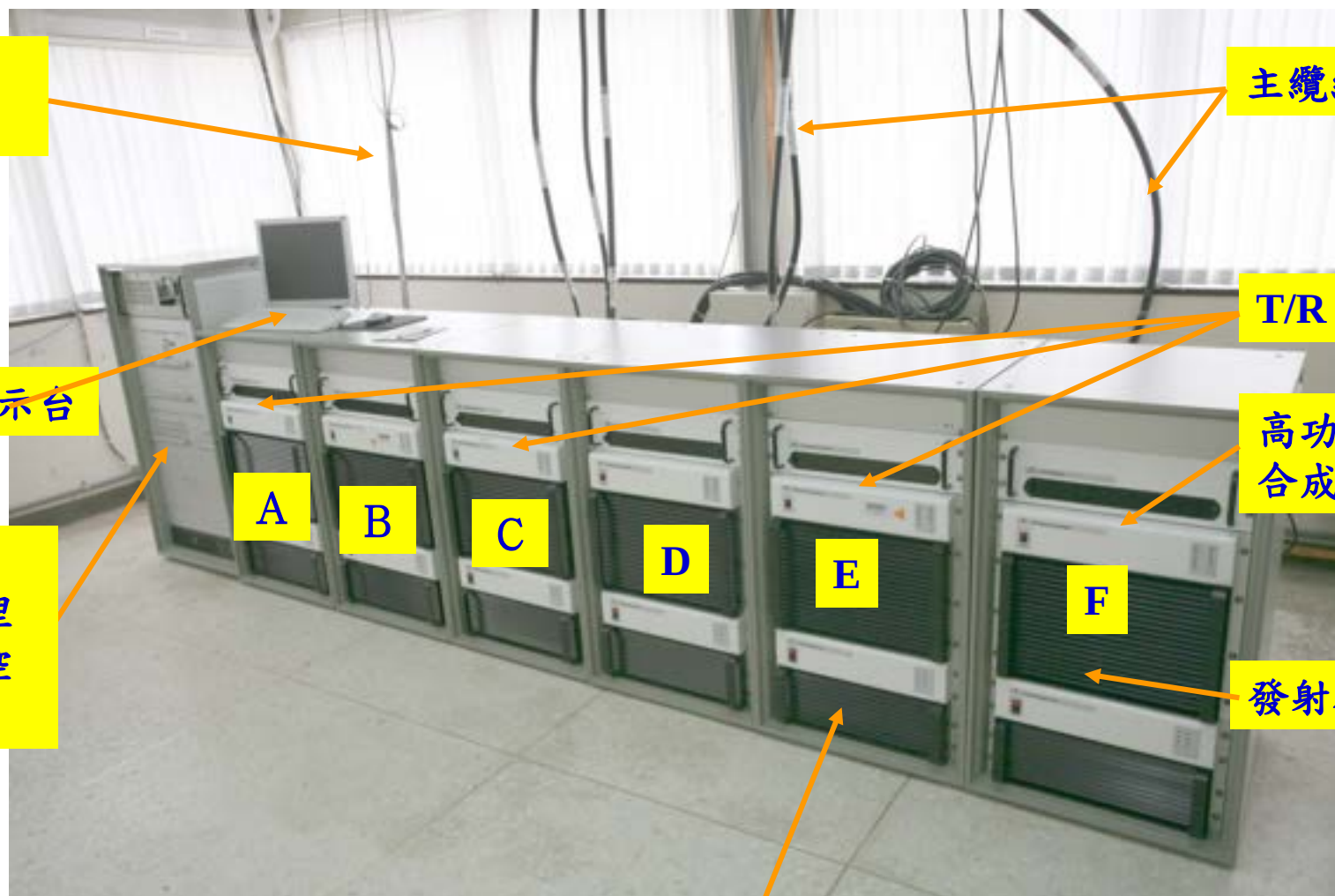
中壢VHF雷達天線陣列







中壢特高頻52MHz雷達系統(1)



相位控制
控制纜線

主纜線

T/R 切換器

控制顯示台

高功率
合成器

A

B

C

D

E

F

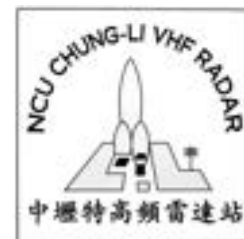
發射機

接收機、
資料處理
與雷達控
制系統

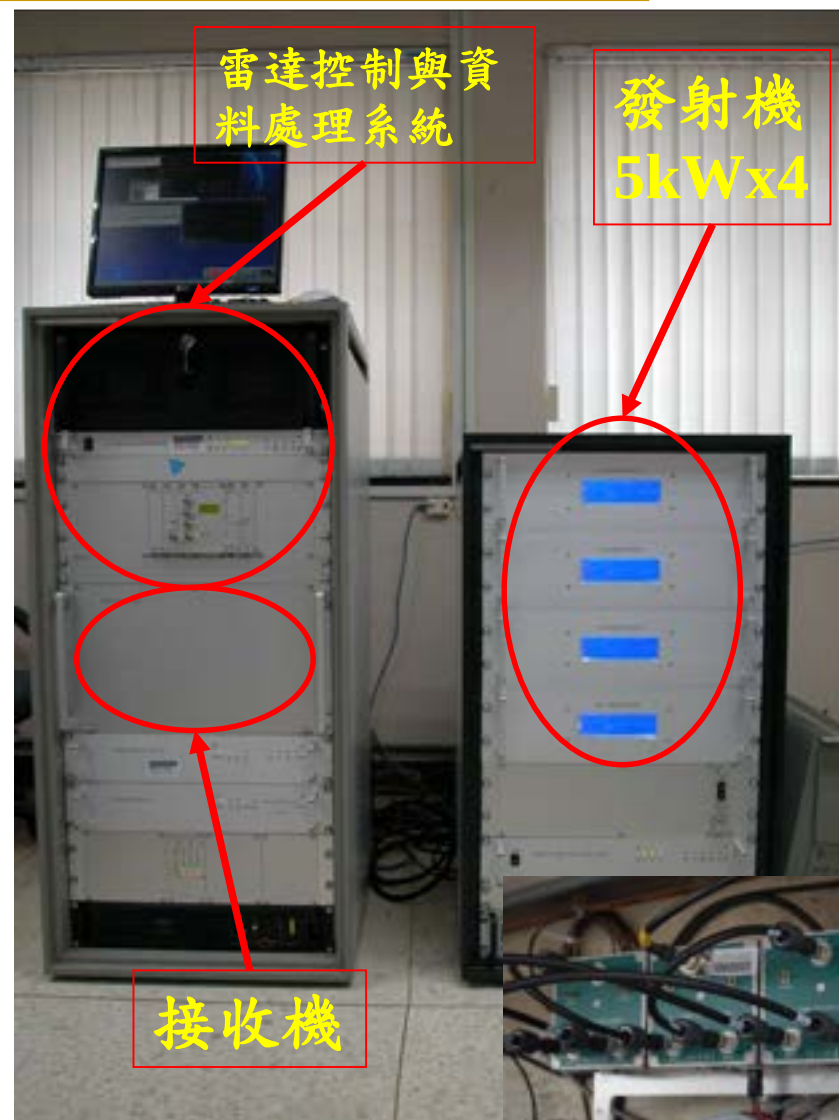
電源供應器



中壢特高頻52MHz雷達系統(2)



中心頻率 Central Frequency	52 MHz
放大器型式 Type of Amplifier	固態功率放大器 Solid state power amplifier
峰值總功率 Total peak power	$\geq 20\text{kW}$ @ 7.5% duty cycle
單一抽屜功率模組峰值功率 Peak envelope RMS power per Module	$\geq 5\text{ kW}$ @ 7.5% duty cycle per unit chassis
功率模組數目 Number of Power Module	4
脈波寬 Maximum pulse length	1~200 μs
最大電壓駐波比 Maximum VSWR	≤ 1.5
狀態顯示內容 Status Readouts Items	輸出功率 (Output power 單位 dBm) 輸出相位 (Output phase 單位 Degrees) 功率放大器溫度 (PA temperature 單位 $^{\circ}\text{C}$)
警告 Alarm conditions	VSWR 過高，風扇故障.





中壢特高頻52MHz雷達 天線系統參數



電離層天線陣列

型式
結構

由32(4x8)根八木天線組成3個矩形陣列，
以各陣列中心相距邊長18, 20, 20公尺形成一等腰三角形，
頂點方向為南偏西17°

半功率波束寬
波束指向
天線地理面積

橢圓型波束7.4° x 15°
以天頂角41°向北 (垂直250 km高的地磁磁力線)
3X648 m²

ST天線陣列

型式
結構

由64(8x8)根八木天線組成3個正方形陣列，
以各陣列中心相距邊長45, 45, 40公尺形成一等邊三角形，
頂點方向為南偏西22.3°

半功率波束寬
波束指向
天線地理面積

近圓形波束7.4° x 7.4°
垂直，及以17°向北，東，南，西傾斜
3X1600 m²

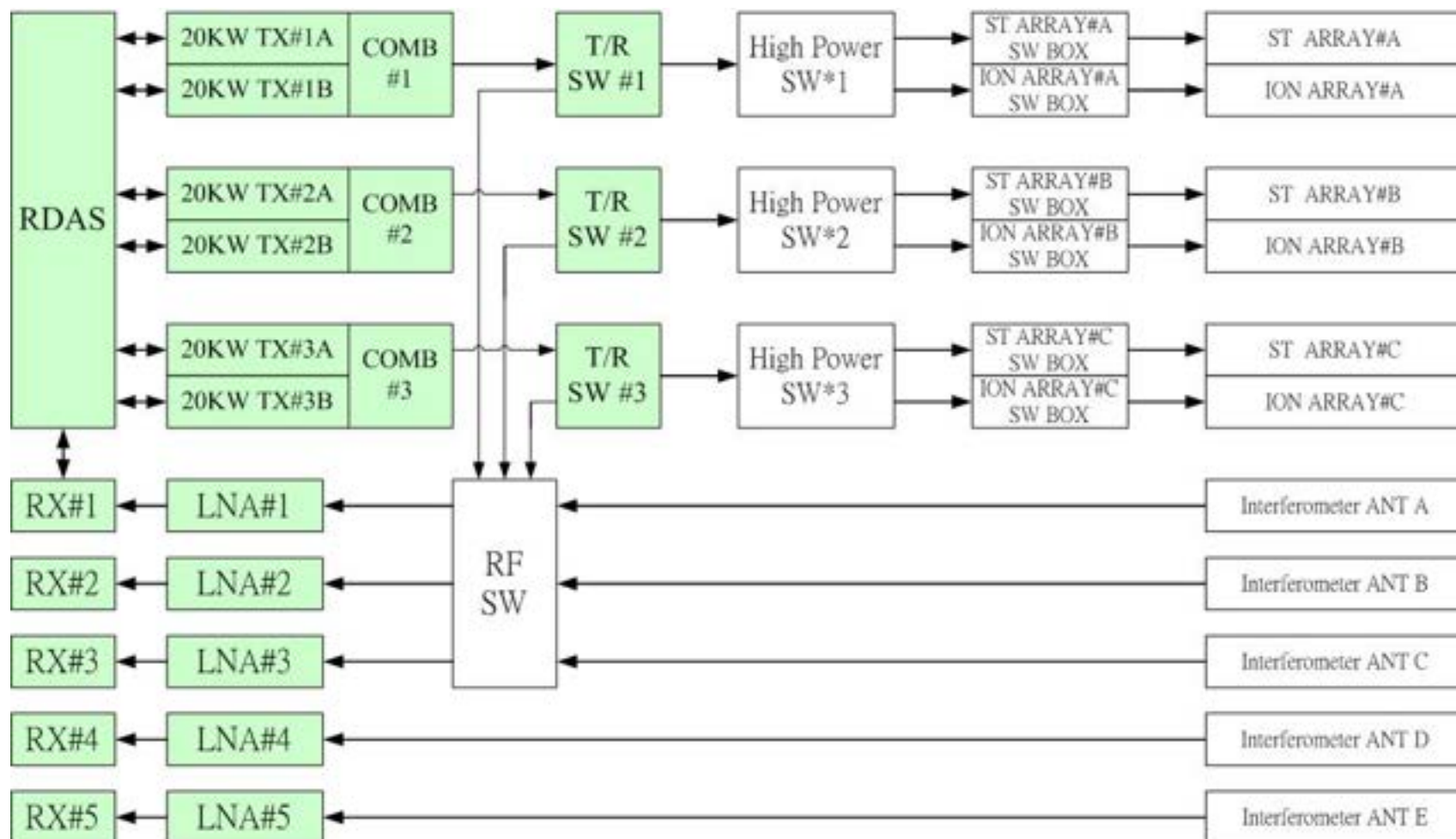
干涉天線陣列

型式
結構
半功率波束寬
波束指向

5根八木天線
不等間距十字型
~55°
垂直

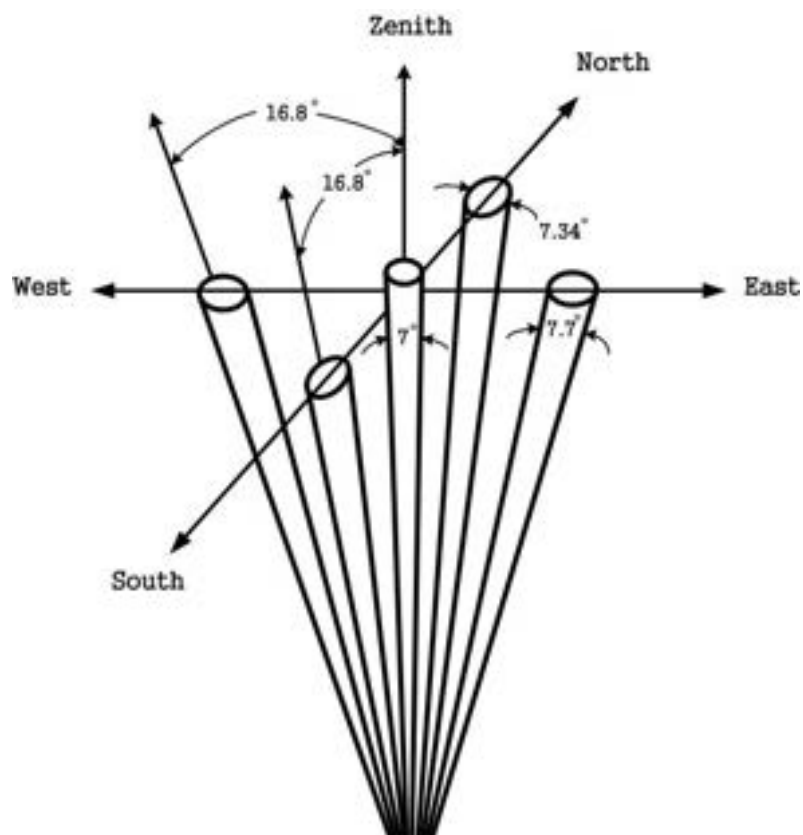
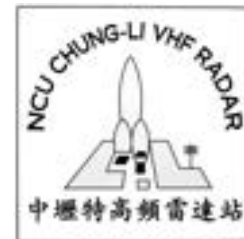


系統功能方塊圖





ST天線陣列波束指向



Gain of Vertical Beam = 26.5 dBi
Gain of Oblique Beam = 25.9 dBi



中壢特高頻30 MHz雷達系統



控制顯示台



資料處理
與雷達控
制系統

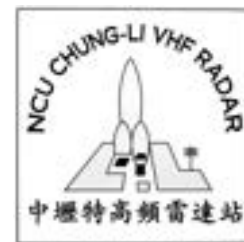
接收機

發射機

電源供應器



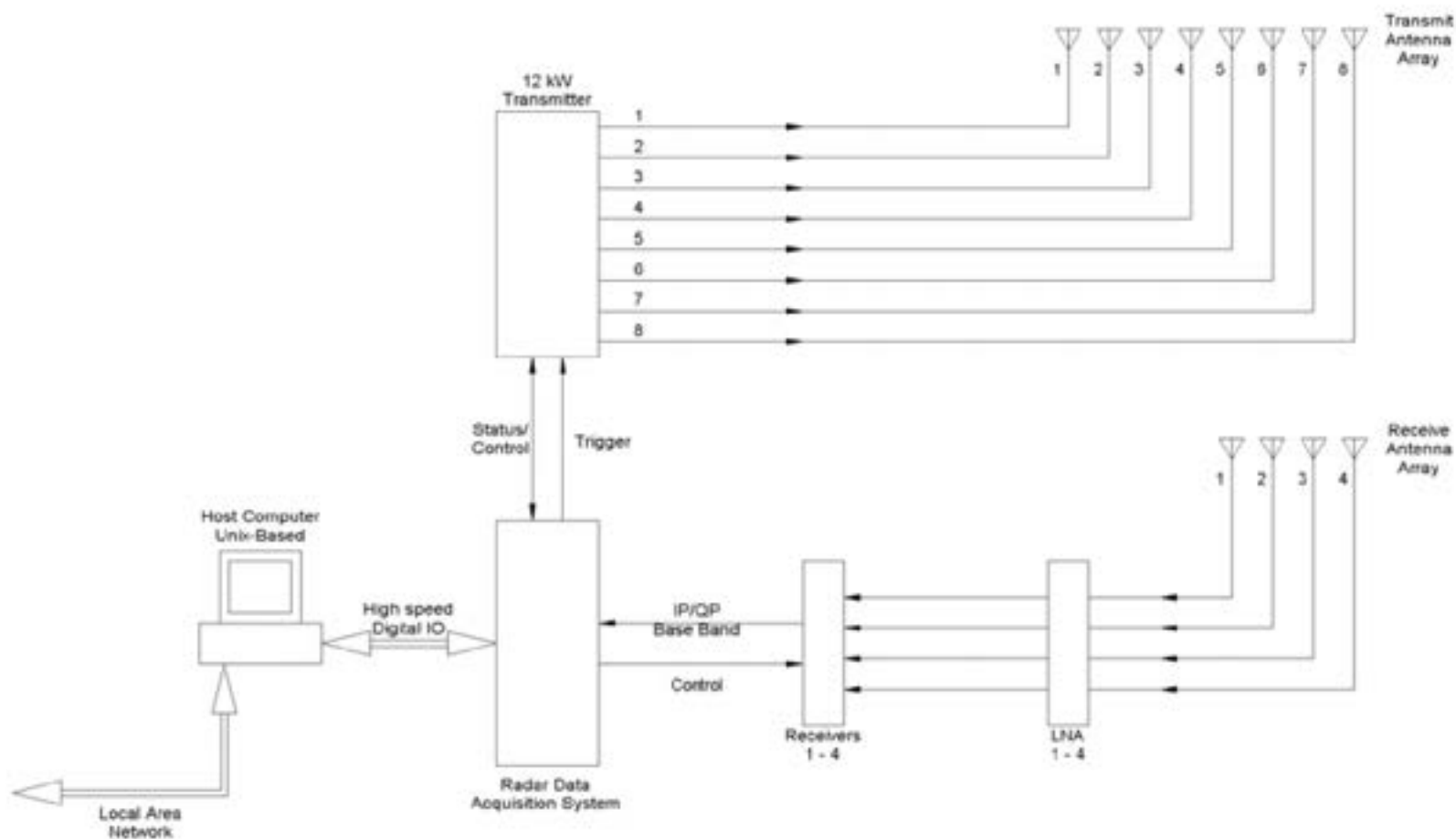
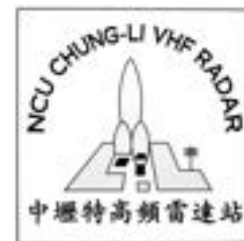
中壢30 MHz 雷達系統參數



座標	地理緯度: 24.9° N ;地理經度: 121° E ; 地磁緯度: 13.3° N ;地磁經度: 189.3° ; 地磁傾角: 35° N
中心頻率	30 MHz (雷達波長10 m)
雷達型式	Bistatic 雷達
尖峰發射功率	12 kW
接收機通道	5
發射脈波長	2 ~ 200 μ s at 0.1 ms step
最大同相積分數	4096
脈波發射週期	100 ms – 10 ms
最大Nyquest頻率	~160 Hz
最大工作週期	7.5%
天線陣列	
天線單元	4-element 八木天線
發射天線陣列	8x1 線性陣列, 長56 m, 半功率波束寬 約 8°x64°
接收天線陣列	3x3 方形陣列, 邊長25 m
波數指向	fixed zenith
接收機頻寬	匹配濾波器: 0.5MHz, 0.25 MHz, 125KHz, 62.5 KHz
相位加碼型式	Complementary Code with 2-64 elements Barker Code with 7, 9, 11, 13 elements



系統功能方塊圖





52 與 30 MHz 八木天線



30 MHz 天線

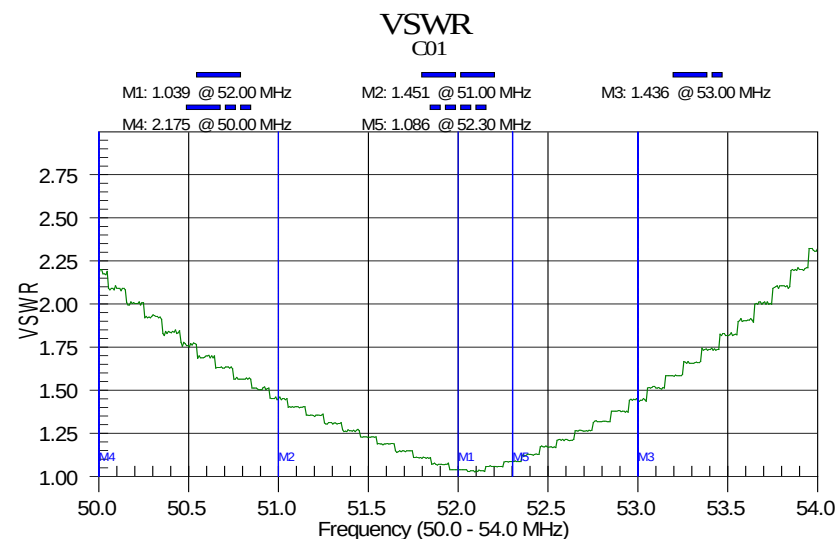
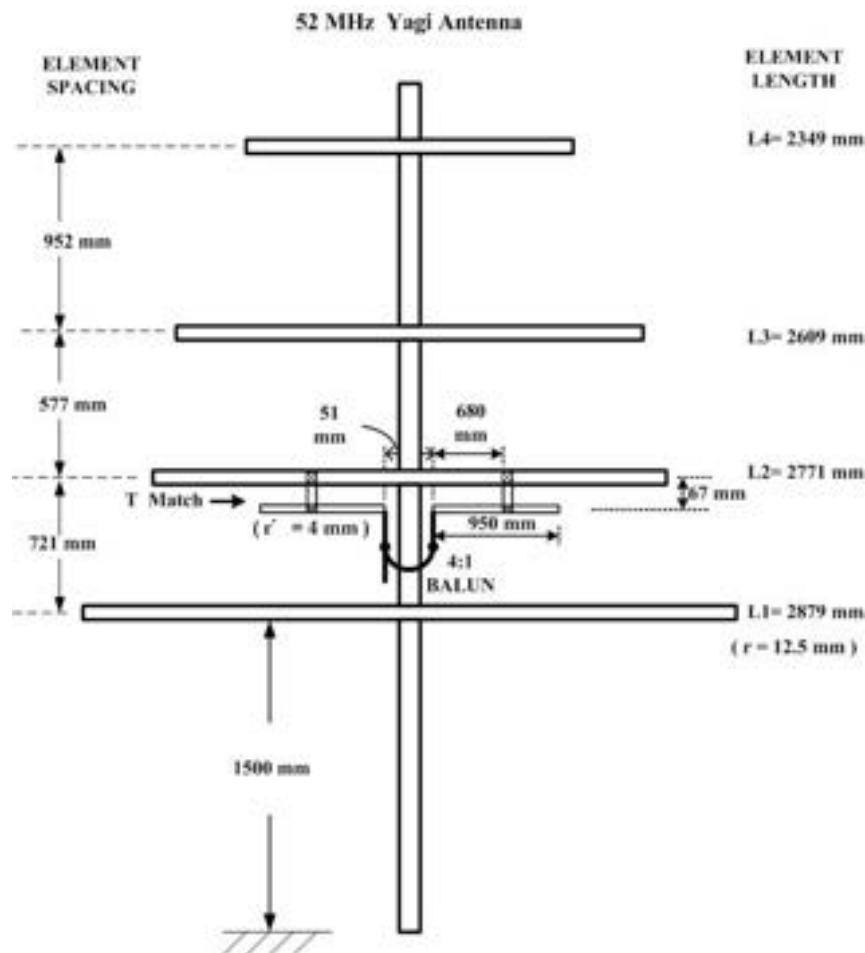


52 MHz 天線





52MHz雷達新天線 頻寬改良設計



Resolution: 517
Std:
Date: 07/23/2006
Model: S331C

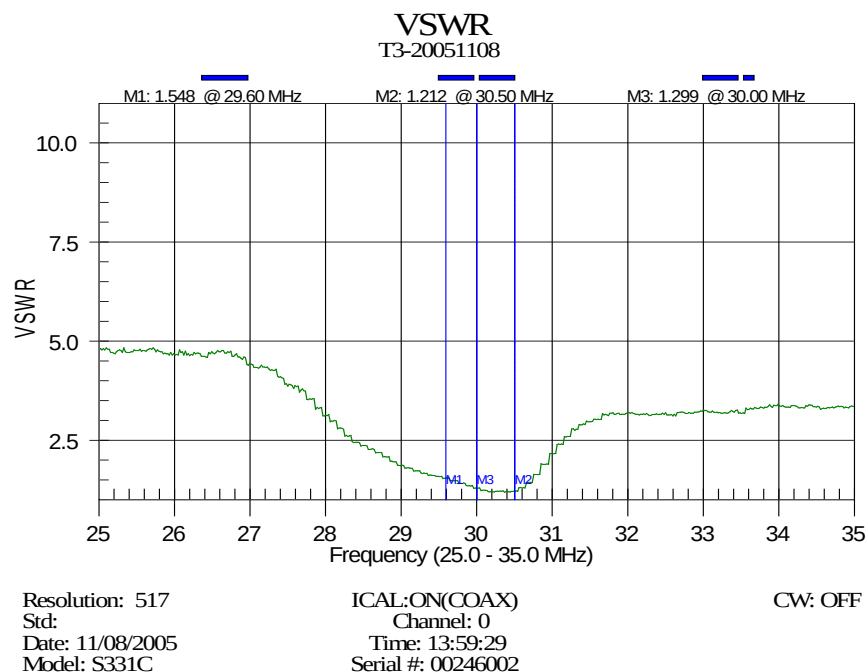
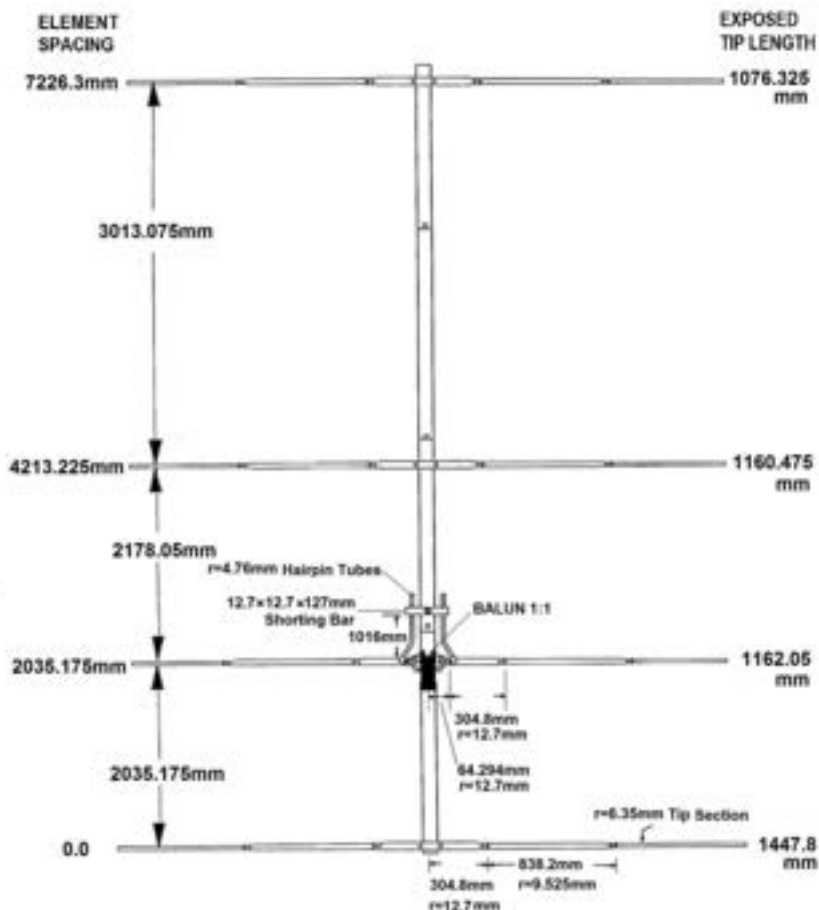
ICAL: ON(COAX)
Channel: 0
Time: 09:52:00
Serial #: 00246002

CW: ON

Bandwidth > 2MHz @ SWR < 1.5



30MHz雷達舊M2天線



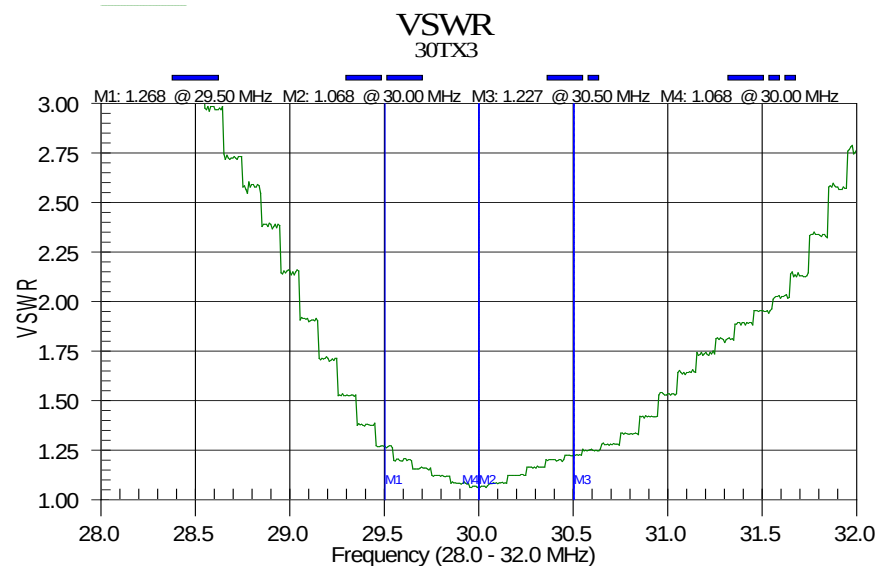
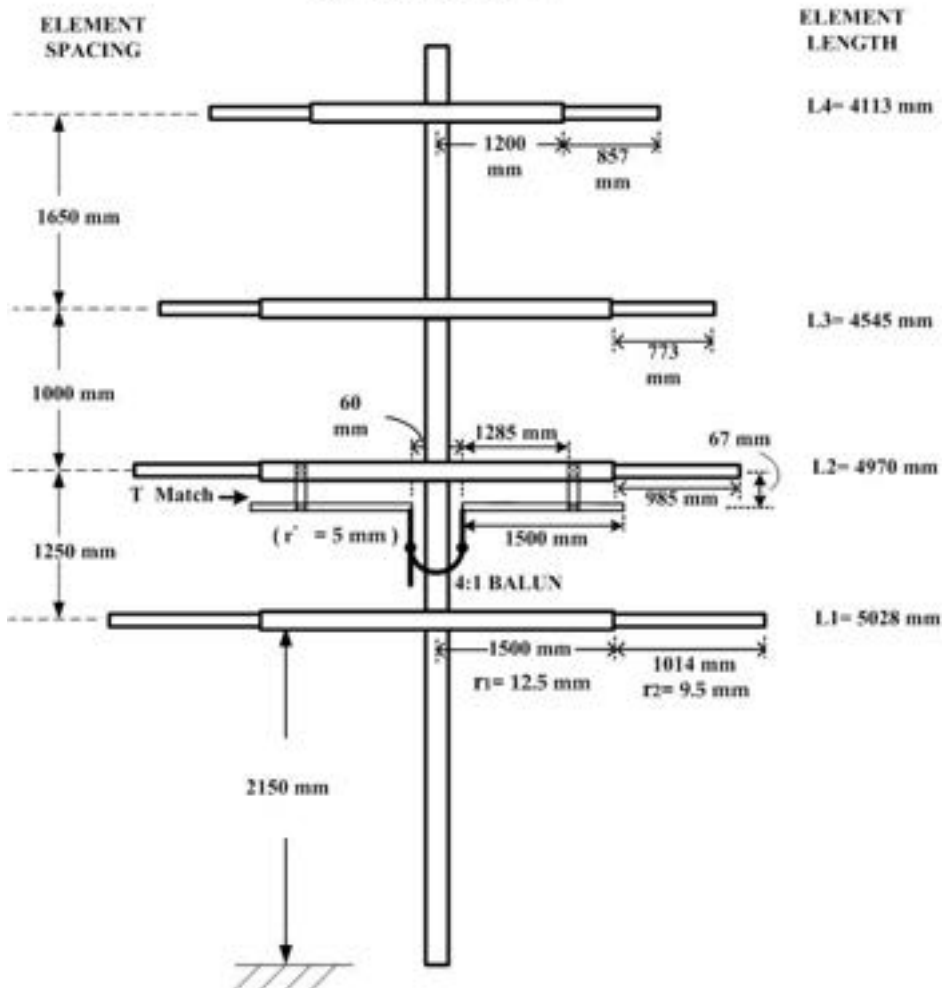
Bandwidth<1MHz @ VSWR<1.5



30MHz雷達新天線



30MHz Yagi Antenna



Bandwidth > 1MHz @ VSWR < 1.5



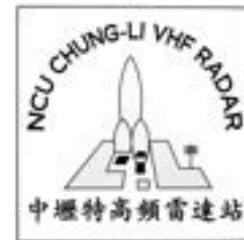
曾參與之國內外重大科研計畫



- 一、國際中層大氣(MAP)探測計畫(1985-1990)
- 二、國際UARS衛星低層太空風場測量計畫(1992)
- 三、日本SEEK-1與SEEK-2探空火箭計畫(1996與1998)
- 四、中華衛星一號傳播通道實驗計畫(1999至2006)
- 五、印度INTAR大氣環流觀測計畫(2005)
- 六、福爾摩沙三號電離層不規則體探測計畫(2006迄今)
- 七、探空火箭2至7號次軌道太空探測計畫(2003至2010)
- 八、國際WINDS探空火箭中性風測量計畫(2007)
- 九、中央氣象局剖風儀(Windprofiler)建置計畫(2009-迄今)
- 十、探空火箭9號電離層不規則體探測計畫(2014/03/26)
- 十一、中央氣象局閃電監測系統建置計畫(2012-迄今)
- 十二、台灣海洋科技中心測海技術研究(2015-迄今)
- 十三、中央氣象局電離層監測系統(2018-迄今)



中央大學雷達站之觀測參數



▲ 低層大氣參數的量測

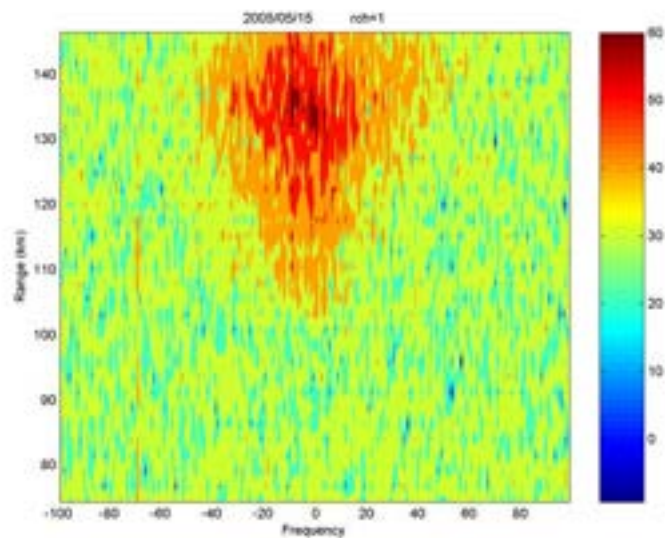
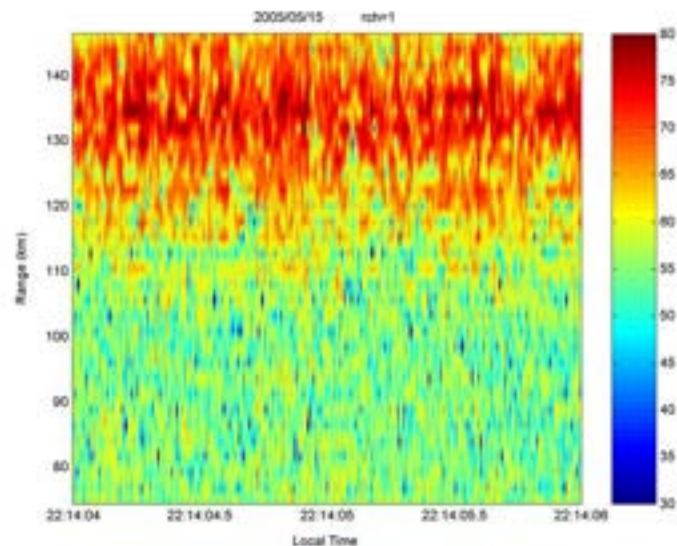
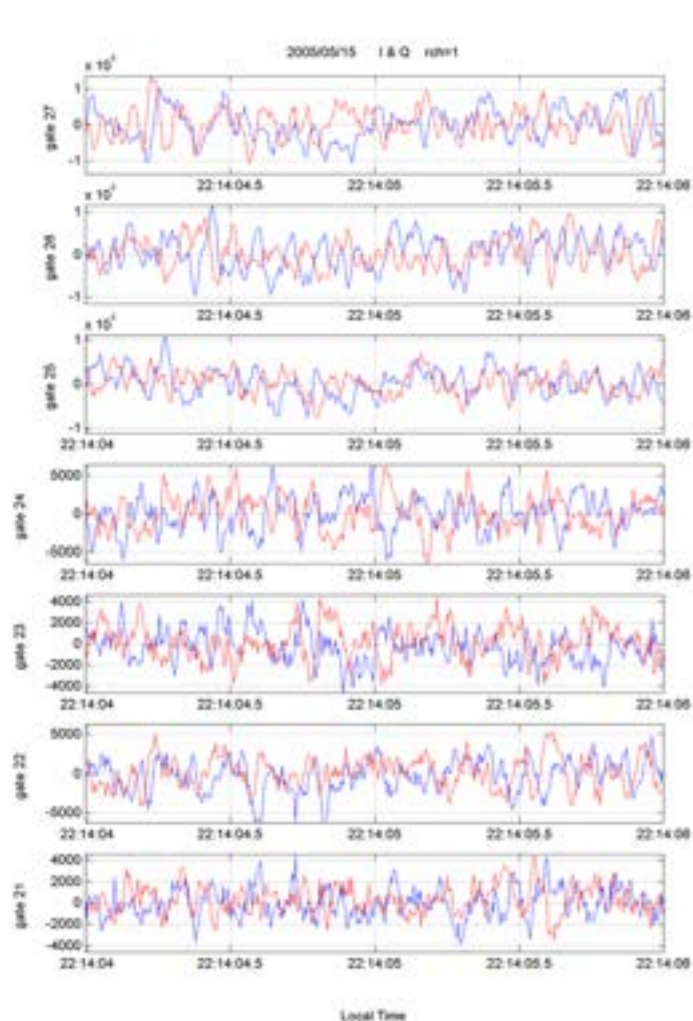
- 大氣溫度與水汽隨高度變化之觀測
- 大氣水平與垂直風場剖面的觀測
- 大氣亂流強度之觀測
- 大氣降雨的觀測
- 對流層頂的觀測等

▲ 高層大氣與太空(電離層)參數的量測與定位

- 流星現象之觀測與定位
- 中氣層(70-100公里)風場的觀測
- 電離層電子密度不規則體之結構，漂移與定位
- 電離層電漿亂流強度的觀測
- 電離層最大電子密度(高頻電波最高可通訊頻率)的觀測
- 背景高頻電波的監測(1-30MHz)，等

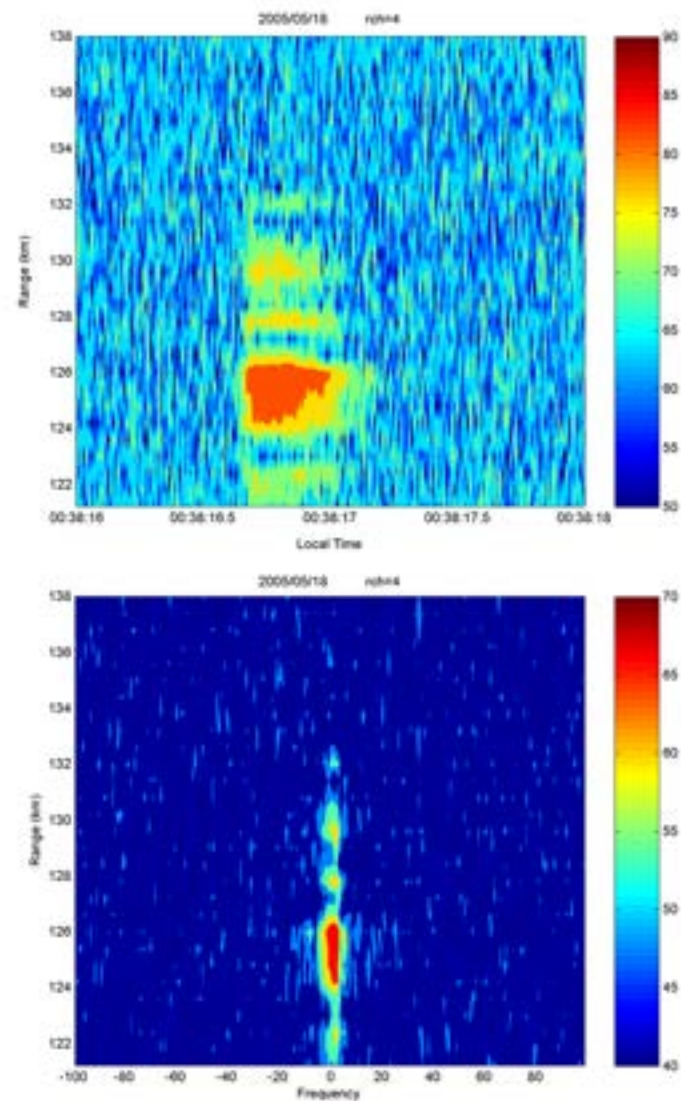
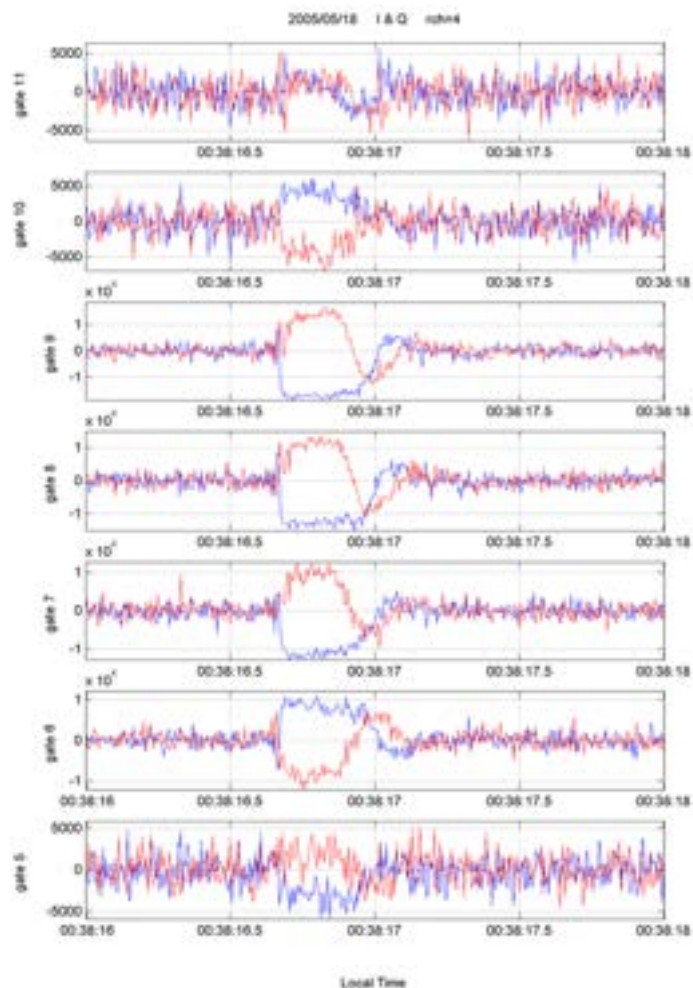


雷達訊號(電離層)



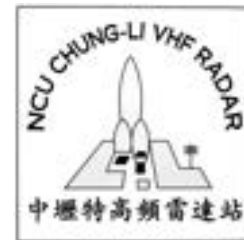


雷達訊號(流星)



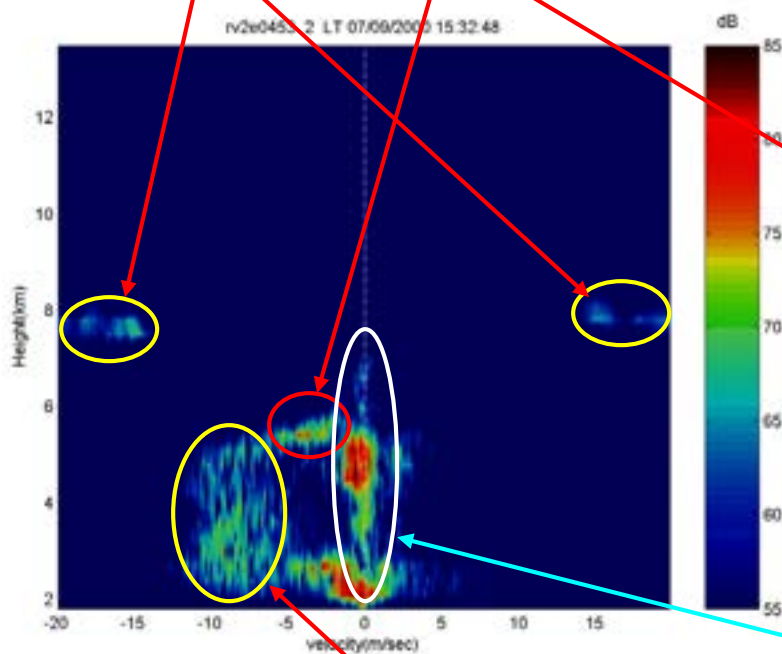


雷達訊號(降水)

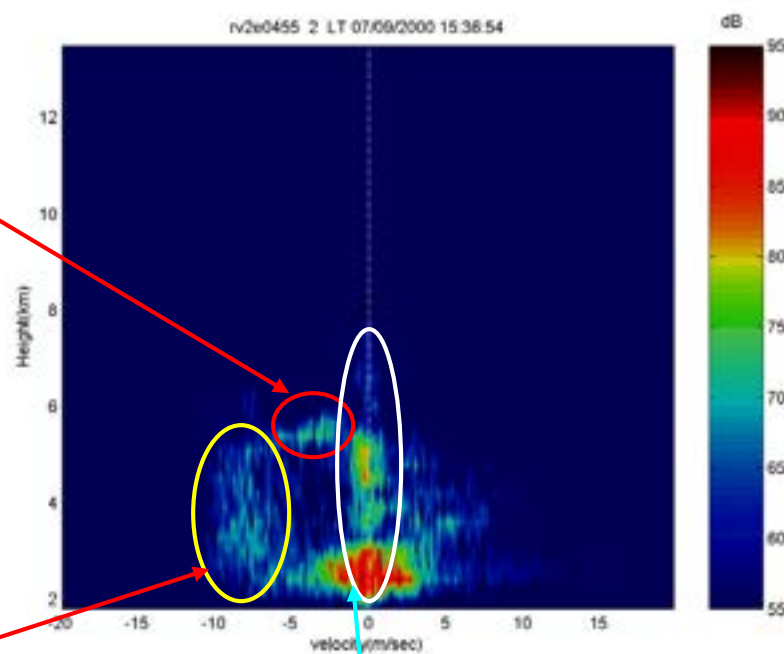


飛機訊號

溶解層訊號



降水回波訊號



大氣回波訊號



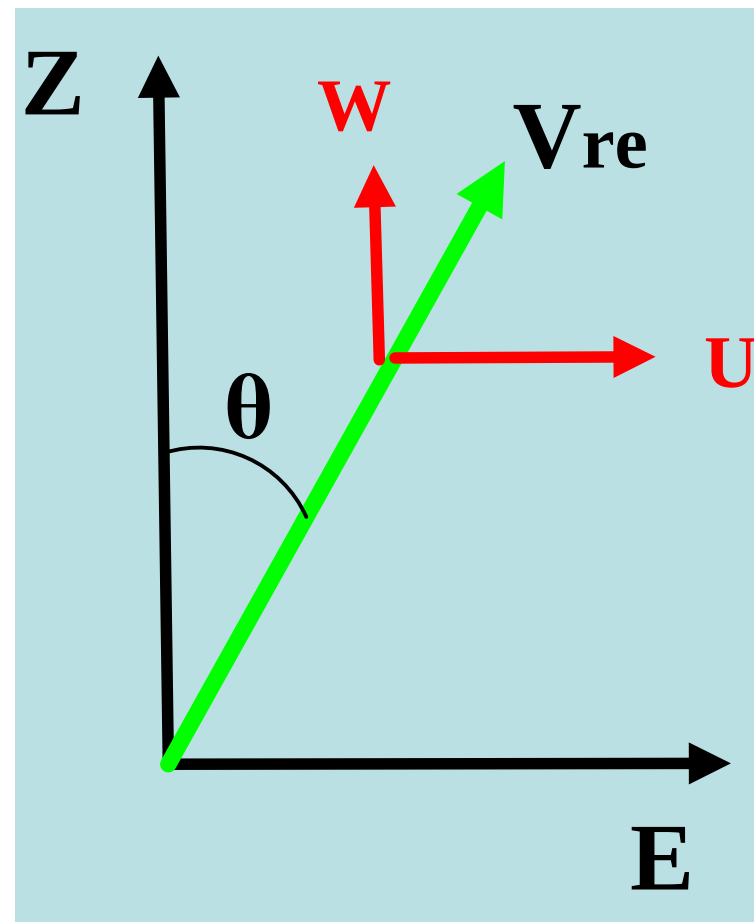
大氣水平風場估算



$$V = \frac{(V_{rn} - W \cdot \cos(\theta))}{\sin(\theta)}$$

$$U = \frac{(V_{re} - W \cdot \cos(\theta))}{\sin(\theta)}$$

W是雷達垂直指向觀測到的速度
V_{re}, V_{rn}是雷達傾斜指向觀測的徑向速度
 θ 是雷達波束傾角
V是水平風速的向北分量
U是水平風速的向東分量

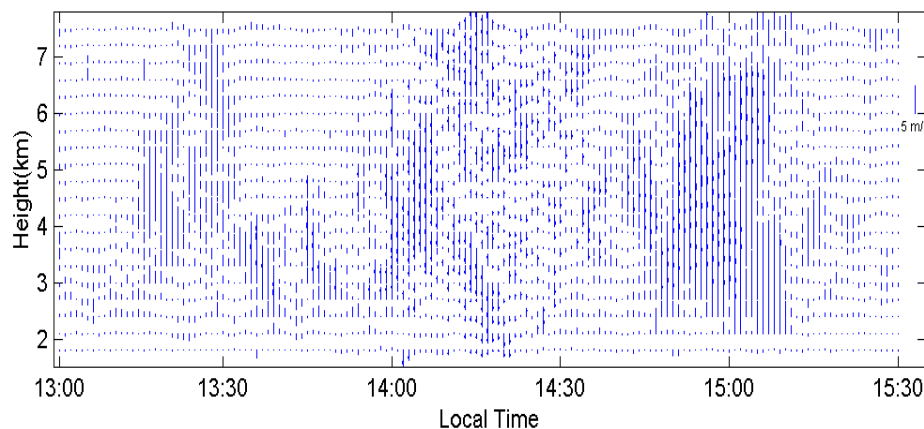




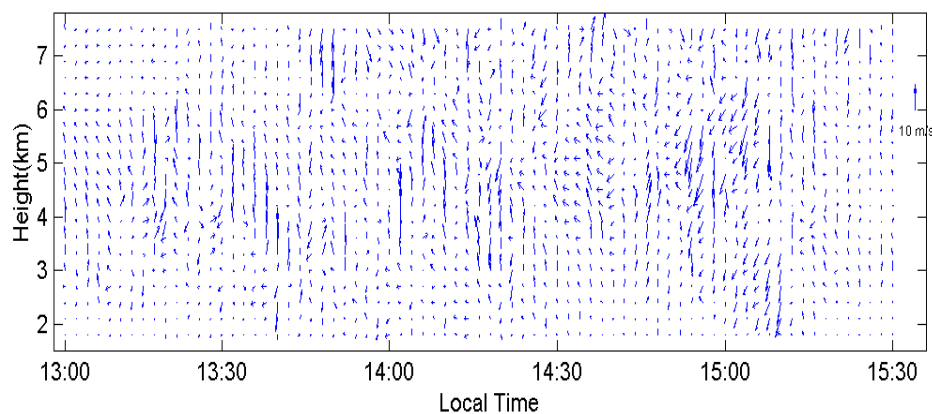
大氣風場與降雨觀測實例



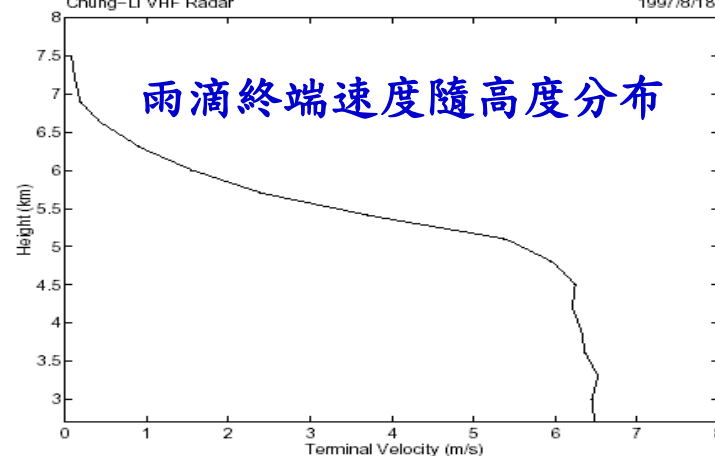
Vertical wind velocity (1999/09/14)



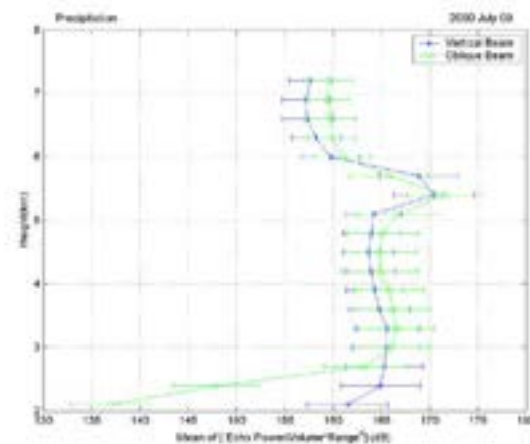
Horizontal wind velocity (1999/09/14)



Chung-Li VHF Radar 1997/8/18



降雨強度隨高度分布

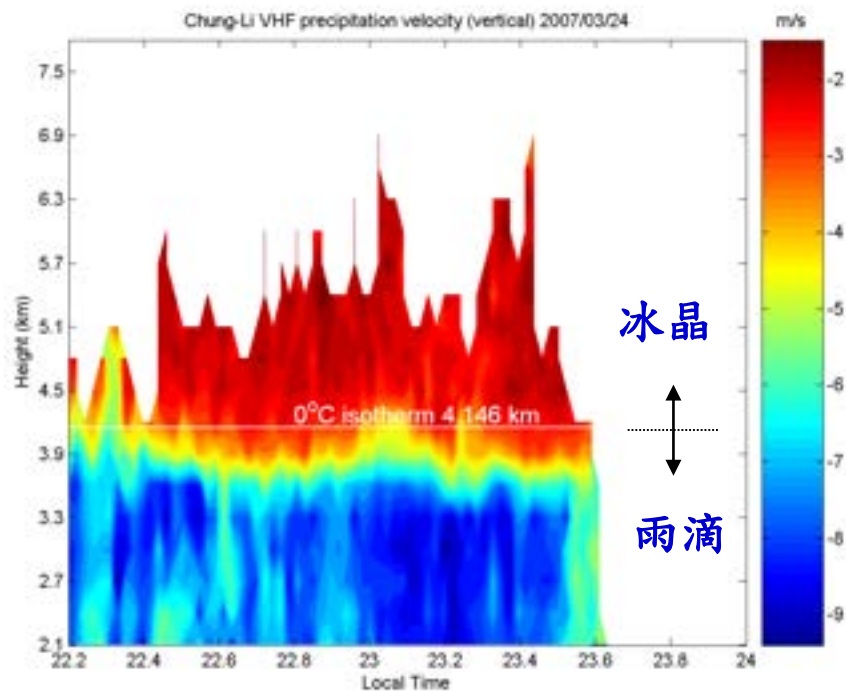




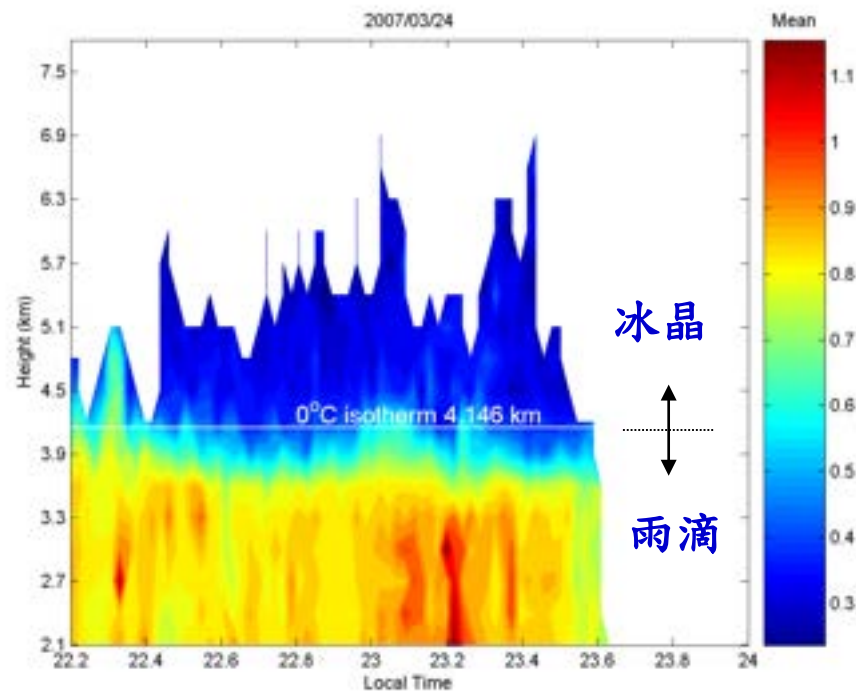
大氣降雨觀測實例



降水顆粒下墜速度隨高度之分布



降水顆粒大小隨高度之分布

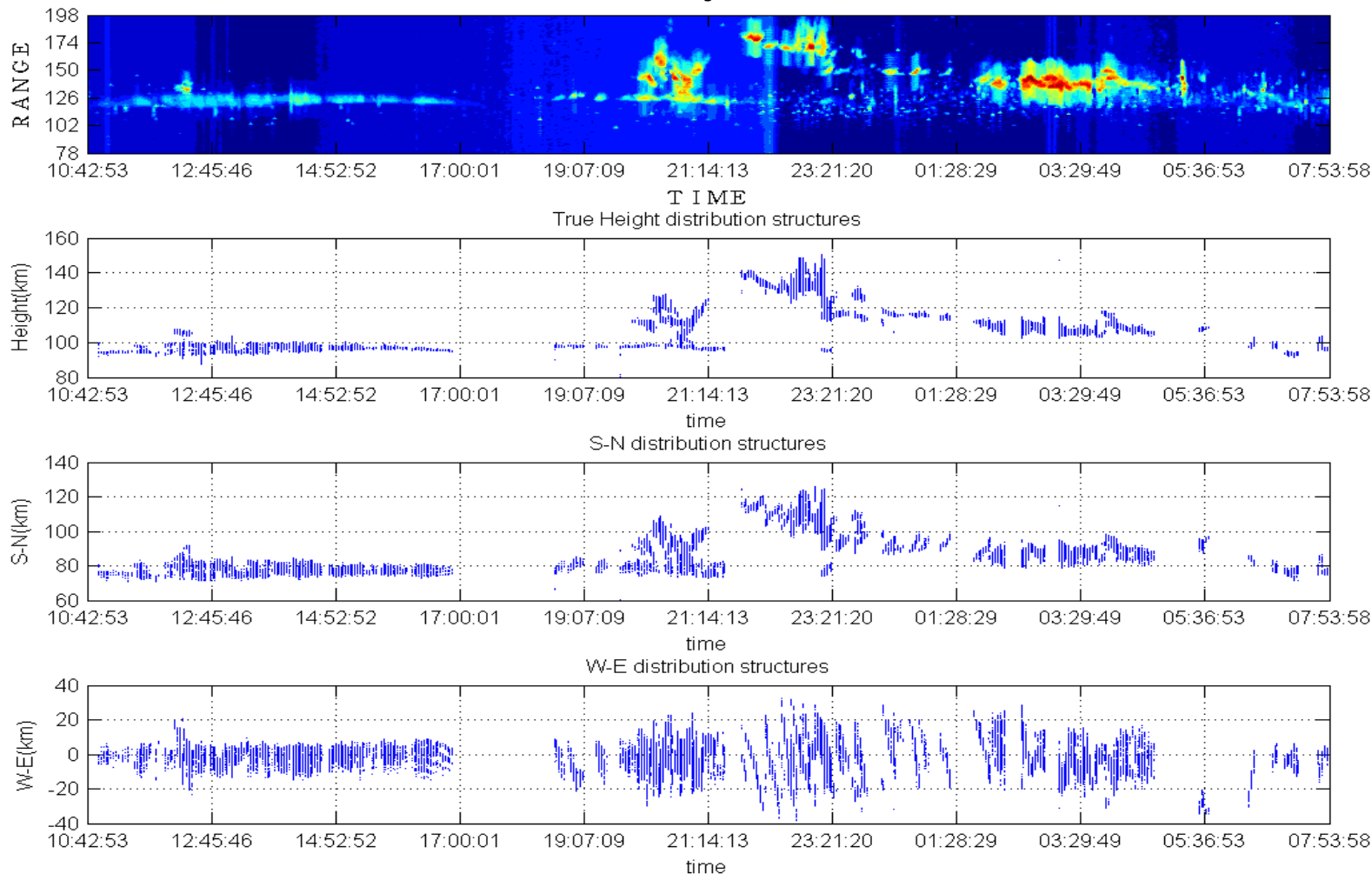




電離層E域電子密度不規則體的 觀測與三維定位

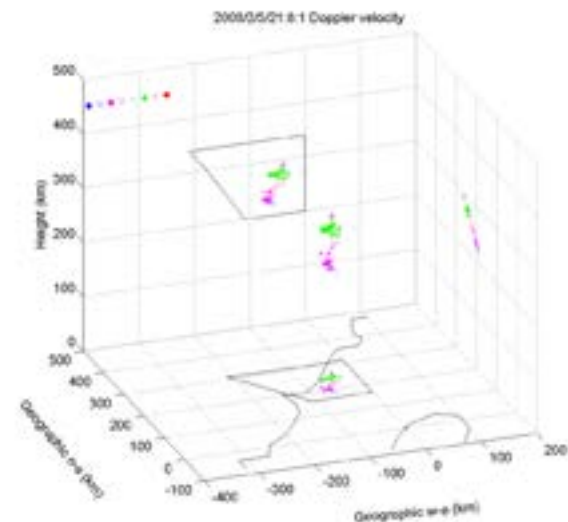
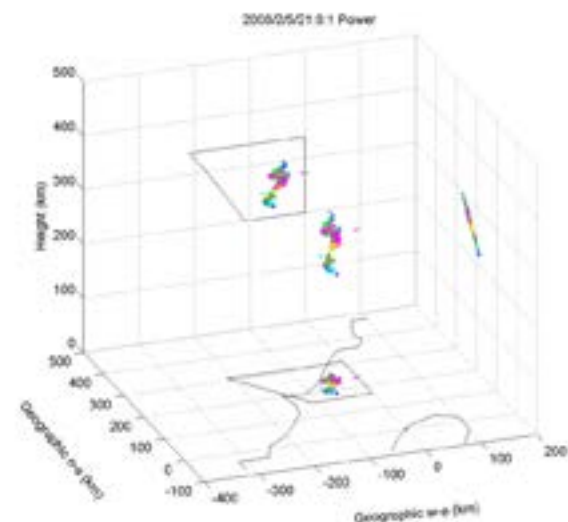
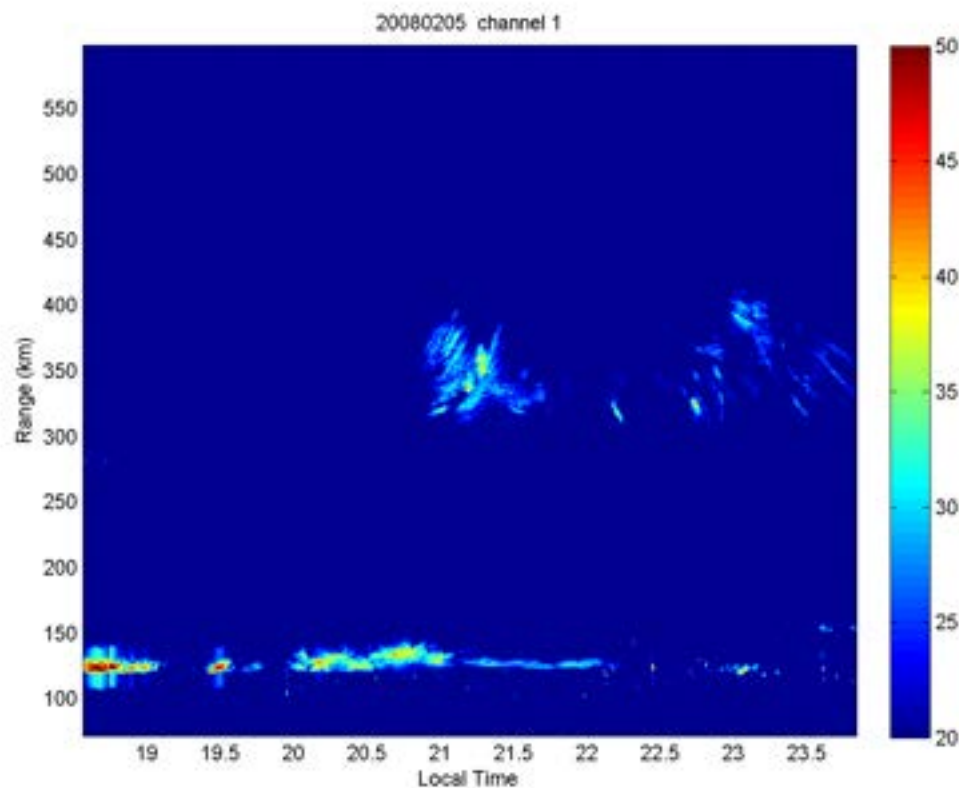
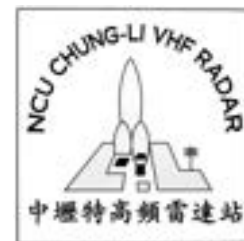


19-20 July 2009



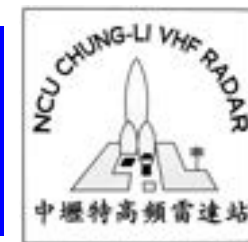


電離層F域電子密度不規則體的 觀測與三維定位



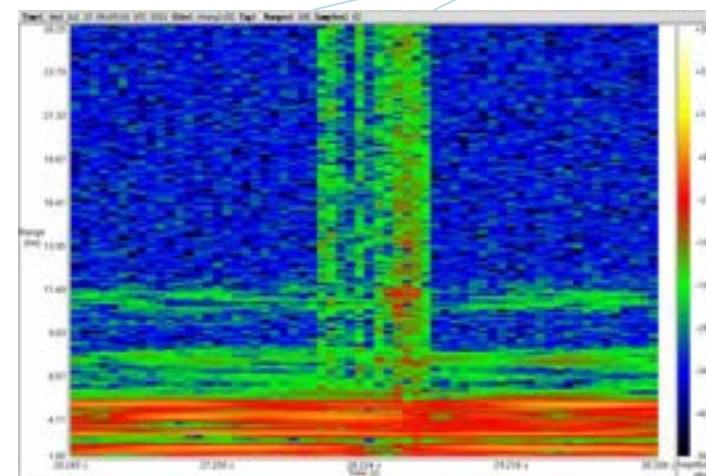
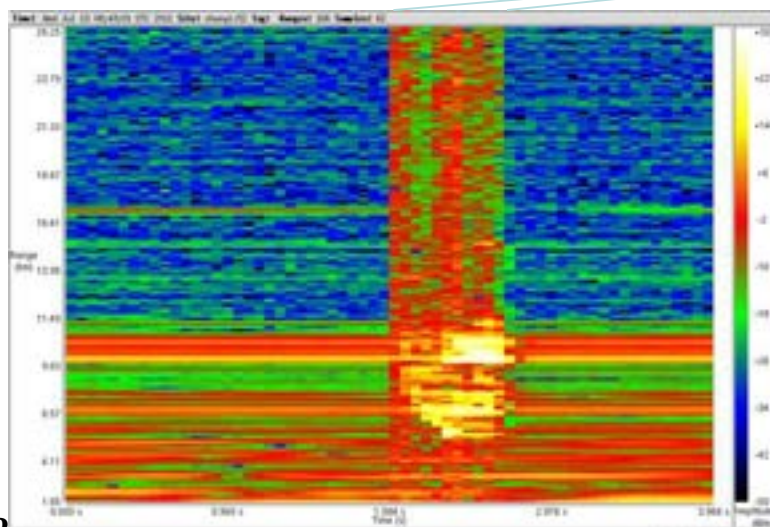
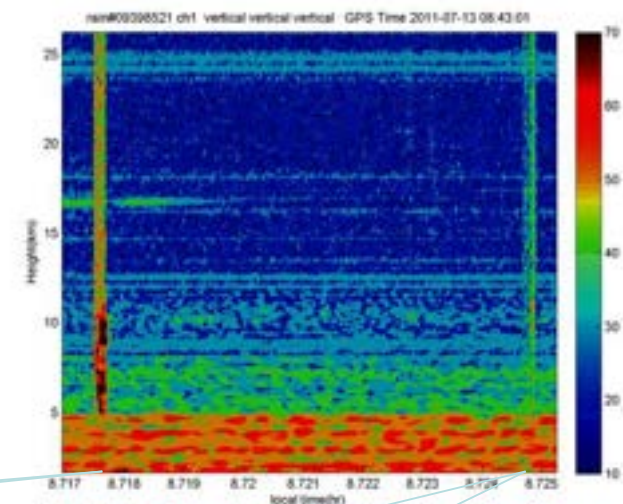


閃電主動偵測系統觀測參數



- 中壢特高頻雷達於2011年7月13日觀測低層大氣時，亦觀測到閃電事件。(ST陣列)

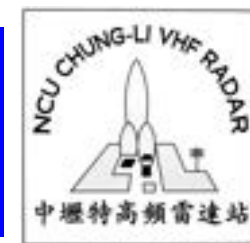
脈波寬Pulse width	1 μ s(空間解析度150m)
發射週期Interpulse period	250 μ s
同相積分coherent integration	256 times
時間解析度time resolution	64 ms





主動式閃電偵測系統簡介

室內部分



中心頻率 Central Frequency	52 MHz
放大器型式 Type of Amplifier	固態功率放大器 Solid state power amplifier
峰值總功率 Total peak power	$\geq 20\text{kW}$ @ 7.5% duty cycle
單一抽屜功率模組峰值功率 Peak envelope RMS power per Module	$\geq 5\text{ kW}$ @ 7.5% duty cycle per unit chassis
功率模組數目 Number of Power Module	4
最大工作週期 Maximum Duty Cycle	10% for 5 kW per Module
脈波寬 Maximum pulse length	1~200 μs
最大電壓駐波比 Maximum VSWR	≤ 1.5
狀態顯示內容 Status Readouts Items	輸出功率 (Output power 單位 dBm) 輸出相位 (Output phase 單位 Degrees) 功率放大器溫度 (PA temperature 單位 $^{\circ}\text{C}$)
警告 Alarm conditions	VSWR 過高，風扇故障.





新屋氣象站

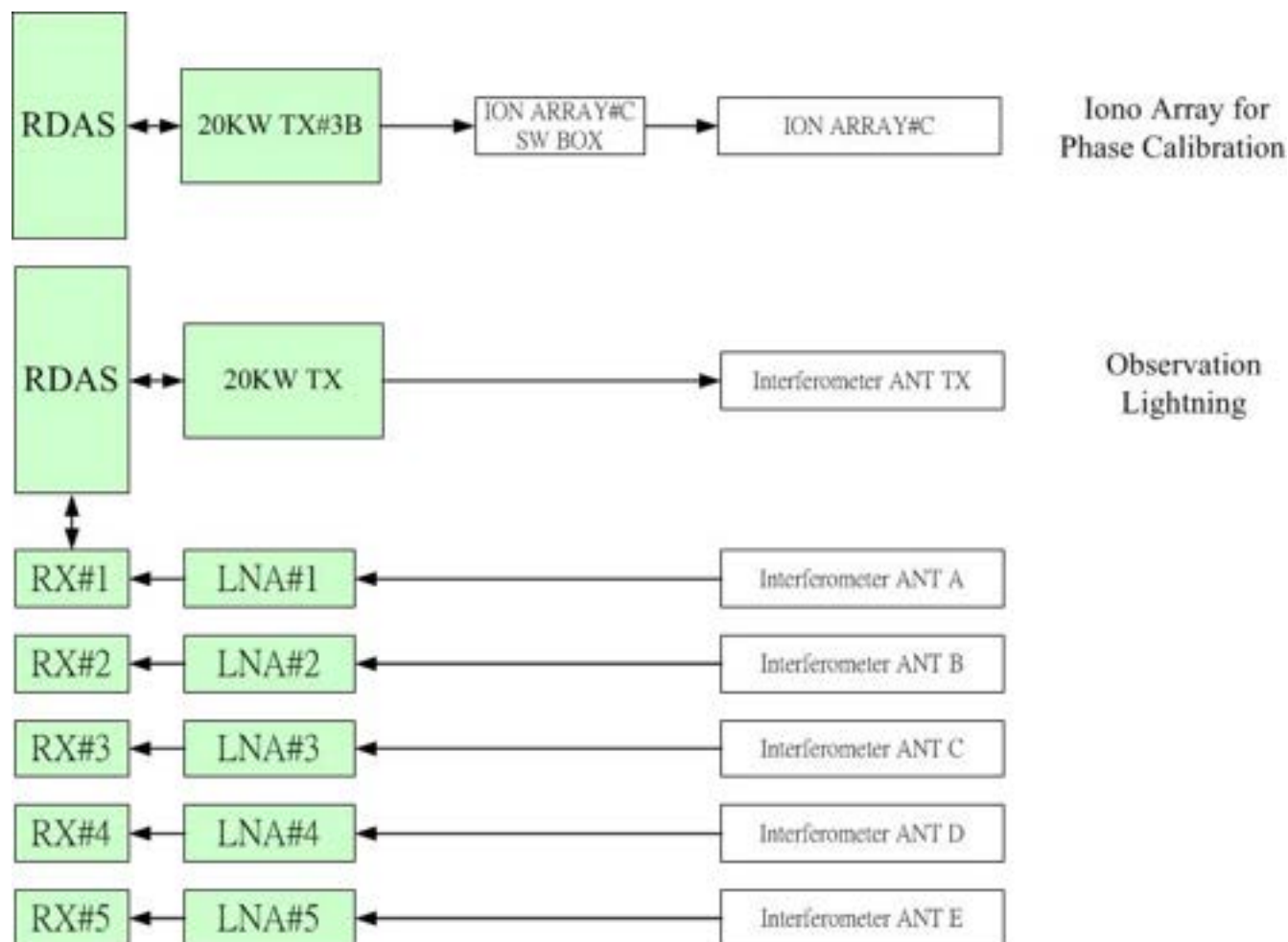
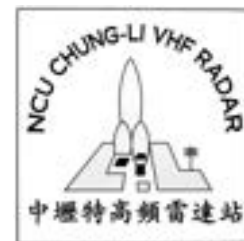
主動式閃電偵測實驗





主動式閃電偵測系統方塊圖

相位校正與觀測系統





主動式閃電偵測系統簡介

不等間距十字型干涉天線陣列



- 系統相位校正觀測模式

以一座4x8電離層天線陣列發射，五根獨立天線接收

- 閃電觀測模式

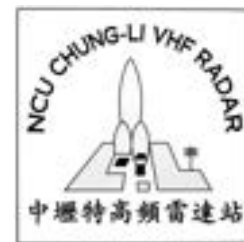
以一根獨立天線發射，五根獨立天線接收，用以全方位觀測定位





52MHz雷達

接收系統相位校準技術



•接收系統包含天線、電纜線、內部系統，經由利用電離層場列不規則體(Field-aligned Irregularities)回波訊號與國際地磁參考模型(IGRF)，進行系統相位校正

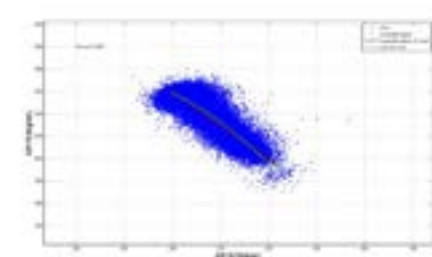
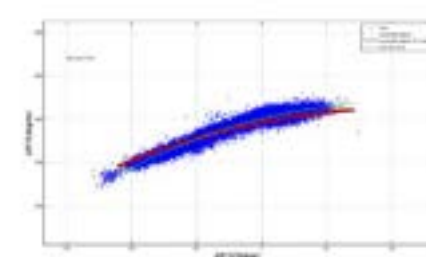
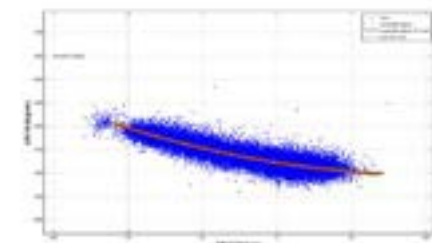
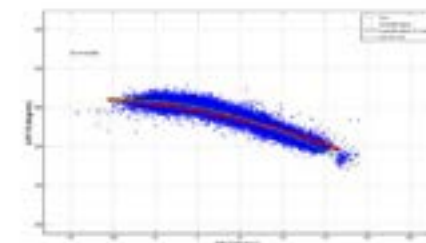
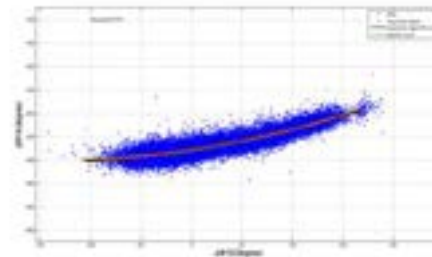
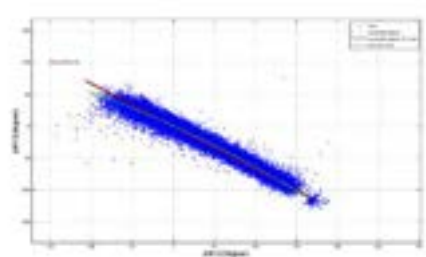
天線1與天線2相位差校正量
 $\Delta\psi_{12} = 37.1^\circ$

天線1與天線3相位差校正量
 $\Delta\psi_{13} = -66^\circ$

天線1與天線4相位差校正量
 $\Delta\psi_{14} = 79.5^\circ$

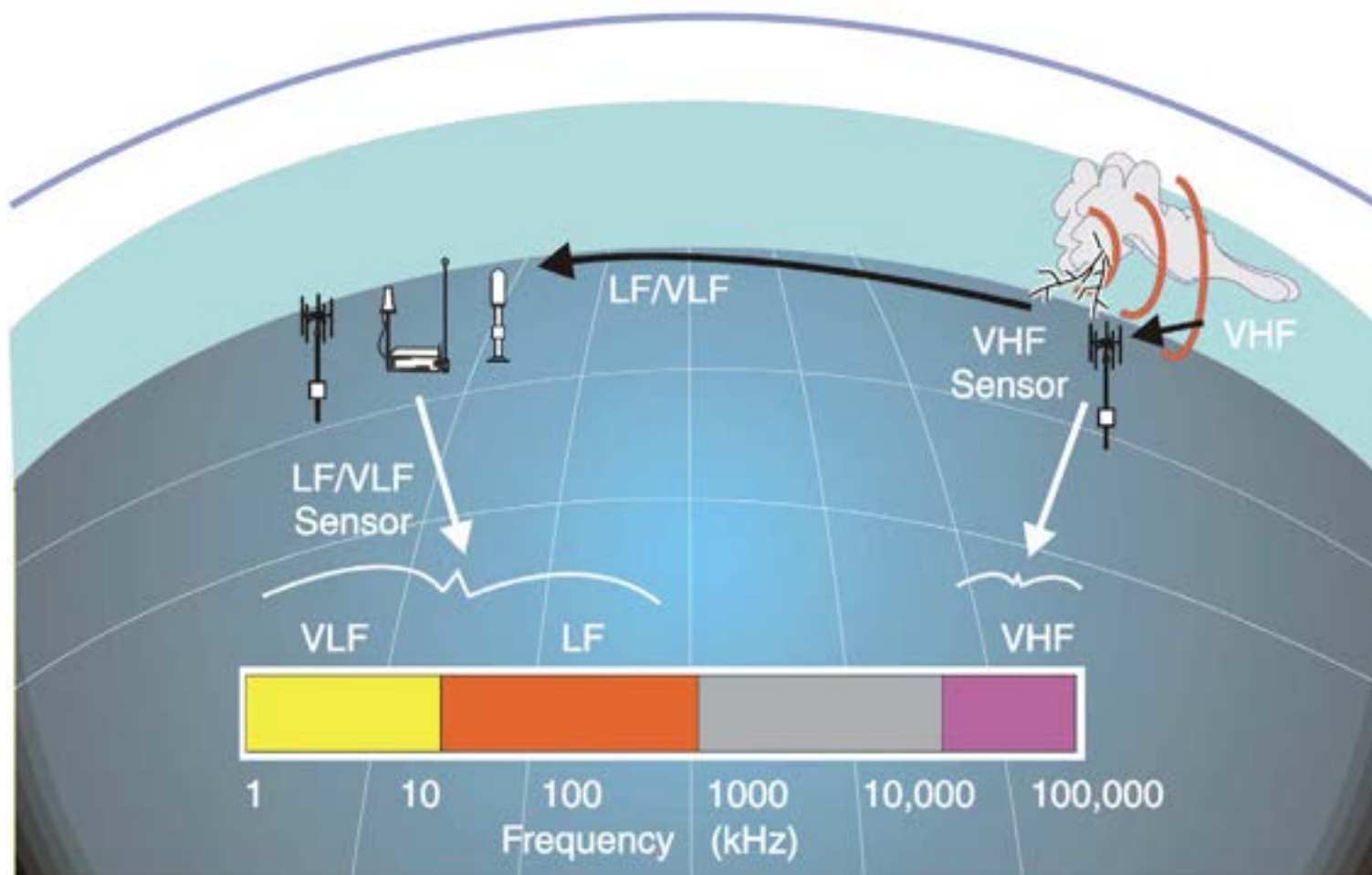
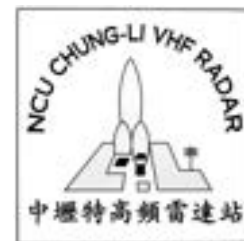
天線1與天線5相位差校正量
 $\Delta\psi_{15} = -85.4^\circ$

(中央校園雷達站天線陣列)





閃電射頻信號輻射





閃電三維結構干涉法測量示意圖

被動式接收



雲間閃電

雲對地閃電

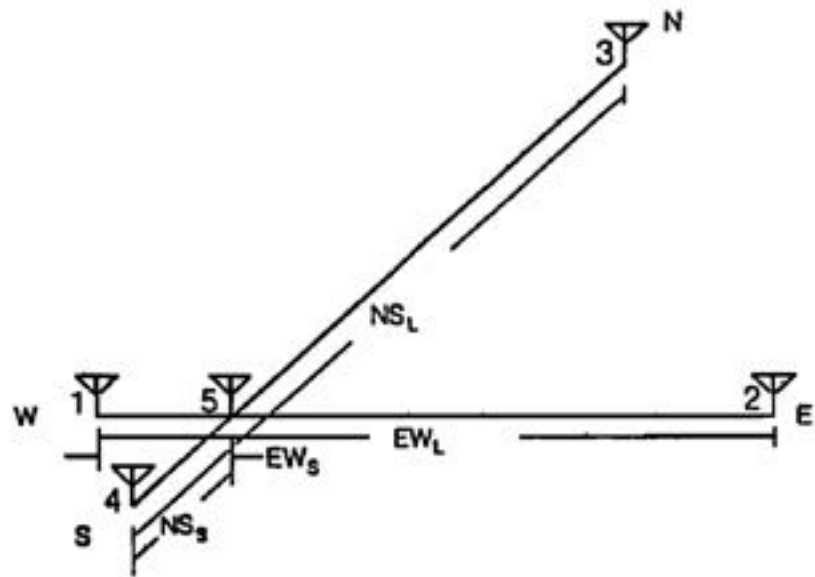


Figure 1. Configuration of the interferometer antennas. Five antennas were shared to produce short (0.5λ) and long (4λ) baselines in orthogonal directions.

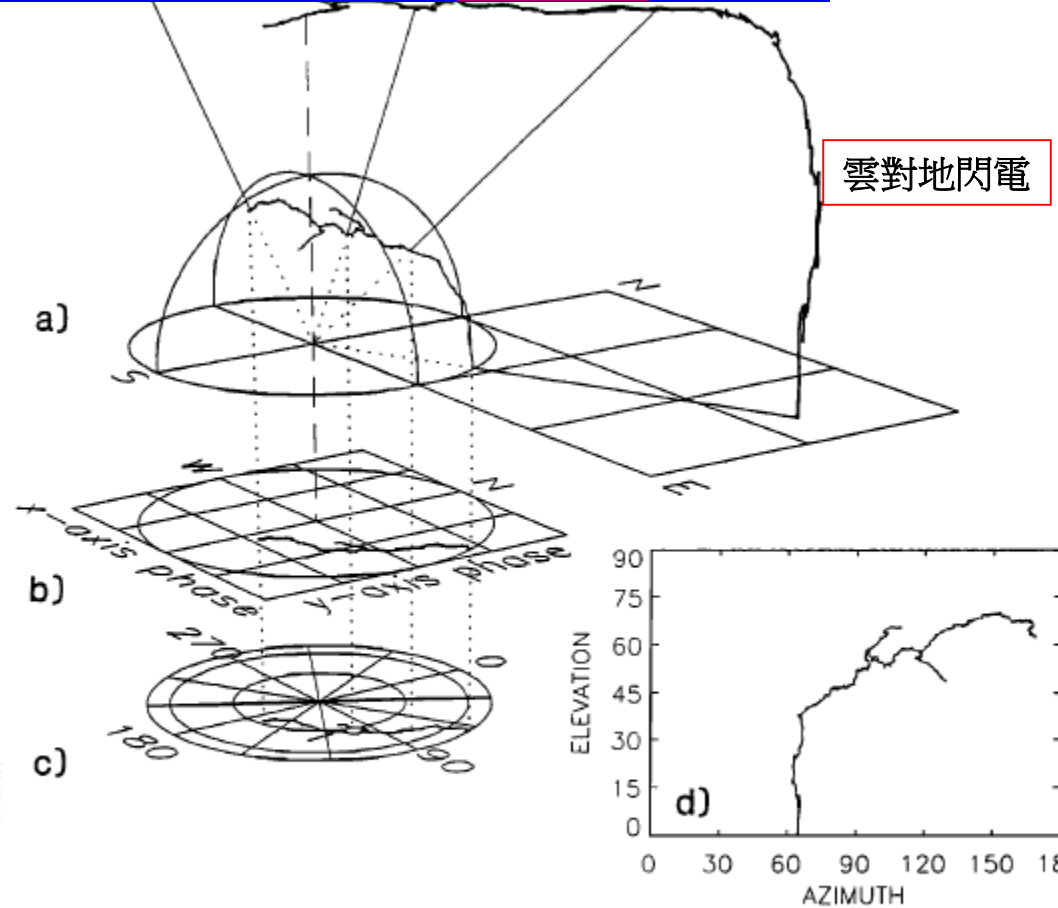


Figure 2. Interpretation of the interferometer measurements. (a) Radiation from the three-dimensional lightning channels can be considered as originating on a unit radius celestial hemisphere. (b) The normalized phase measurements from orthogonal baselines correspond to the projection of the sources from the celestial hemisphere onto the plane of the antenna array. (c),(d) In turn, the phase values indicate the azimuth and elevation of the sources.



十字型不等間距干涉天線陣列 雷達空間干涉定位法



- 方位角 $\phi = \tan^{-1} \left(\frac{\frac{\Phi_{12}}{\pi} C_{13} - \frac{\Phi_{13}}{\pi} C_{12}}{\frac{\Phi_{15}}{\pi} C_{14} - \frac{\Phi_{14}}{\pi} C_{15}} \cdot \frac{3C_{14} + 2C_{15}}{3C_{13} + 2C_{12}} \right)$

- 天頂角

$$\theta = \sin^{-1} \left(\frac{\frac{\Phi_{12}}{\pi} C_{13} - \frac{\Phi_{13}}{\pi} C_{12}}{\sin \phi (3C_{13} + 2C_{12})} \right) = \sin^{-1} \left(\frac{\frac{\Phi_{15}}{\pi} C_{14} - \frac{\Phi_{14}}{\pi} C_{15}}{\cos \phi (3C_{14} + 2C_{15})} \right)$$

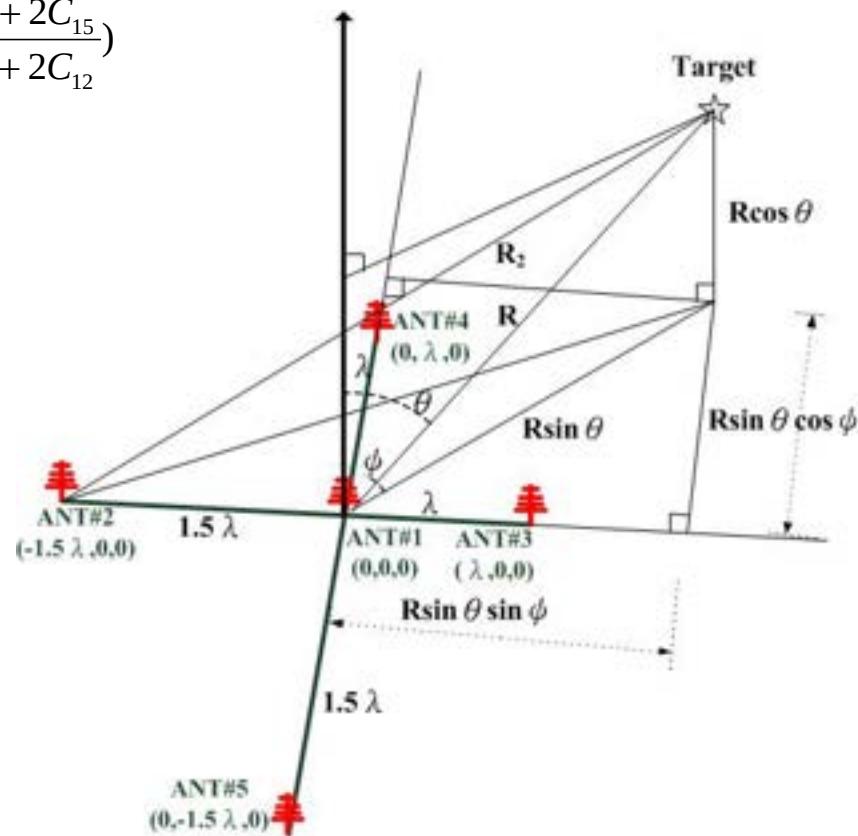
$$C_{12} = \frac{\Phi_{12}^2}{4\pi^2} - 2.25$$

$$C_{13} = \frac{\Phi_{13}^2}{4\pi^2} - 1$$

$$C_{14} = \frac{\Phi_{14}^2}{4\pi^2} - 1$$

$$C_{15} = \frac{\Phi_{15}^2}{4\pi^2} - 2.25$$

- 距離: 由雷達距離閘 (Range Gate) 決定。





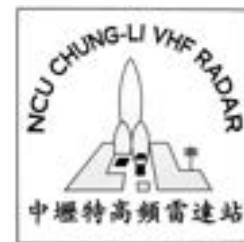
主動式閃電偵測系統 觀測參數



- **PRF 4000 Hz**
- **同相積分(Number of Coherent Integration) 8 次**
- **發射脈波長(pulse length) $1\mu\text{s}$**
- **空間解析度(range resolution) 150 m**
- **時間解析度(time resolution) 2 ms**
- **發射功率 20 KW**
- **使用單根天線發射**
- **五根干涉天線陣列接收**

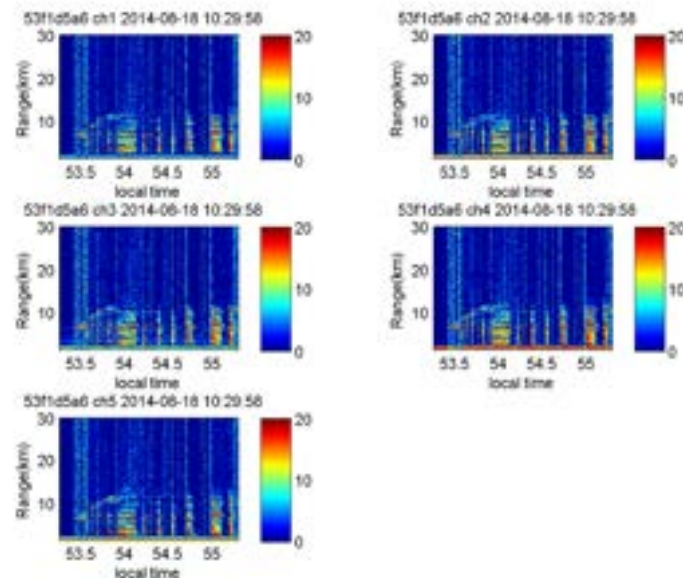
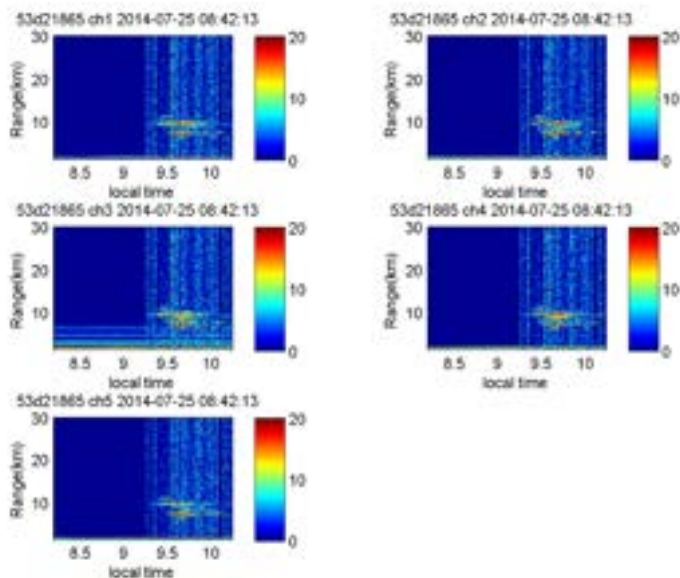


2014年主動式閃電偵測訊號



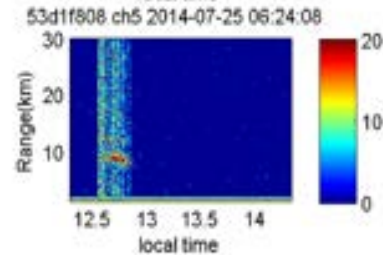
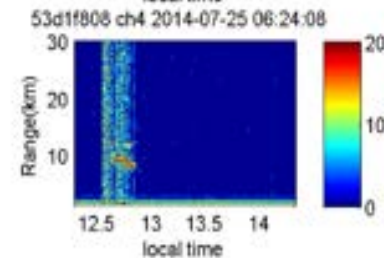
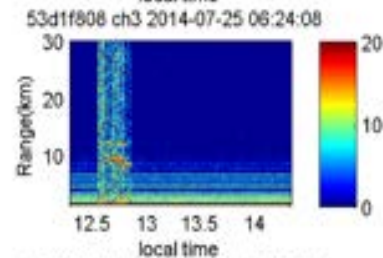
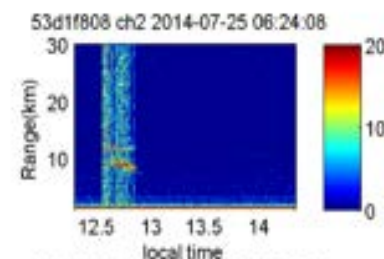
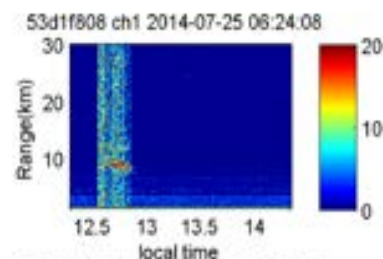
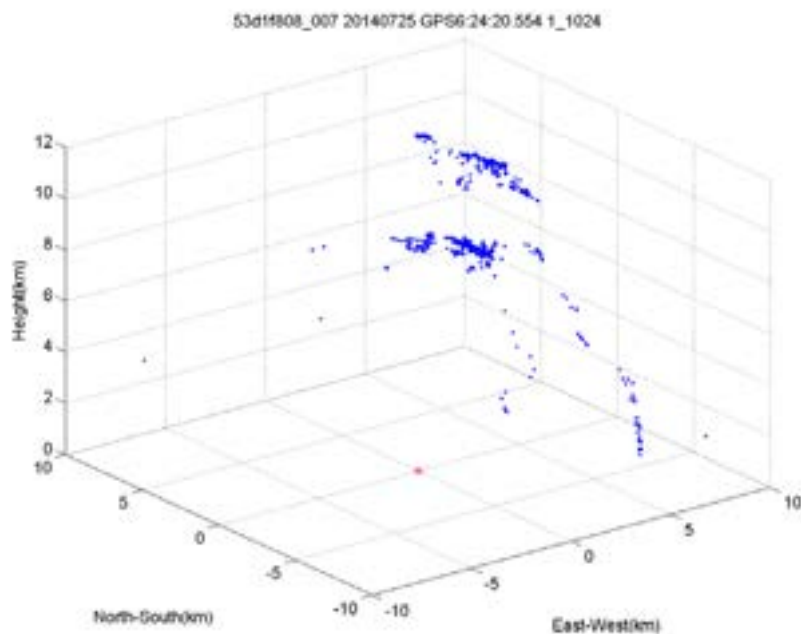
7月25日 GPS08:25

8月18日 GPS10:29



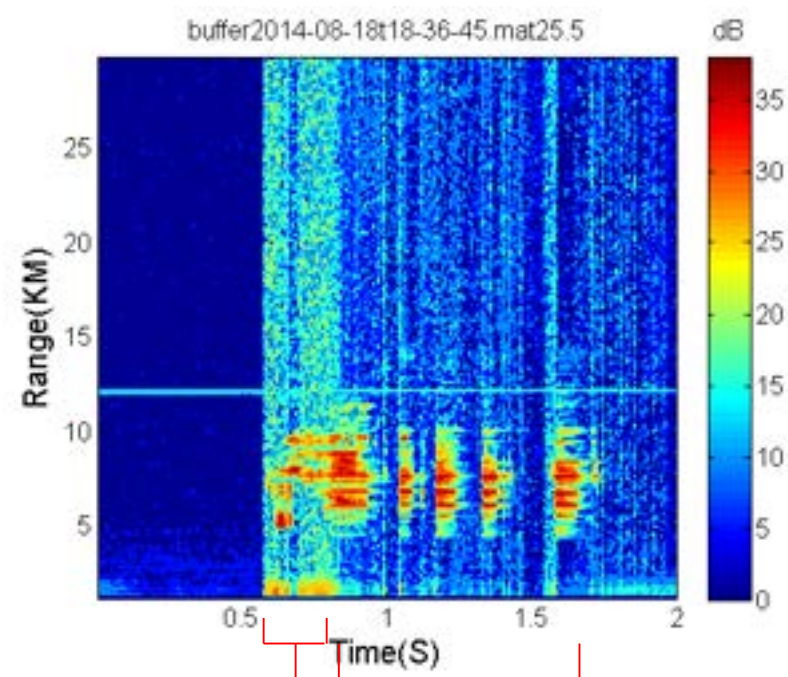
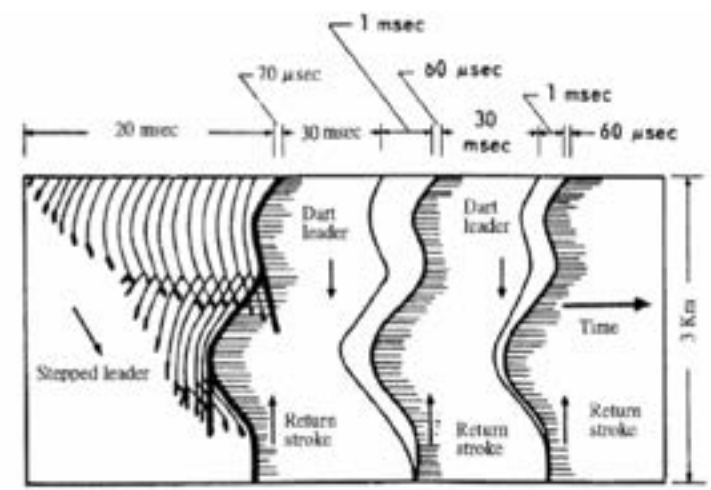
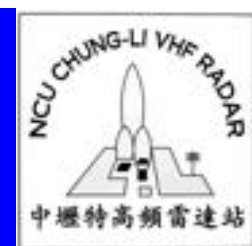


主動式閃電偵測訊號 3D結構圖



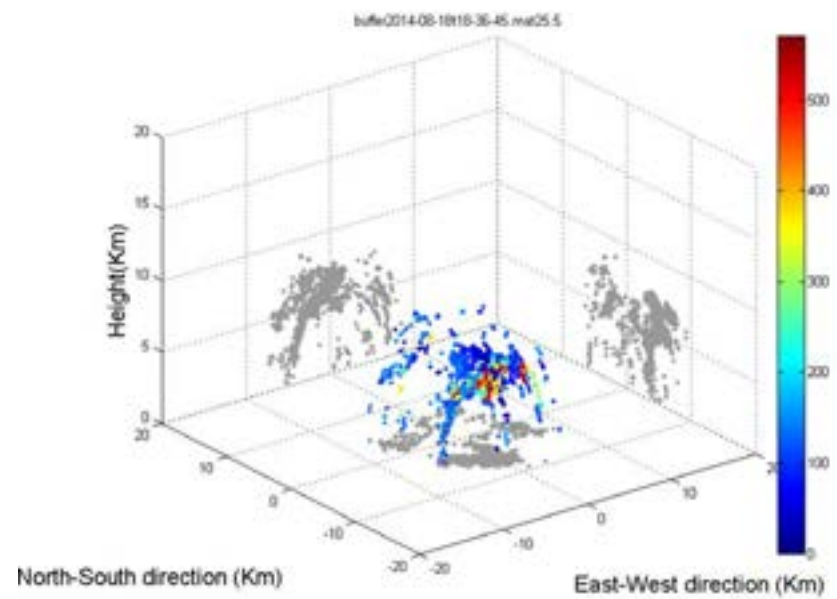


階梯先導/激射導流/回擊 stepped leader/Dart Leader/Return Stroke



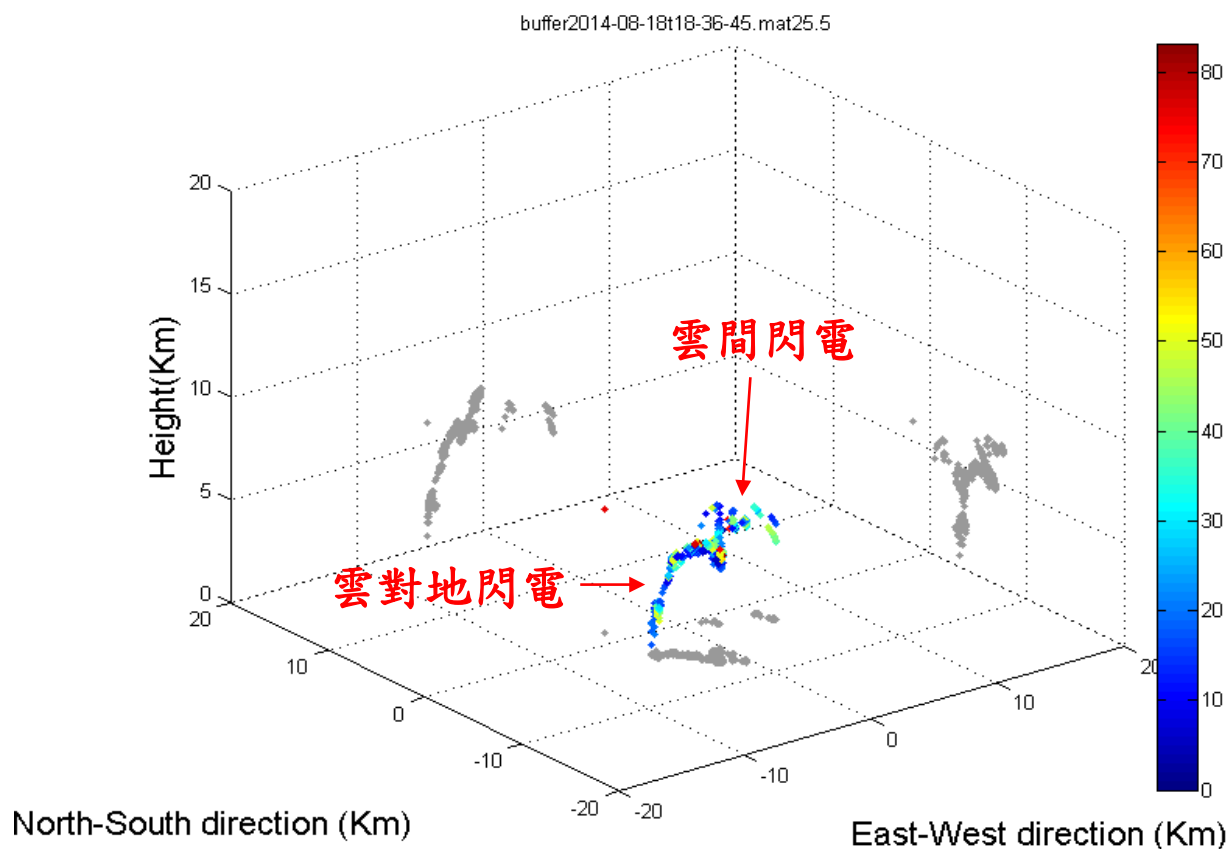
階梯先導

激射導流與回擊



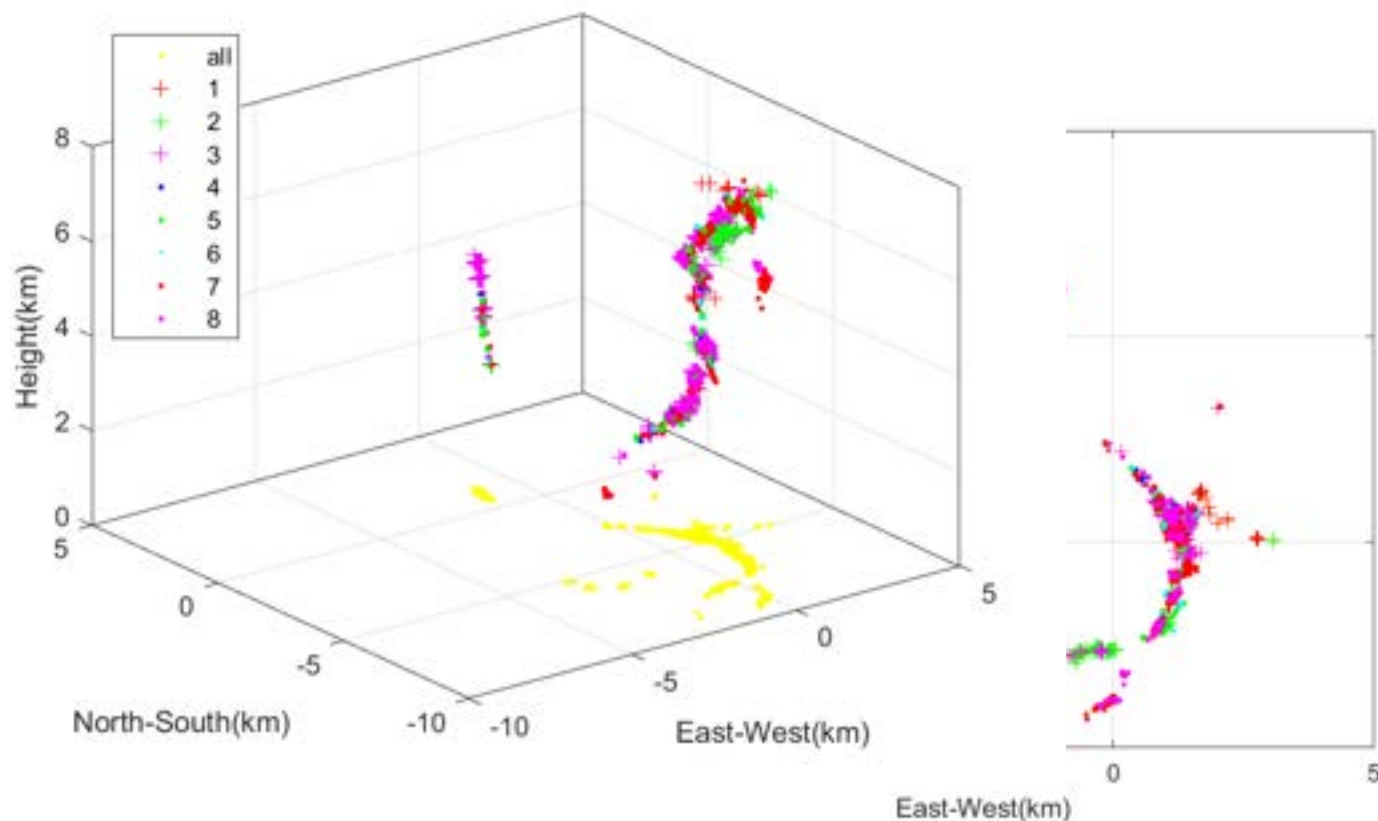
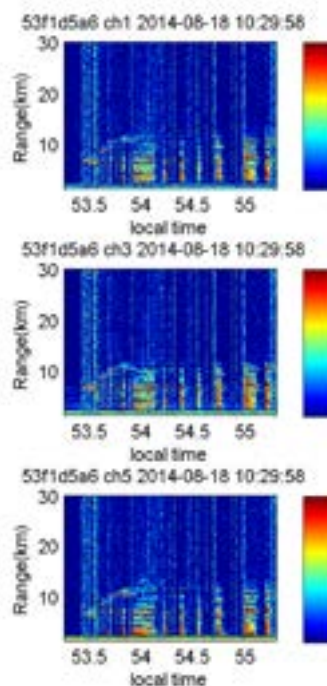
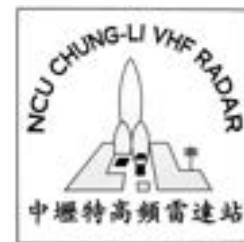


雲間閃電/雲對地閃電 IC Lightning/CG Lightning





主動式閃電偵測雲地閃訊號





中壢雷達研究群 研究方向與成果



* 低層大氣回波功率方向靈敏性(Asspect Sensitivity)的研究

— 首創利用雷達垂直波束觀測大氣方向靈敏性與亂流結構

* 特高頻雷達降水的觀測與研究

— 首先發現在伴隨強烈上升氣流的降雨環境下，亂流回波有顯著衰竭現象

* 大氣重力波的特高頻觀測與研究

— 針對大氣重力波在臨界層的傳播特性以及重力波譜的變化進行數值模擬

* 大氣閃電的特高頻雷達觀測與研究

— 首創利用特高頻雷達進行閃電的定位與特性研究

* 雷達新技術的理論開發與研究

— 開發出傾斜SAD測風法，發展剖風儀回波機制統一理論等

* 特高頻雷達測海技術的理論開發與研究 *

— 發展特高頻雷達測海架構、天線系統與計算海浪、洋流場技術等



中壢雷達研究群

研究方向與成果(續)



* 電離層 E 域形態一(Type 1)電子密度不規則體的研究

— 首先發現形態一不規則體沿東西向延伸尺度可以超過12公里，推翻理論的預測
同時亦發現形態一不規則體空間結構可以以薄層或團狀方式存在

* 電離層 E 域似週期(Quasi-Periodic, QP)回波之研究

— 首先提出QP回波與大氣重力波的南北向傳播無關

* 波束加寬效應(Beam Broadening Effect)之研究

— 首度理論證明水平風向會影響波束加寬效應，並首先推導出電離層場列(Field-aligned)不規則體的波束加寬都卜勒頻譜

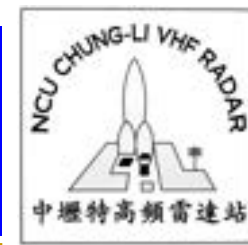
* 流星的觀測與研究

— 確認在電離層E域內的流星尾電漿密度不規則體亦具場列特性，同時發生機制有可能為中性亂流的激發結果



中壢雷達研究群

研究方向與成果(續)



* 頻率域雷達干涉法(FDI)之理論研究

— 首先理論證明亂流的非均向性以及回波訊雜比均會影響FDI的觀測結果

* 電離層 E 域形態二(Type 2)電子密度不規則體方向靈敏性之研究

— 首創利用雷達干涉法重建不規則體的三維空間分布並估計三公尺不規則體之方向靈敏性(或方向角Aspect Angle)

* 開發出測量電離層 Es 層極化電場強度的技術

— 首創利用雷達干涉法與波束加寬效應理論測量電離層Es薄層內之垂直向極化電場之技術

* 電離層 E 域電子密度不規則體三維結構之重建與辨識

— 首創利用雷達干涉法重建不規則體的三維空間結構，並分辨其結構形態為層狀或團狀或其他形狀



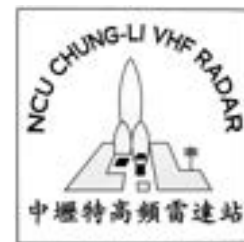
結論



- 一、中央大學雷達站擁有全台灣唯一一座可以從事大氣與太空環境監測與研究的高頻／特高頻大型陣列雷達。
- 二、中央大學太空科學團隊不但參與國內外眾多大型科研計畫，並擁有實務經驗，可提供相關單位研參。
- 三、目前正進行多項大型科研計畫，包括福爾摩沙衛星三號電離層探測計畫，全國剖風儀網建置計畫，全國閃電偵測網建置計畫，測海技術研究計畫、電離層監測系統計畫等，對於提升我國太空科技學術水準以及大氣觀測應用方面有具體的貢獻。



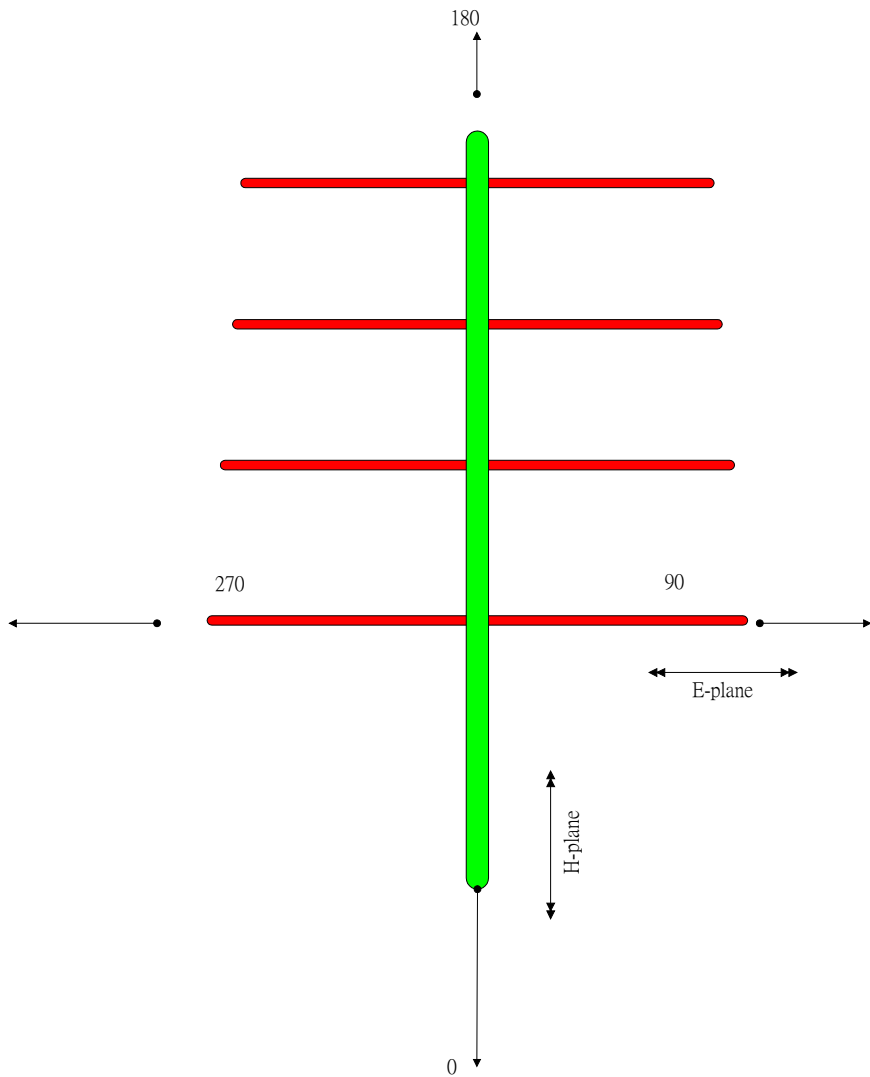
END



天線場型模擬與實驗參數



天線陣列中所使用八木天線之座標系統，E平面
延著90°-270°方向，H平面延著0°-180°方向





單一Yagi天線電場場型



$$E_0(\theta, \phi) = \sum_{m=1}^4 \frac{I_m}{I_2} F_m(\theta) e^{i\beta d_{m2} \sin \theta \sin \phi}$$

I_m : 第 m 支元件終端電流

F_m : 第 m 支元件場型因子

d_{m2} : 第 m 支元件和第2支元件之垂直距離

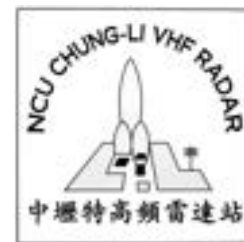
β : $2\pi/\lambda$ (λ 為波長)

θ : 天頂角

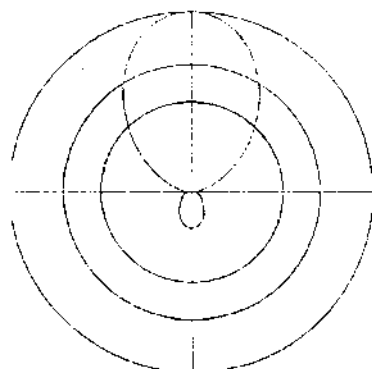
ϕ : 方位角



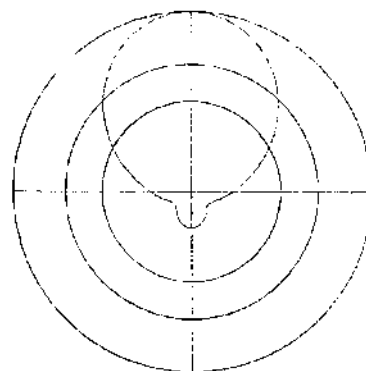
中壢特高頻雷達 單Yagi原始場型模擬



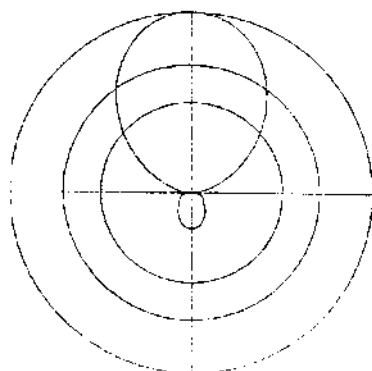
八木天線電場場型



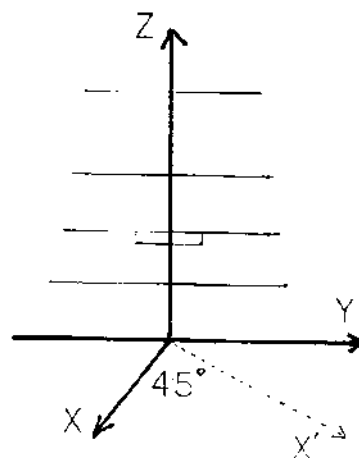
Y-Z PLANE



X-Z PLANE

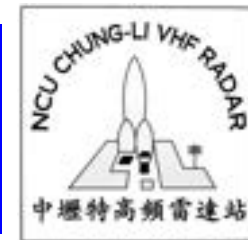


X'-Z PLANE

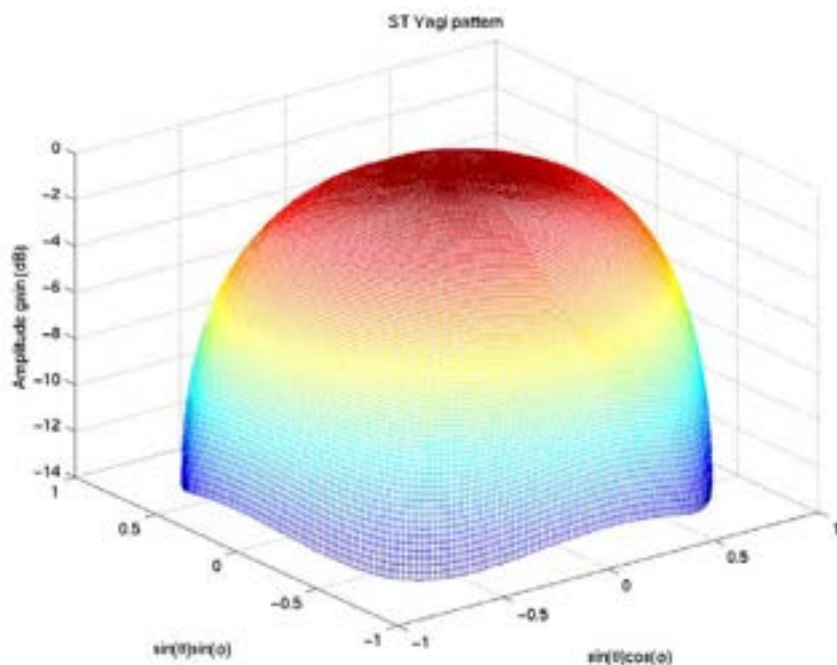




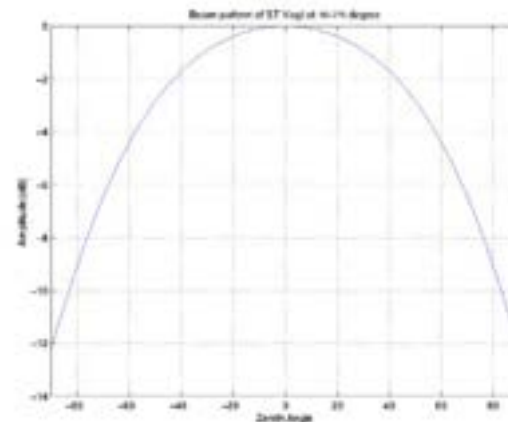
ST天線所使用八木天線空間場型圖



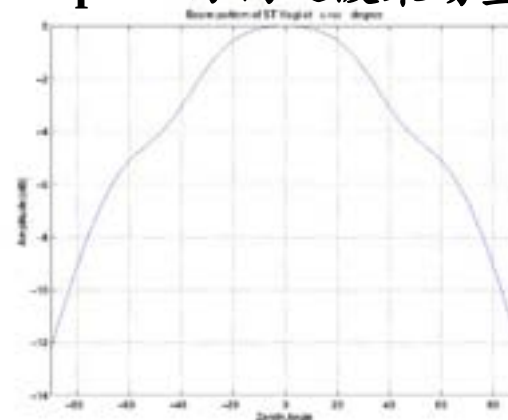
顯示主波束非常寬



ST天線所使用八木天線沿
E-plane之波束場型圖



ST天線所使用八木天線沿
H-plane方向之波束場型





天線陣列之輻射場型



$$E(\theta, \phi) = |f(\theta, \phi) \times F(\theta, \phi)|$$

E:總場型

f:元件場型(element pattern)

F:陣列場型因子(array factor)



$N \times N$ 之等距天線陣列因子 (array factor)



$$F(\theta, \phi) = \frac{\sin \frac{N\phi_1}{2} \cdot \sin \frac{N\phi_2}{2}}{N \sin \frac{\phi_1}{2} \cdot N \sin \frac{\phi_2}{2}}$$

θ : 天頂角

ϕ : 方位角

$$\phi_1 = \frac{2\pi}{\lambda} d \sin \theta \cos \phi + \delta_1$$

$$\phi_2 = \frac{2\pi}{\lambda} d \sin \theta \sin \phi + \delta_2$$

d : 兩相鄰天線之距離

$\delta_1 \delta_2$: 兩相鄰天線之相位差



中壢特高頻雷達ST天線陣列 原始場型模擬

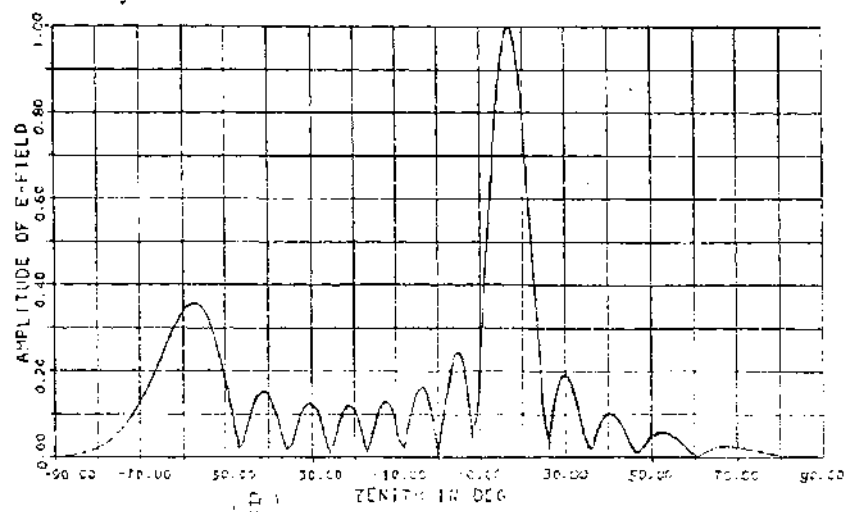
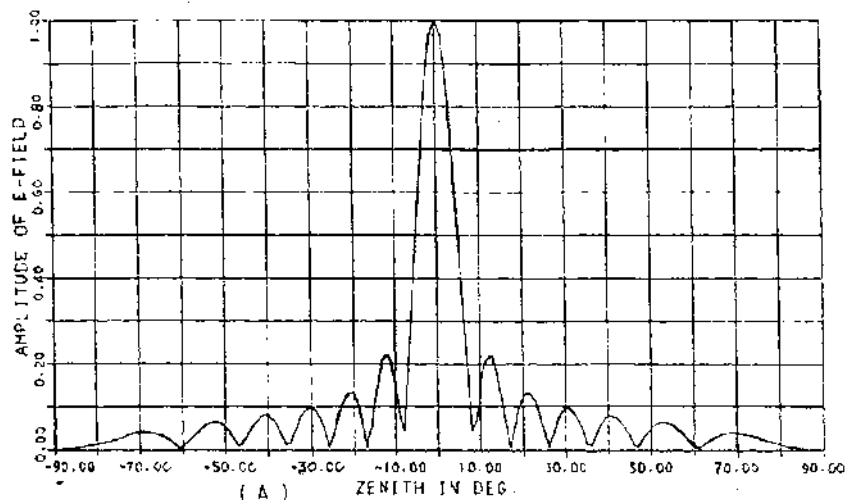
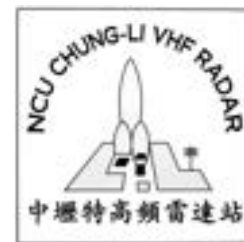


FIG. 7



中壢特高頻雷達ST天線陣列 原始場型模擬



8 * 8 ARRAY

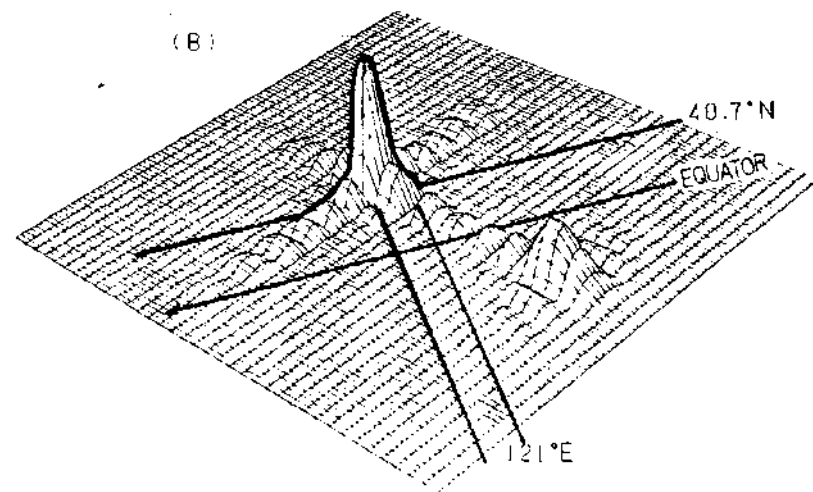
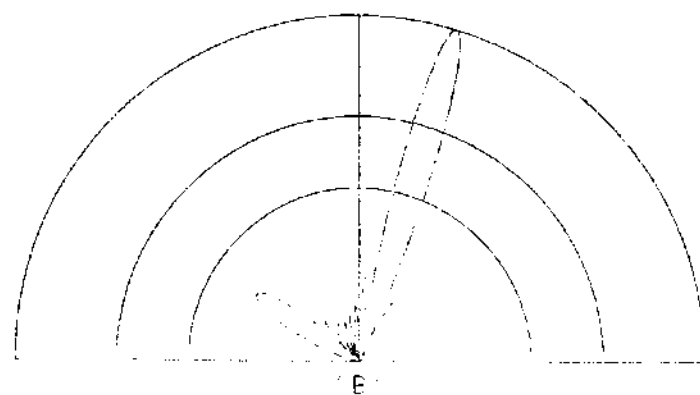
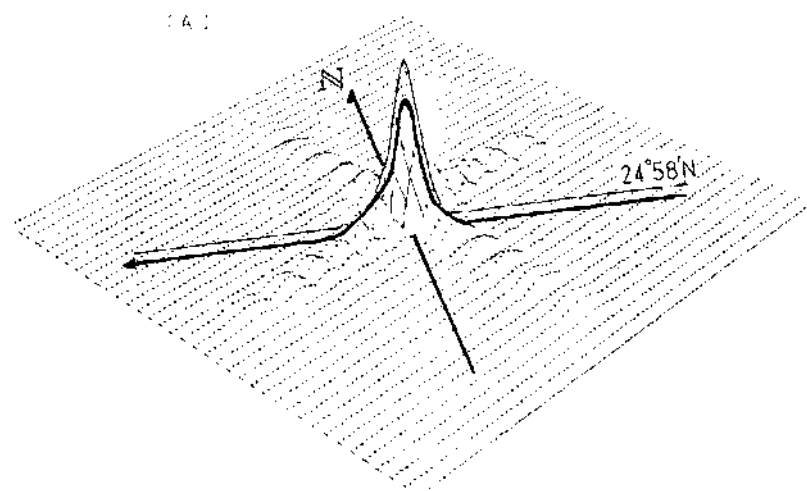
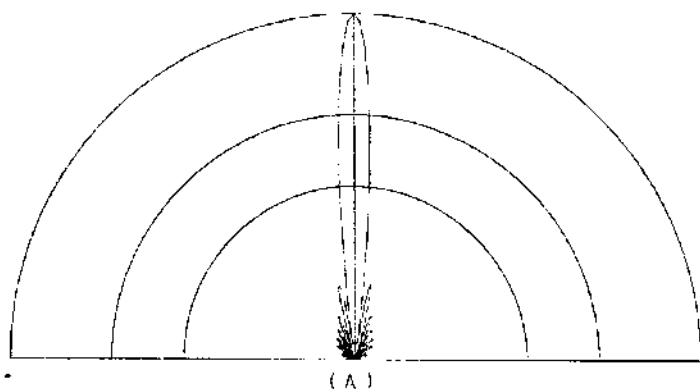
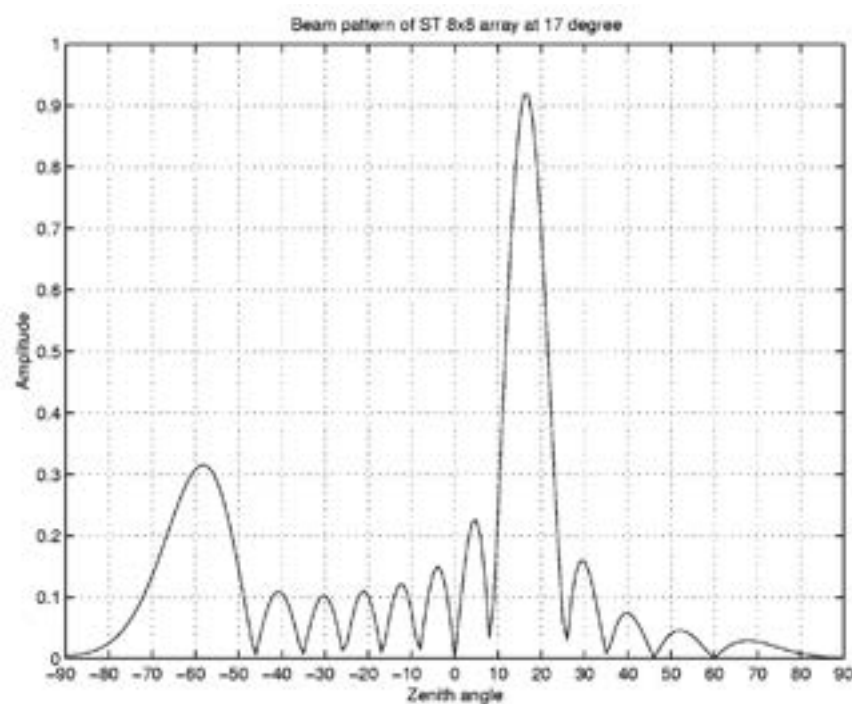
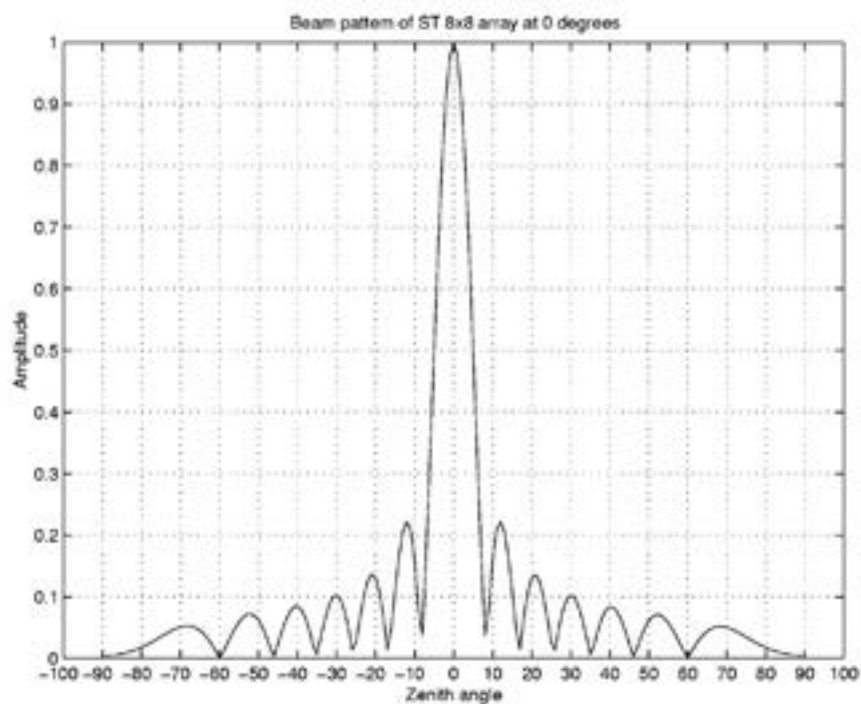


FIG. 1

FIG. 2

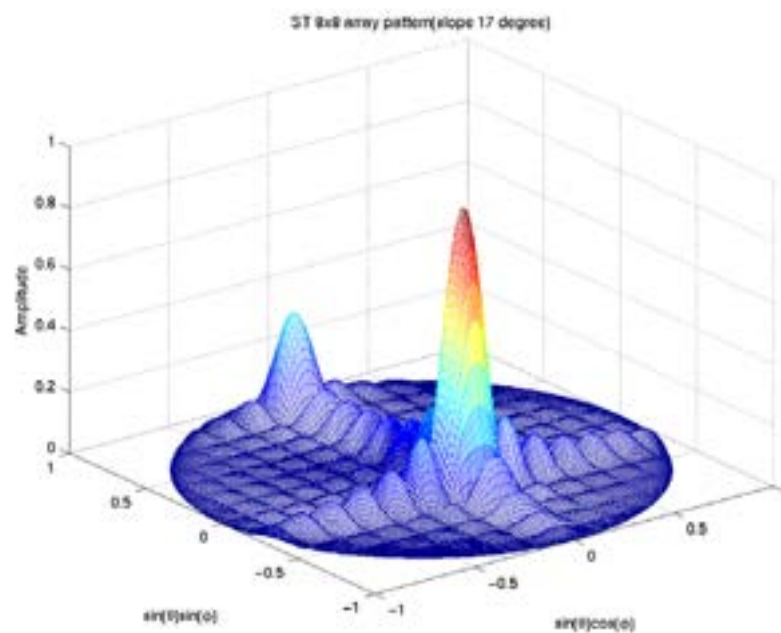
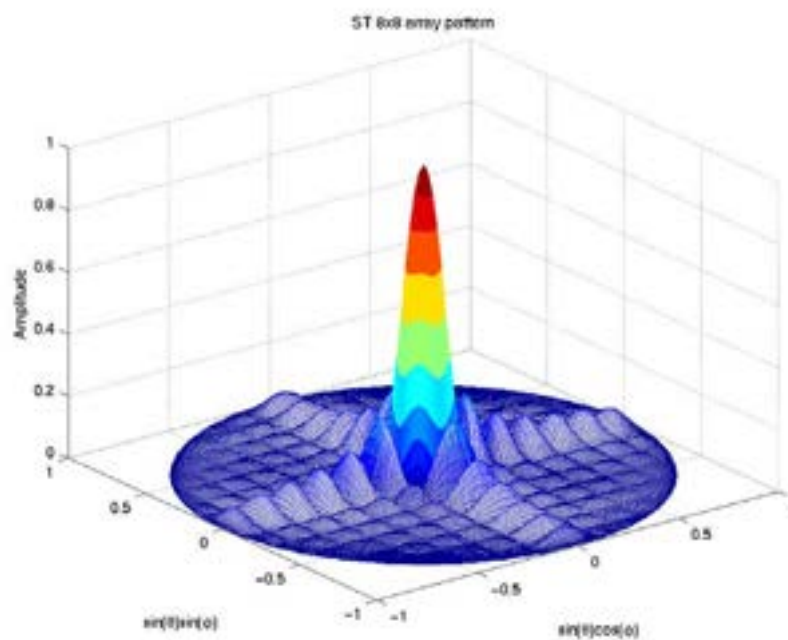
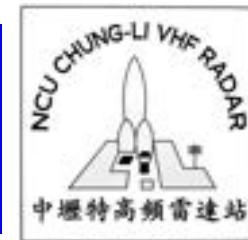


中壢特高頻雷達ST天線陣列 場型模擬





中壢特高頻雷達ST天線陣列 場型模擬





MxN之等距天線陣列因子 (array factor)



$$AF(\theta, \phi) = \frac{1}{M} \cdot \frac{\sin \frac{M}{2} kdx(\sin \theta \cos \phi - \sin \theta_0 \cos \phi_0)}{\sin \frac{1}{2} kdx(\sin \theta \cos \phi - \sin \theta_0 \cos \phi_0)} \\ \cdot \frac{1}{N} \cdot \frac{\sin \frac{N}{2} kdy(\sin \theta \sin \phi - \sin \theta_0 \sin \phi_0)}{\sin \frac{k}{2} dy(\sin \theta \sin \phi - \sin \theta_0 \sin \phi_0)}$$

M為x方向之天線數目，N為y方向之天線數目

M、N均為偶數

θ_0 為主波束天頂角(41°)， ϕ_0 為主波束方位角(180°)

θ 為任意之天頂角

ϕ 為任意之方位角

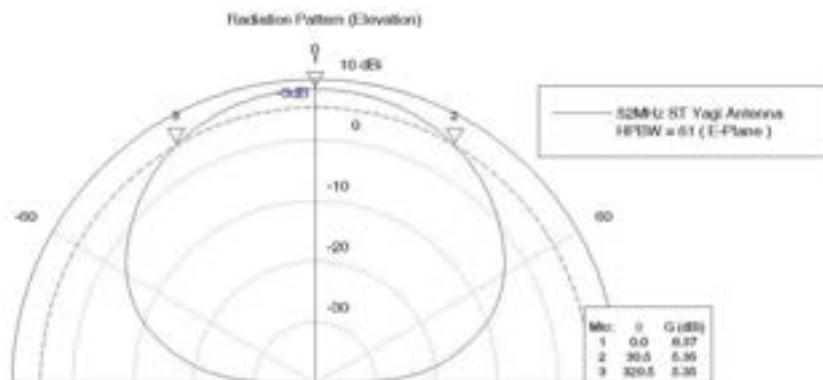
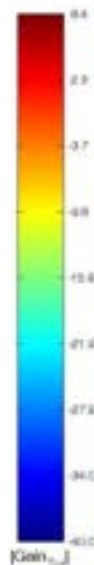
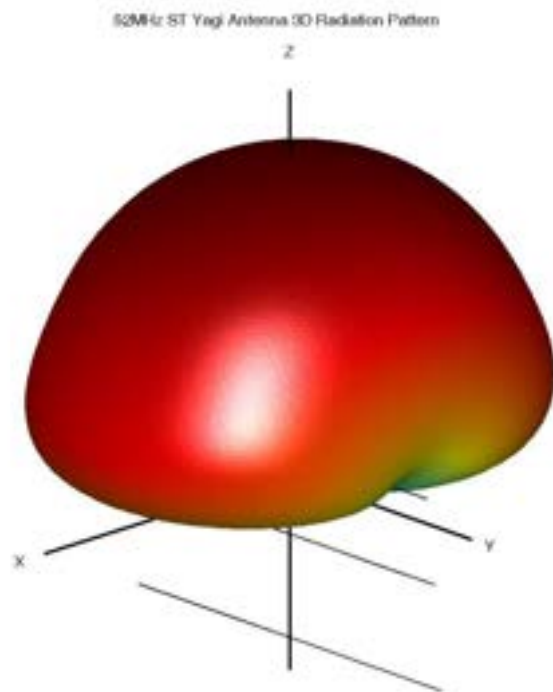


52MHz雷達

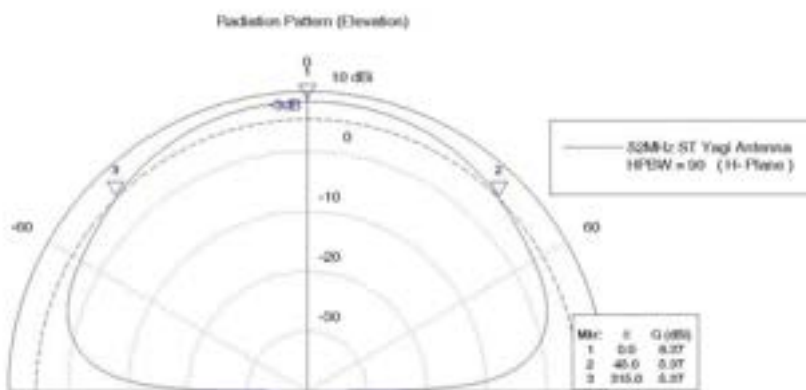
單一八木天線3D場型圖



E plane



H plane



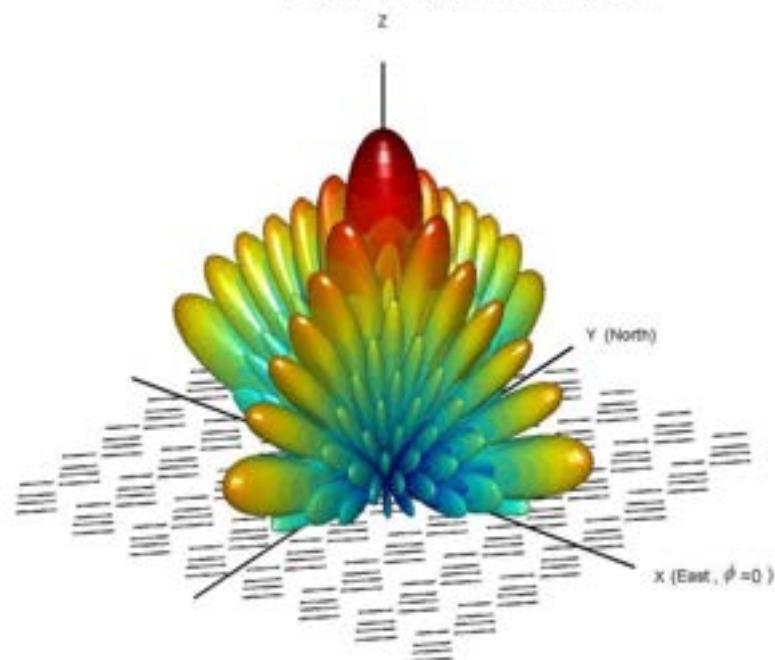


8X8陣列天線3D場型圖

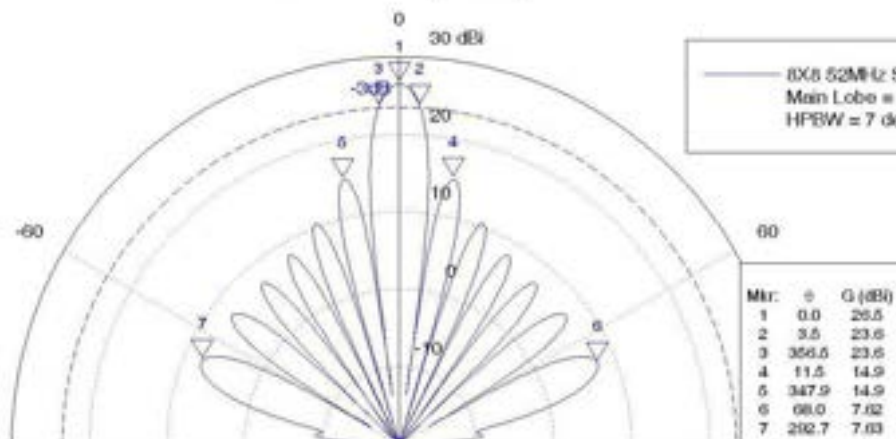
垂直波束



8X8 Yagi Antenna Array 3D Radiation Pattern (52 MHz)

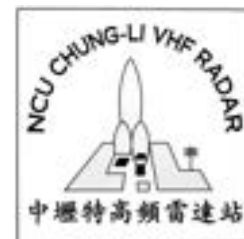


Radiation Pattern (Elevation)

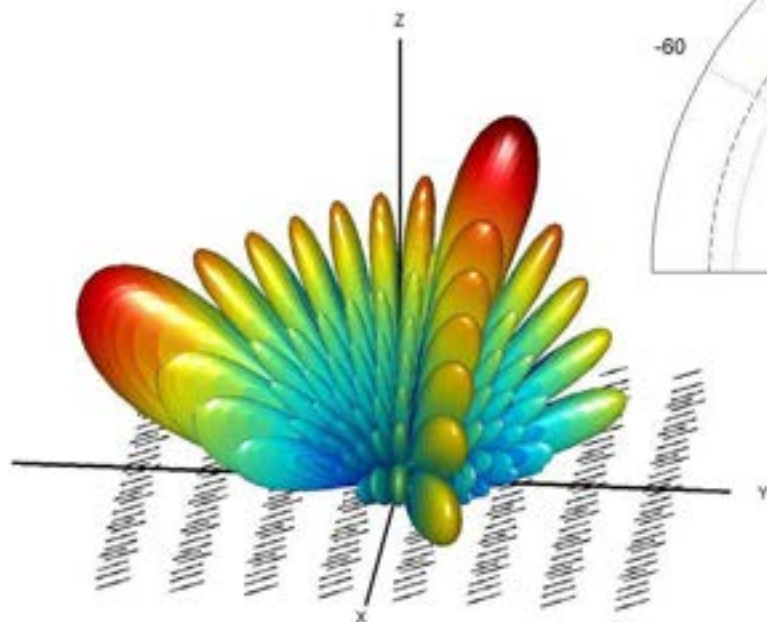




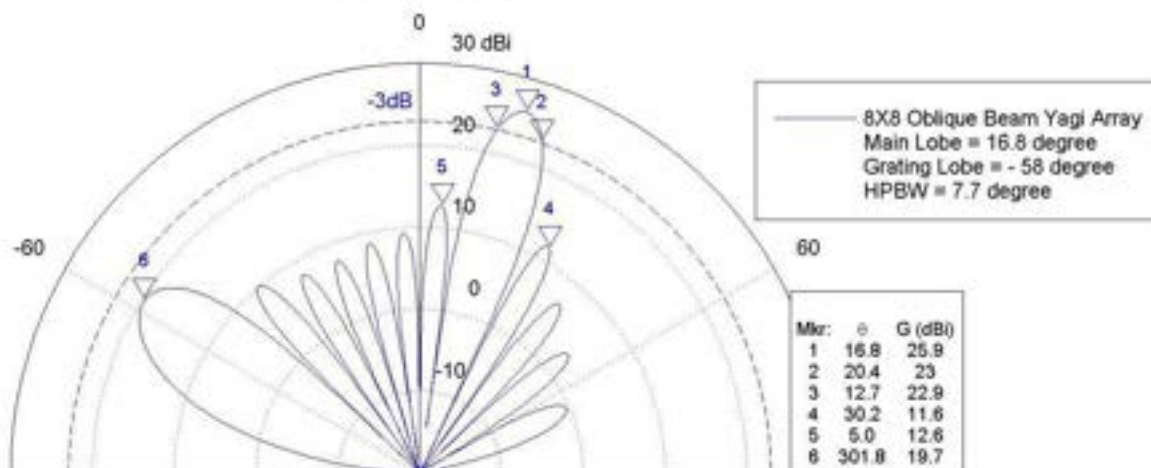
8X8陣列天線3D場型圖 傾斜波束



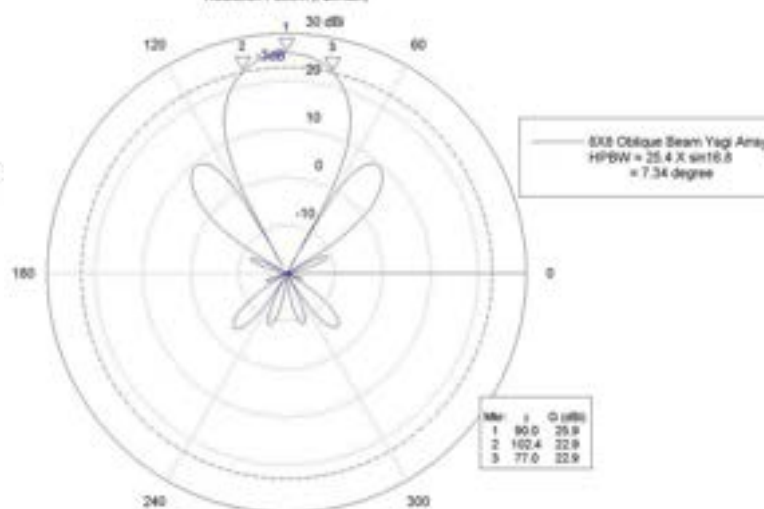
8X8 Yagi Antenna Array 3D Radiation Pattern (Oblique Beam)



Radiation Pattern (Elevation)



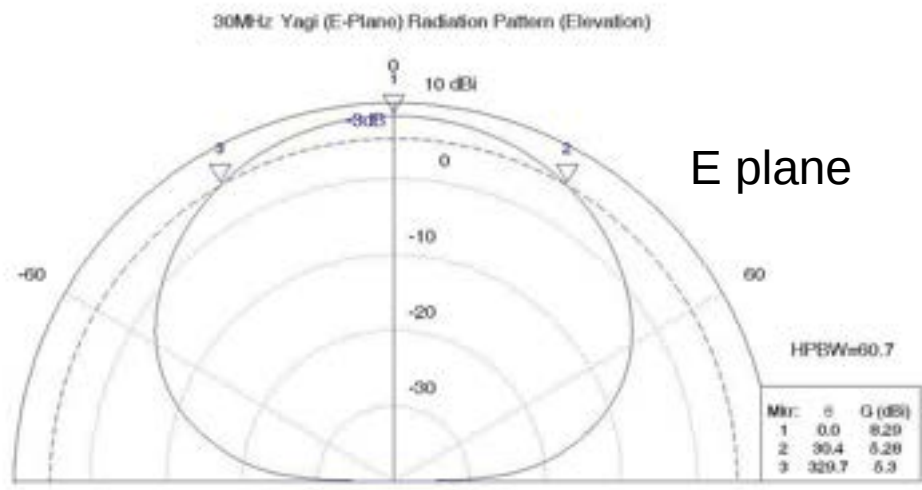
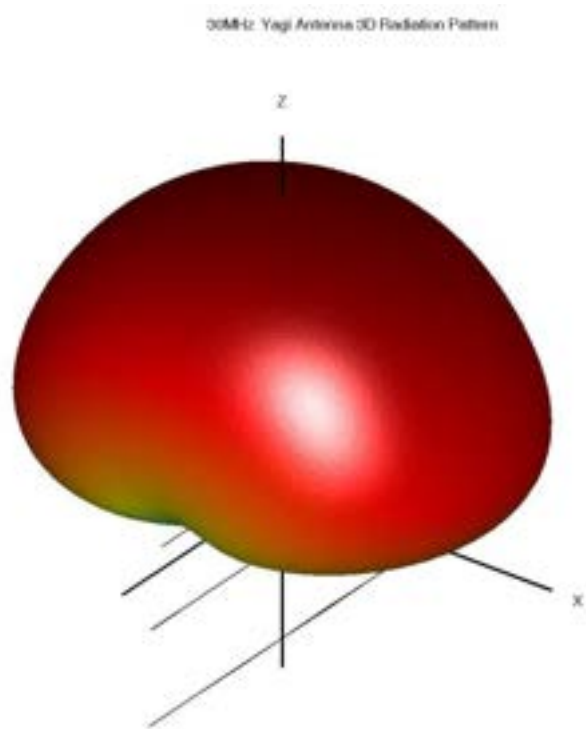
Radiation Pattern (Azimuth)



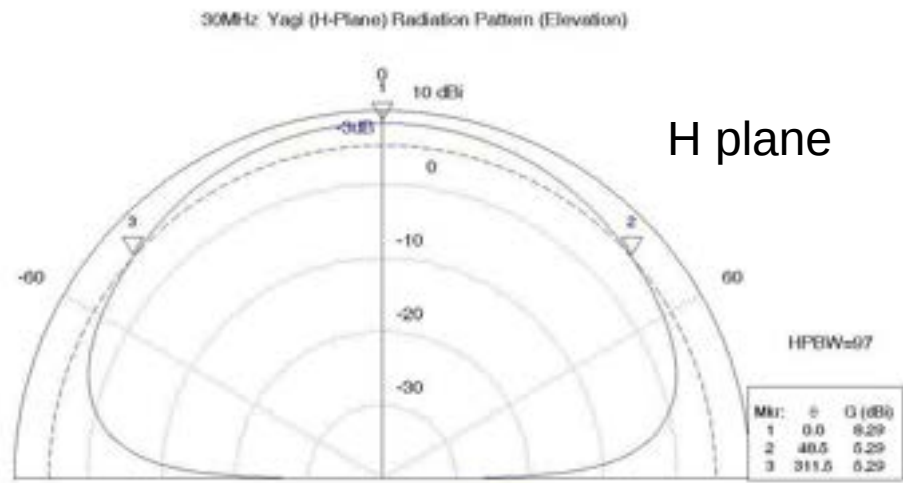


30MHz雷達

單一八木天線3D場型圖



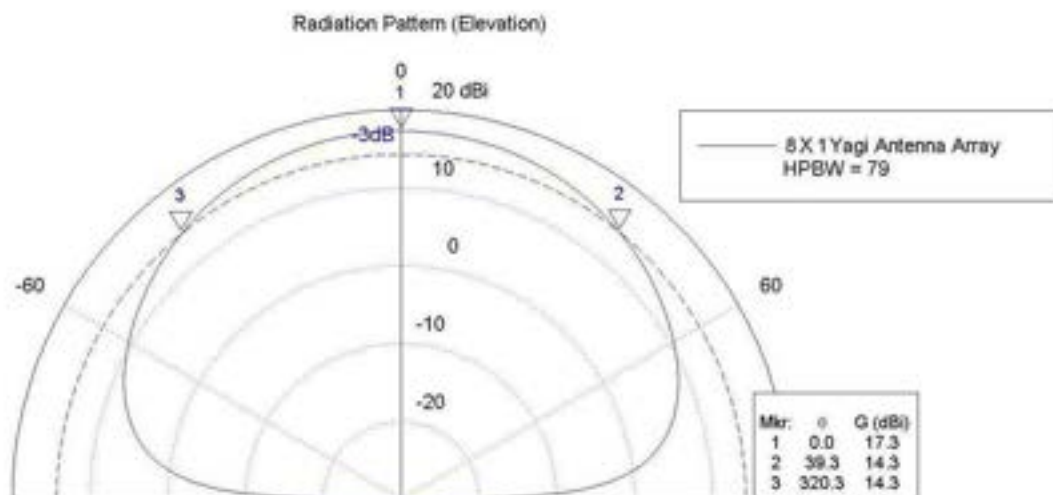
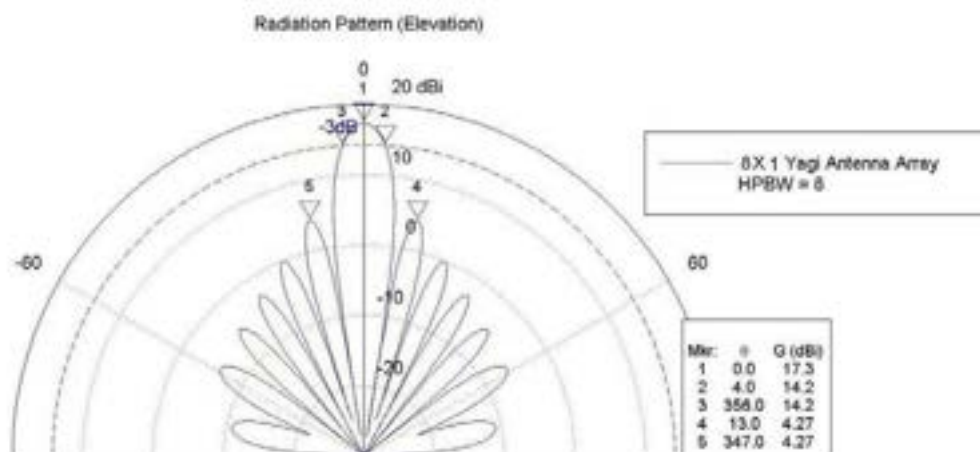
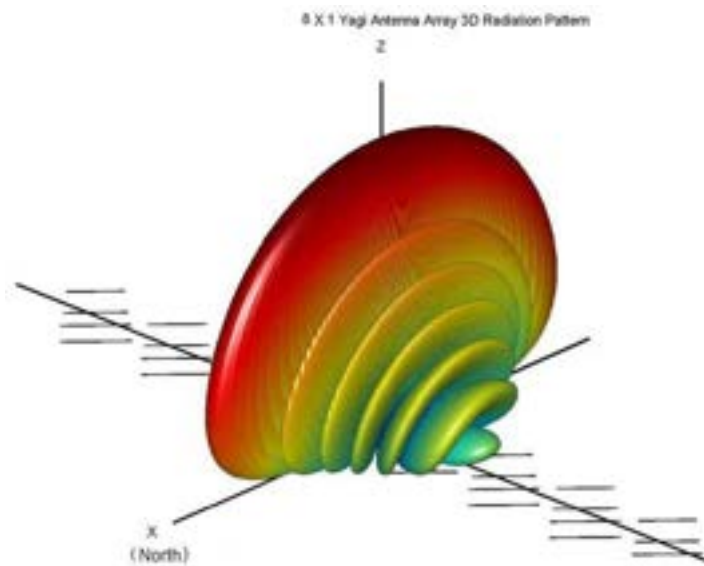
E plane



H plane



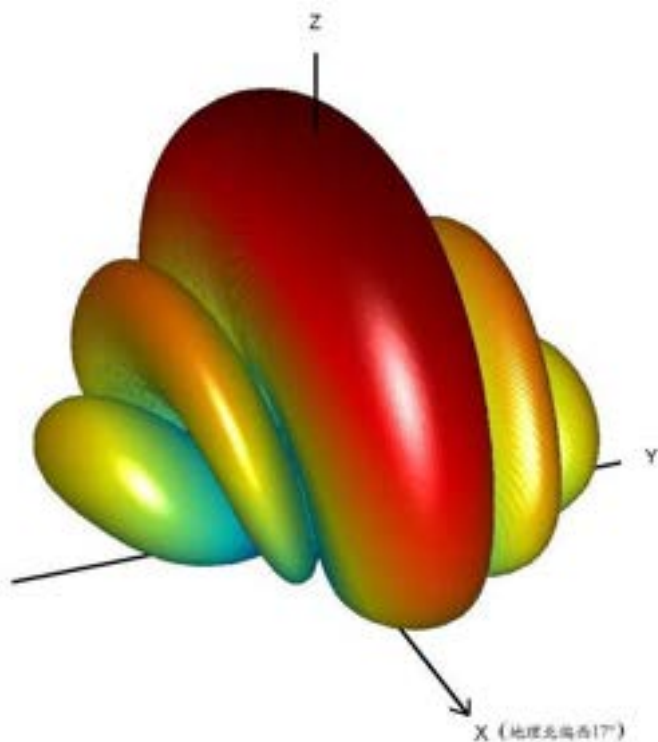
1x8 陣列天線系統



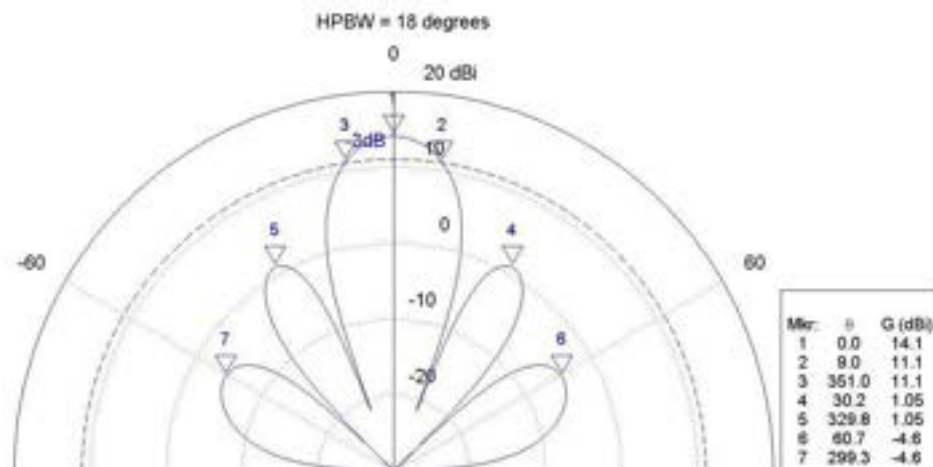


1X4陣列天線3D場型圖

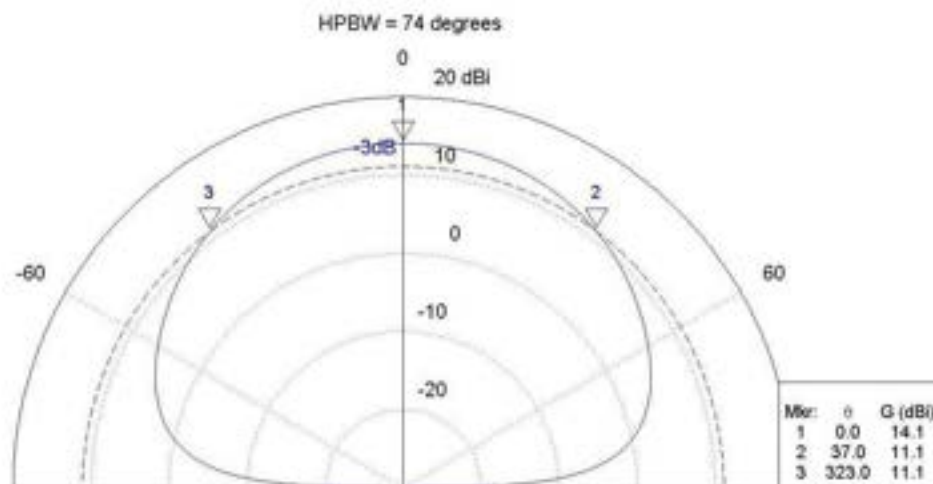
30MHz 4X1 Yagi Antenna Array 3D Radiation Pattern (Antenna Spacing 7m)



30MHz 4X1 Yagi Antenna Array 2D Radiation Pattern (Antenna Spacing 7m)

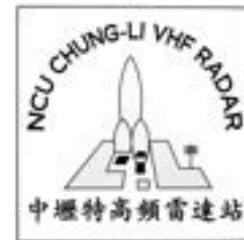


30MHz 4X1 Yagi Antenna Array 2D Radiation Pattern (Antenna Spacing 7m)





雷達系統參數設定



- 包含PRF(1/觀測週期(1/IPP))
- 觀測脈波長
- 最低觀測距離
- 總觀測距離
- 同相積分數
- 每筆觀測時間長度



雷達系統參數設定(續)



- Range Resolution(空間解析度)=Velocity of Light(光速) * Pulse length(脈波長度)/2
- Time Resolution(時間解析度)= Number of Coherent Integration(同相積分次數)*Inter Pulse Period(發射週期)
- Nyquist Frequency(臨界頻率)=1/(2 * Time Resolution)
- Sampling Time(1個檔案觀測時間)= Time Resolution * Trains
- R_{MAX} (最大可觀測距離)= Inter Pulse Period (發射週期) * Velocity of Light /2
- R_{MIN} = Delay Time * Velocity of Light /2
- R_{MAX_exp} = Sampling Range Gates * Pulse length * Velocity of Light /2 + R_{MIN}



52MHz雷達系統 參數設定



radar - [iso2009] iso2009test

File

Comment: No Comment

General Acquisition Parameters

Pulse Repetition Frequency: 200 Hz

Sampling Resolution: 1200 metres

Sampling Start Range: 72000 metres

Sampling Range Interval: 542400 metres

Number of gates: 452

Number of coherent integrations: 2

Number of additional coherent integrations before processing: 1

Experiment scheduling time interval: 123 seconds

Number of digitiser channels to acquire: 8

Data Rate out of the RDRG: 529.63 kB/s

Data Rate to the processors: 529.63 kB/s

Calculate range zero based on pulse? Yes No

Calculate SC TRLO based on pulse? Yes No

Receiver Gain Value

Rx 1: 105 dB

Rx 2: 105 dB

Rx 3: 105 dB

Rx 4: 105 dB

Rx 5: 105 dB

Rx 6: 105 dB

Receiver Filter RFBW

1MHz 500kHz 250kHz 125kHz

radar - [iso2009] iso2009test

File

Frequency Configuration

List of offset frequencies to use for FBI (Hz)

0

Add Remove

Pulse Configuration

Should the transmitter be enabled? Yes No

Number of pulses to use: 1

Pulse length (OFFW) for each pulse address

Pulse 1: 8000 ms

Pulse 2: 8000 ms

Pulse shapes for each pulse address

Pulse 1: Gaussian Square Short Gaussian Triangle Shaped Trapezoid

Pulse 2: Gaussian Square Short Gaussian Triangle Shaped Trapezoid

Pulse amplitude for each pulse address

Pulse 1: 255

Pulse 2: 255

Pulse phase for each transmitter

Tx 1: In phase Out of phase

Tx 2: In phase Out of phase

Tx 3: In phase Out of phase

Tx 4: In phase Out of phase

Tx 5: In phase Out of phase

Tx 6: In phase Out of phase

Perform transmitter phase flipping? Yes No

Pulse code for each pulse address

Pulse 1: 1111100110101

Pulse 2:

Preset Pulse Codes

No codes 2-bit Barker

Set



52MHz雷達系統 參數設定



radar - [iso02009] iso02009test

File

Preset Pulse Codes

No codes
2-bit Barker
3-bit Barker
4-bit Barker (a)
4-bit Barker (b)
5-bit Barker

Set

Beam Steering Configuration

Beam directions to scan

ionospheric ionospheric ionospheric

Add Beam

Time to dwell on each beam direction

128 seconds

Acquisition time per azimuth scan cycle

129.0 seconds

Data Processing Parameters

Raw Data Processor (processor 0)

Should the raw data be dumped to disk?

Yes No

POWER Processor (processor 1)

Enable the POWER processor?

Yes No

Should the Power processor take data before or after extra integration?

Before After

Number of data points to analyse per POWER record

256

Time required to produce a POWER record

2.6

FDT Processor (processor 2)

Enable FDT data processor

Yes No

Should the FDT process data before or after extra integration?

Before After

Acquisition time per analysis cycle

2.6 seconds

Options to pass to the FDT plugin

Add Beam

The number of points required to perform each FDT analysis

256

Number of receivers in FDT analysis

3

Number of post-analysis averages for FDT analysis

1

FFT window type

None Hamming

All Processors

Options to pass to all plugins

DEBUG plugin:1

Add Beam



52MHz雷達資料格式 (raw data)

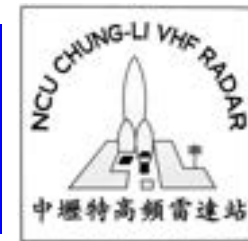


檔頭內容

- **ACQ_TIME 11**
- **BEAM_DIRECTION west**
- **BEAM_LIST vertical north east south west**
- **CHANNELS 6**
- **EXPTAG all_beams**
- **FREQUENCY 52000000**
- **GATES 68**
- **GPS_STATUS LOCK**
24.968206 N 121.186737 E
- **INTEGRATIONS 256**
- **INTERVAL 20400**
- **PLUGIN_PATH**
/home/radar/ud3/lib/processor
- **PRF 6000**
- **PULSE_LENGTH 2000**
- **PULSE_SHAPE square**
- **RANGE 1000**
- **RESOLUTION 300**
- **RESULTS /local0/Results**
- **SITENAME chungli**
- **DATA 1153997681**



雷達資料格式(raw data)



Gate 1

Channel 1 I : 1, 2, 3,,512.....

Channel 1 Q : 1, 2, 3,,512.....

Channel 2 I : 1, 2, 3,,512.....

Channel 2 Q : 1, 2, 3,,512.....

:

Gate 2

:

Gate3

:

:

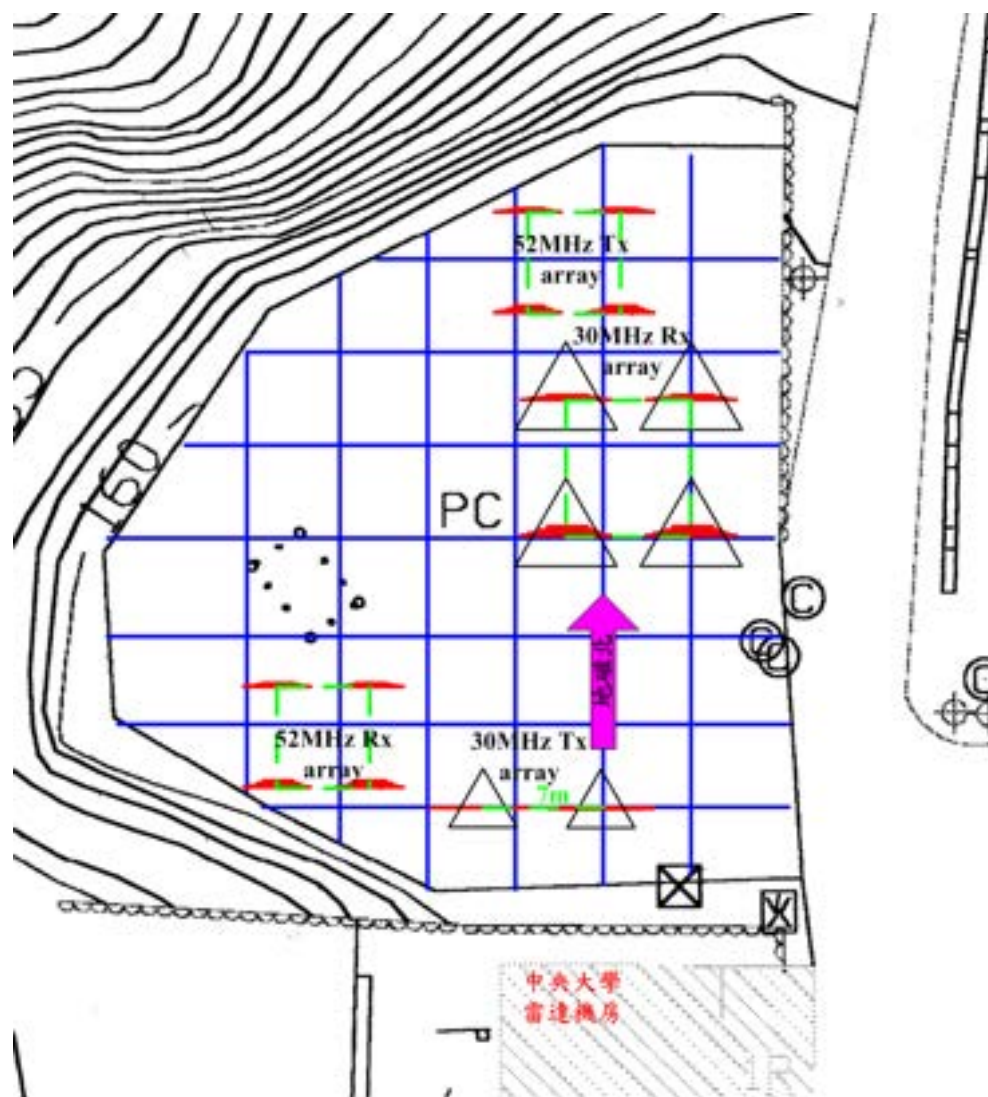
Gate 100

:

:

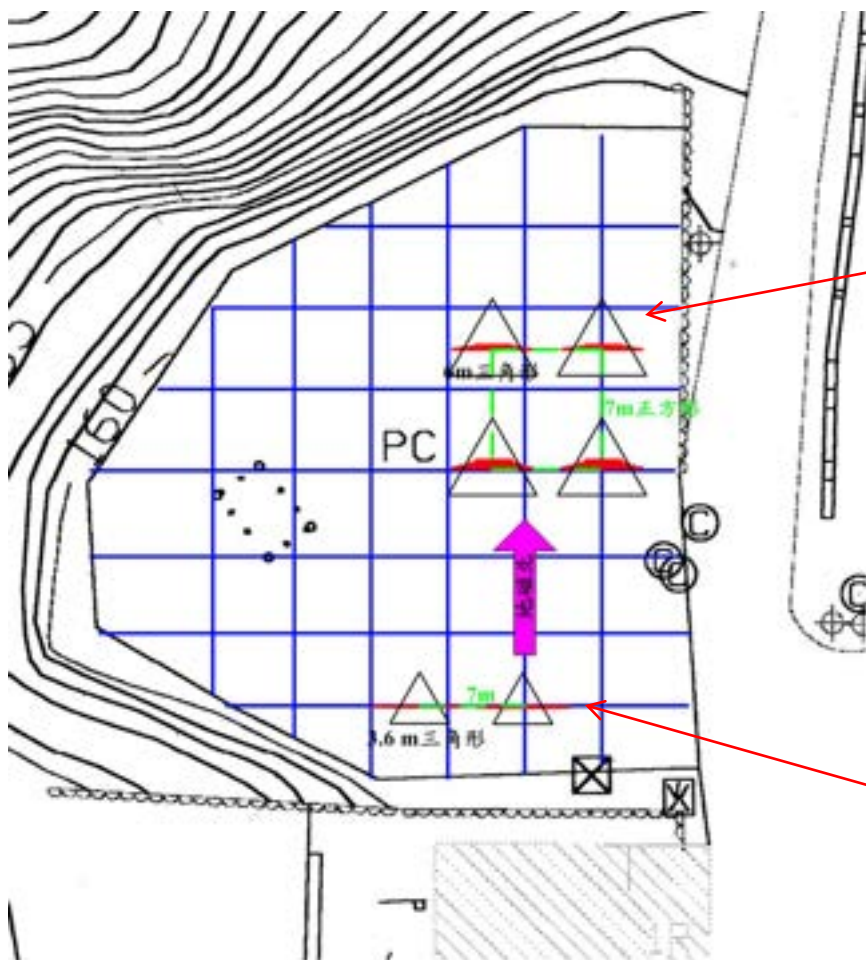


探空七號火箭射場地面觀測設備 30MHz與52MHz雷達天線放列簡圖

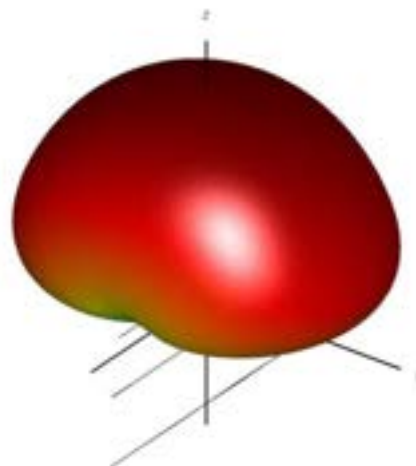




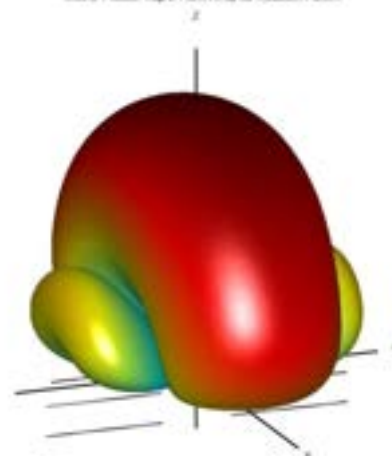
30MHz 雷達架構



30MHz Yagi Antenna 3D Radiation Pattern



30MHz 1-Watch Yagi E-Plane Array 3D Radiation Pattern

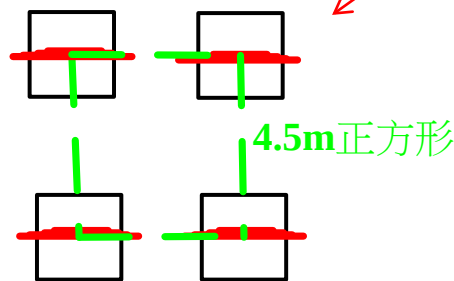
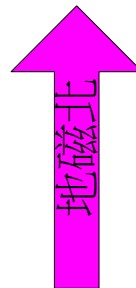
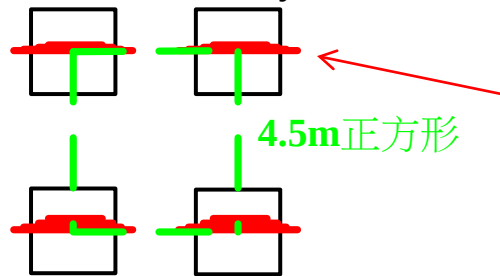




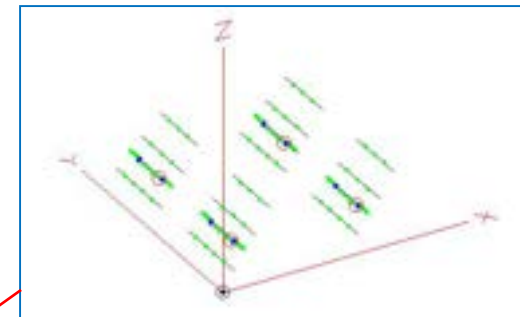
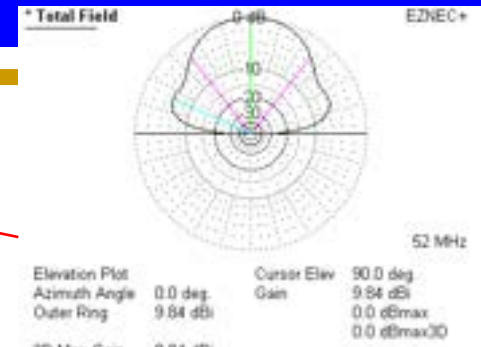
52MHz 雷達架構



Rx antenna array



Tx antenna array





中壢特高頻雷達參與實驗簡介

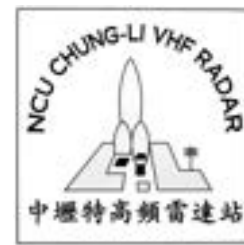


探空火箭實驗(二、三、五、七、九) 與中壢特高頻雷達



實驗過程

(探空七號火箭)

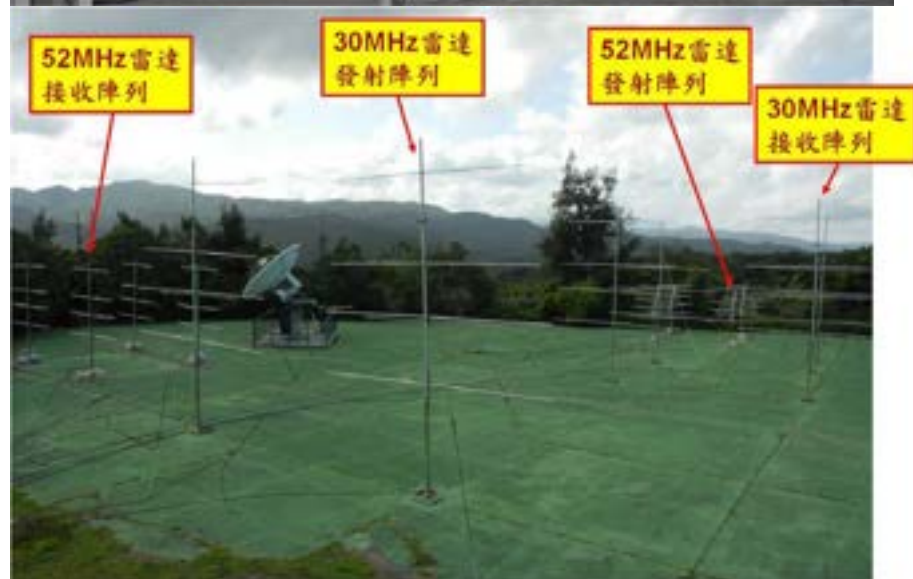


- 一、17:05—中壢電離層觀測儀開始觀測到Es回波
- 二、18:30—**九鵬30與52MHz雷達同步偵測到電漿不規則體回波**
- 三、18:45—**中壢52MHz雷達偵測到電漿不規則體回波**
- 四、19:50—探空火箭七號順利發射
- 五、280秒—火箭達最高點，約289公里
- 六、540秒—火箭落海，探空七號科學酬載實驗結束
- 七、23:34—三頻標識訊號接收機觀測到電離層閃爍現象
- 八、01:48—電離層不規則體消失，全部實驗結束



探空火箭計畫

中大九鵬與太麻里雷達站





九鵬射場中央大學雷達站

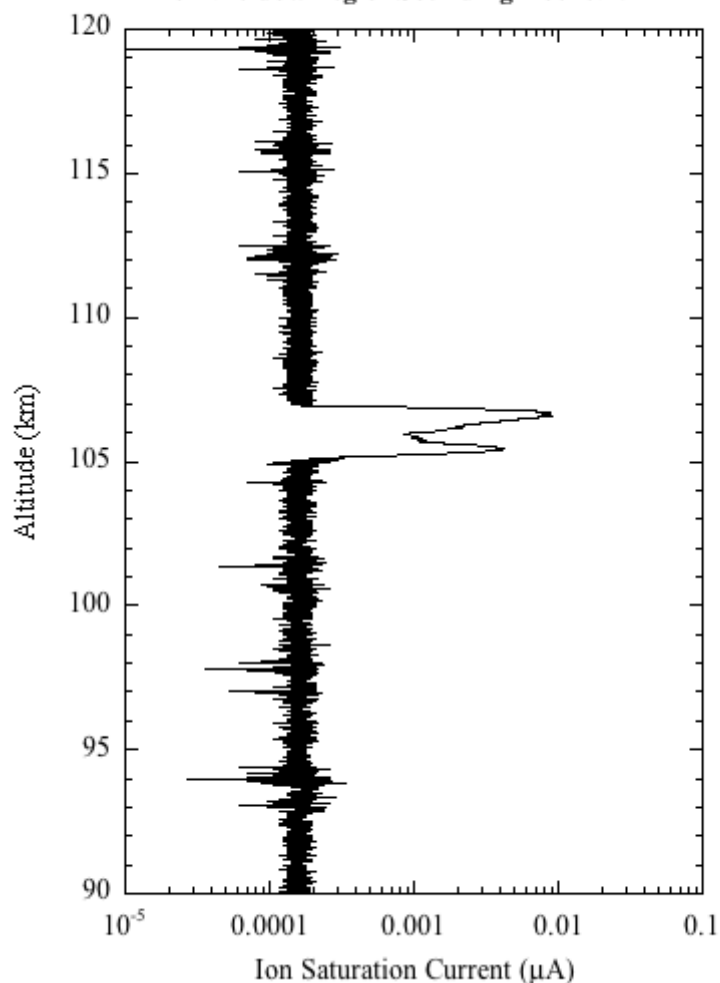




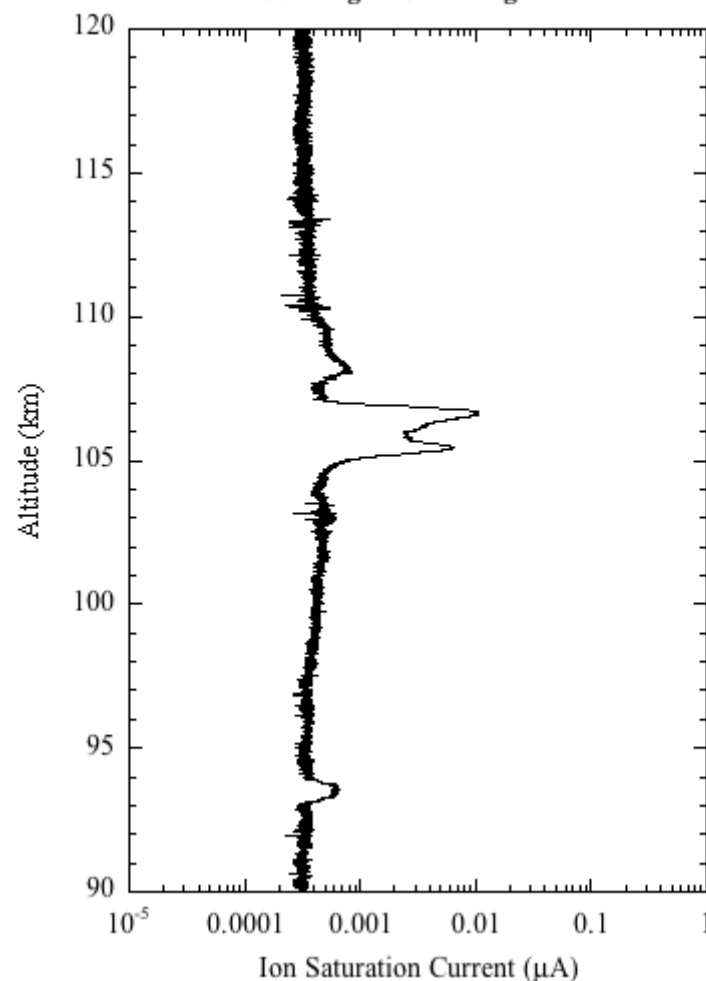
探空火箭七號科學實驗初步結果 —散塊E層(電漿不規則體)—



Preliminary ion saturation current (ISN) measurement
on the downleg of Sounding Rocket VII



Preliminary ion saturation current (IST) measurement
on the downleg of Sounding Rocket VII

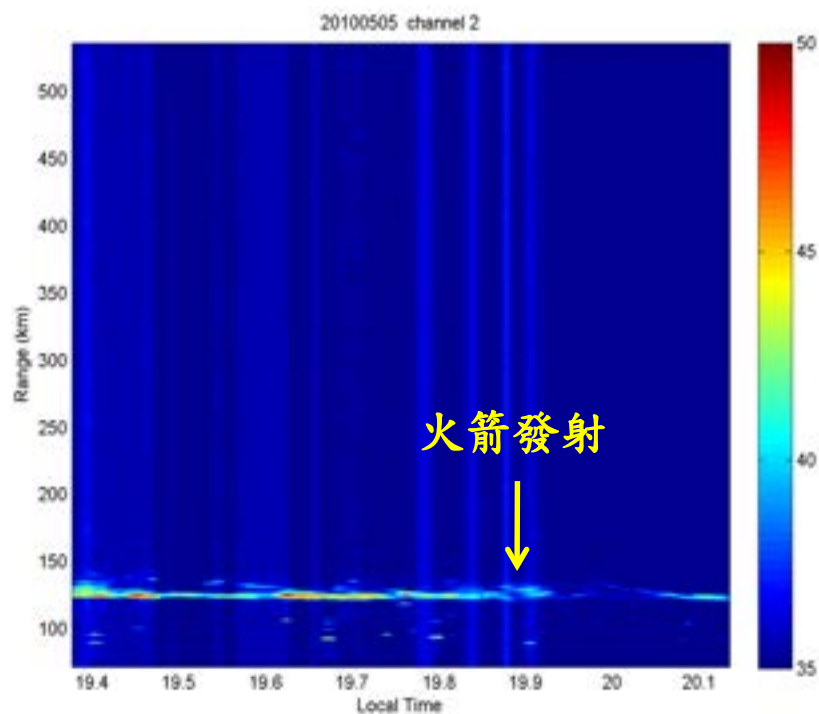




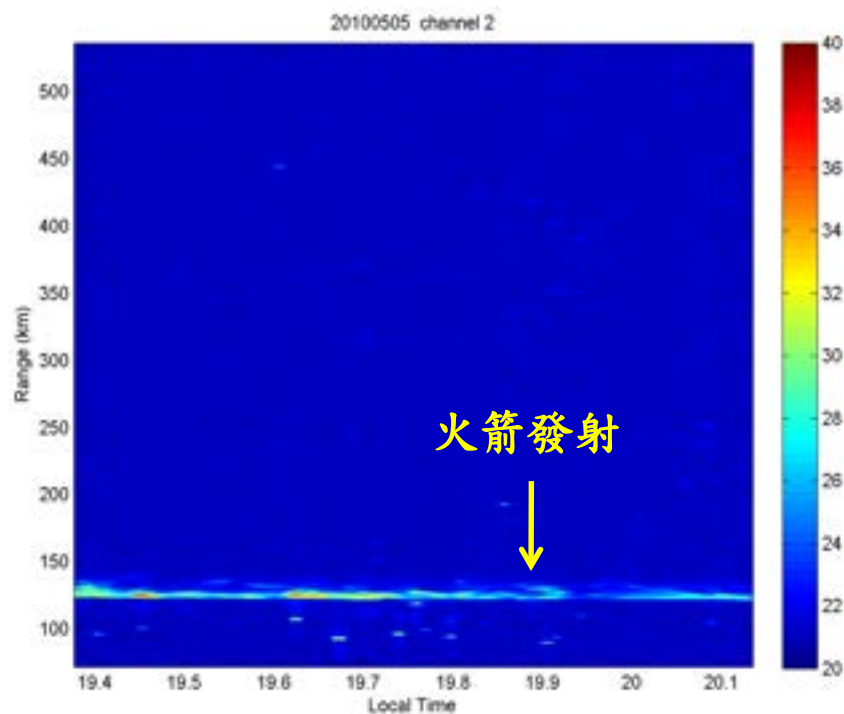
探空火箭七號科學實驗 地面特高頻雷達初步結果



九鵬30 MHz雷達



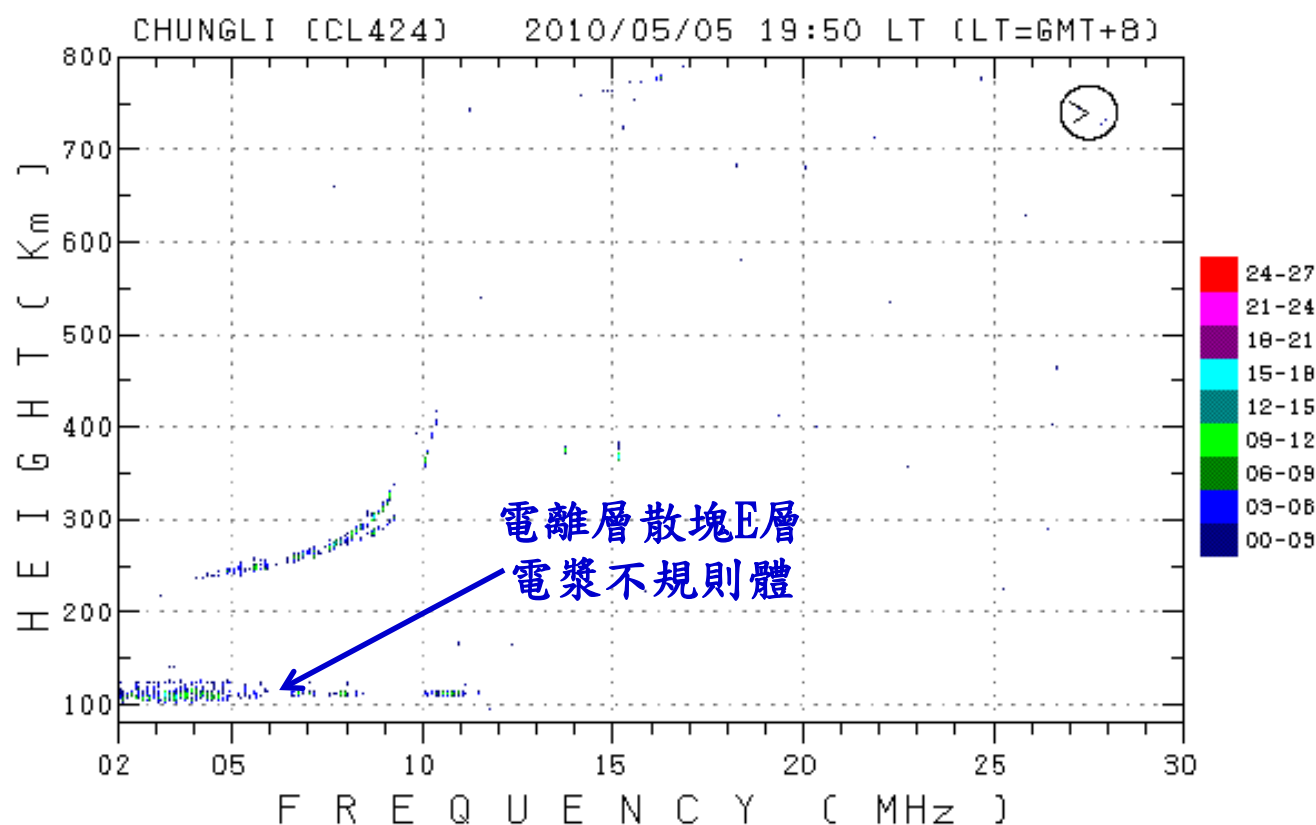
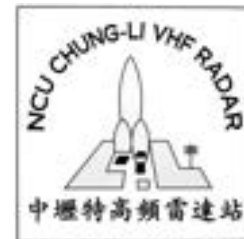
九鵬52 MHz雷達

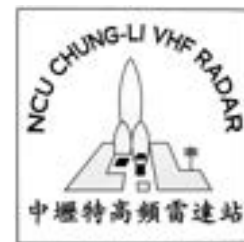




探空火箭七號科學實驗

地面電離層觀測儀初步結果

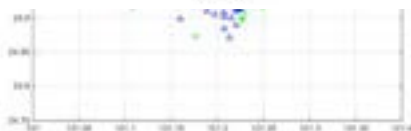
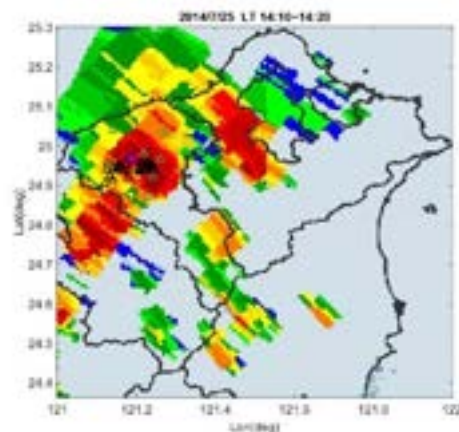
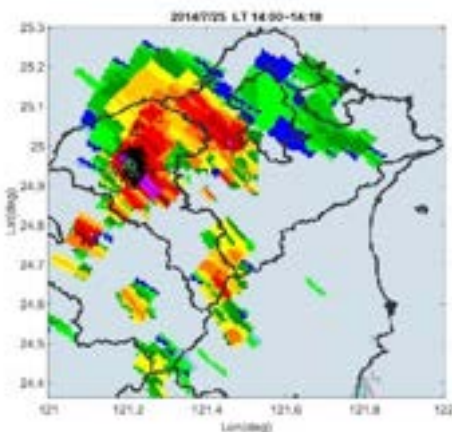
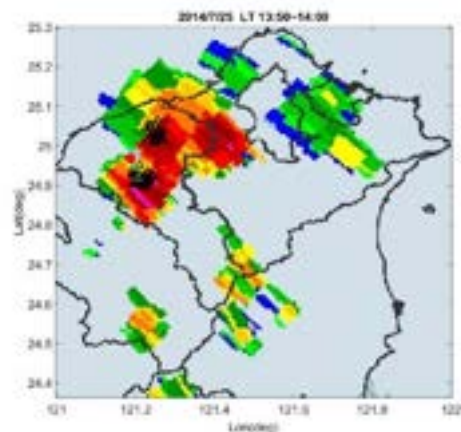
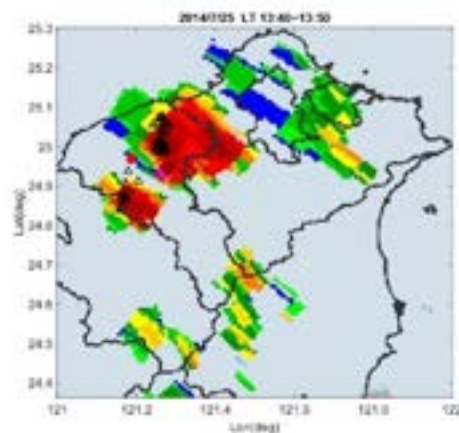
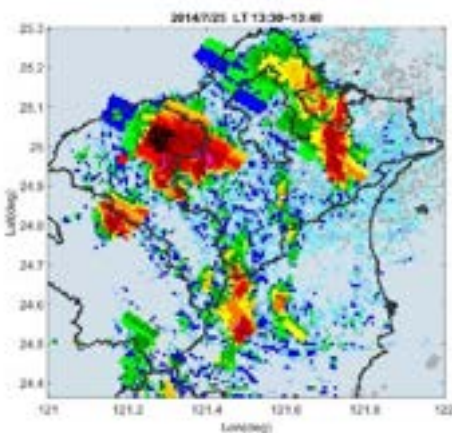
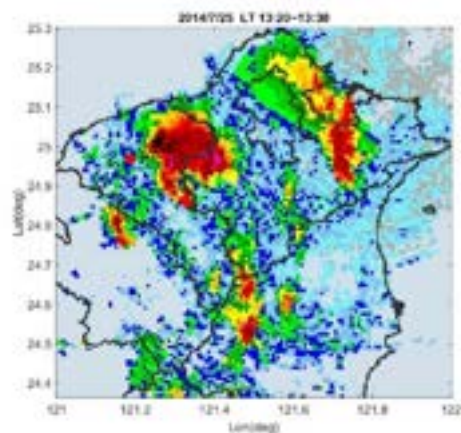




中壢特高頻雷達 閃電觀測實驗

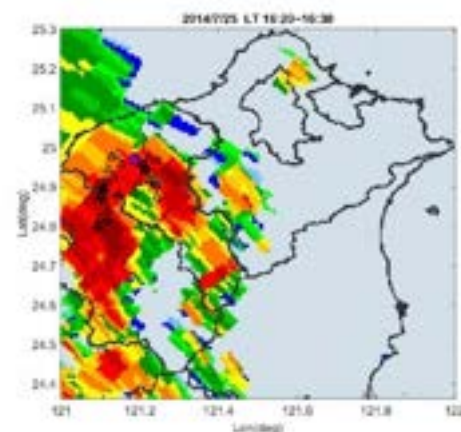
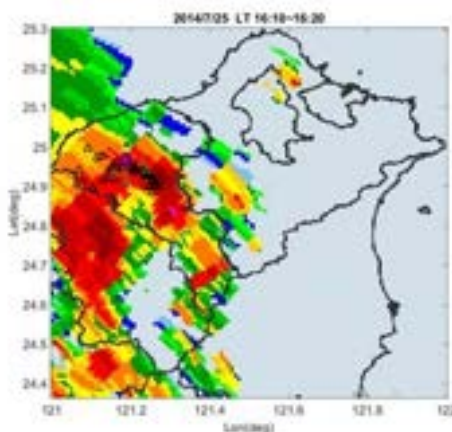
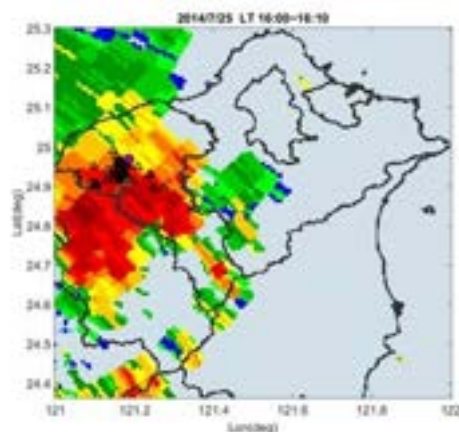
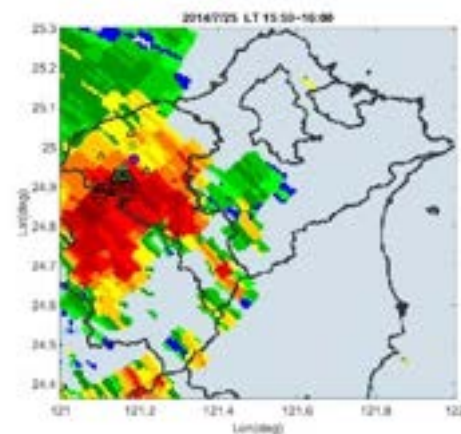
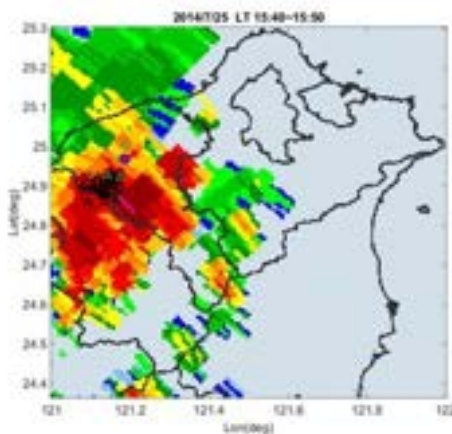
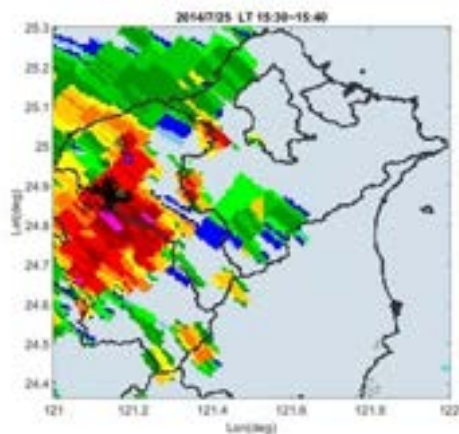
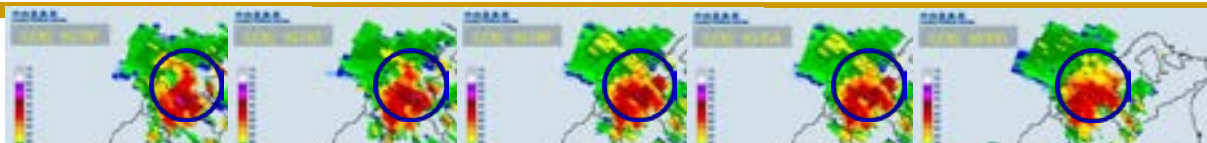


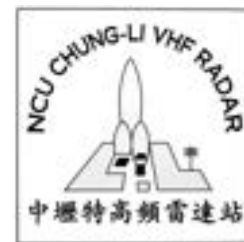
雷雨胞與閃電(1)





雷雨胞與閃電(2)

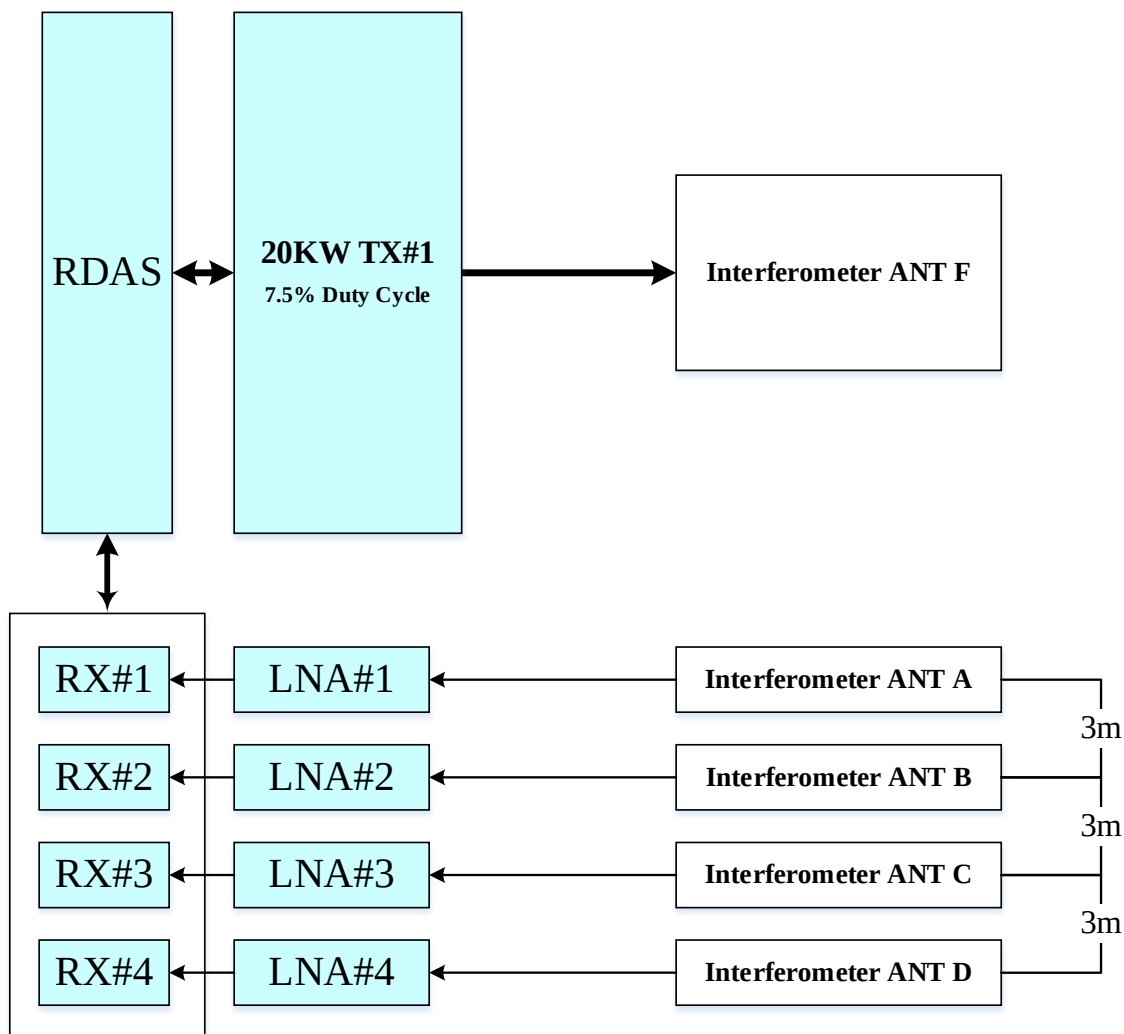




中壢特高頻雷達 測海實驗

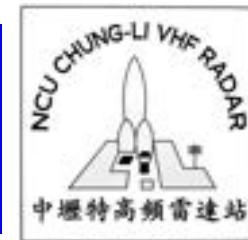


測海雷達系統架構





中壢特高頻52MHz雷達 測海觀測系統



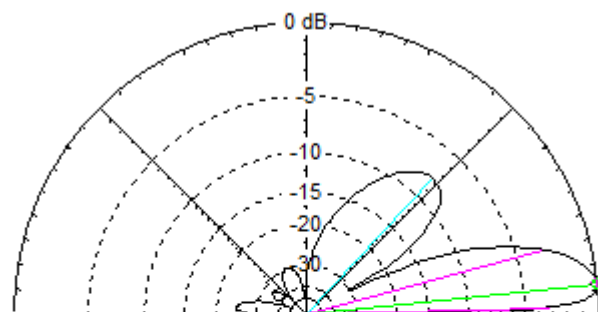


發射天線性能



Total Field

EZNEC+



52 MHz

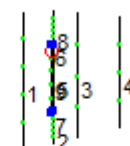
Elevation Plot	Cursor Elev	6.0 deg.
Azimuth Angle	Gain	11.99 dBi
Outer Ring	0.0 dBmax	
	0.0 dBmax3D	
3D Max Gain	11.99 dBi	
Slice Max Gain	11.99 dBi @ Elev Angle = 6.0 deg.	
Beamwidth	13.7 deg.; -3dB @ 1.6, 15.3 deg.	
Sidelobe Gain	4.24 dBi @ Elev Angle = 47.0 deg.	
Front/Sidelobe	7.75 dB	

Elevation Plot	Cursor Elev	1.0 deg.
Azimuth Angle	Gain	7.13 dBi
Outer Ring	-4.86 dBmax	
	-4.86 dBmax3D	

3D Max Gain	11.99 dBi
Slice Max Gain	11.99 dBi @ Elev Angle = 6.0 deg.
Beamwidth	13.7 deg.; -3dB @ 1.6, 15.3 deg.
Sidelobe Gain	4.24 dBi @ Elev Angle = 47.0 deg.
Front/Sidelobe	7.75 dB

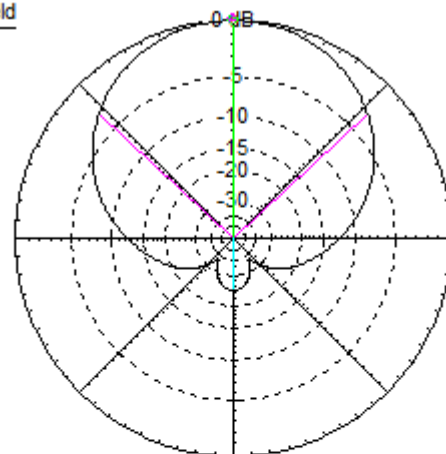
Z

EZNEC+



Total Field

EZNEC+



52 MHz

Azimuth Plot	Cursor Az	90.0 deg.
Elevation Angle	Gain	11.99 dBi
Outer Ring	0.0 dBmax	
	0.0 dBmax3D	

3D Max Gain	11.99 dBi
Slice Max Gain	11.99 dBi @ Az Angle = 90.0 deg.
Front/Back	24.54 dB
Beamwidth	95.0 deg.; -3dB @ 42.5, 137.5 deg.
Sidelobe Gain	-12.55 dBi @ Az Angle = 270.0 deg.
Front/Sidelobe	24.54 dB

Azimuth Plot	Cursor Az	90.0 deg.
Elevation Angle	Gain	7.13 dBi
Outer Ring	0.0 dBmax	
	-4.86 dBmax3D	

3D Max Gain	11.99 dBi
Slice Max Gain	7.13 dBi @ Az Angle = 90.0 deg.
Front/Back	24.17 dB
Beamwidth	95.0 deg.; -3dB @ 42.5, 137.5 deg.
Sidelobe Gain	-17.04 dBi @ Az Angle = 270.0 deg.
Front/Sidelobe	24.17 dB



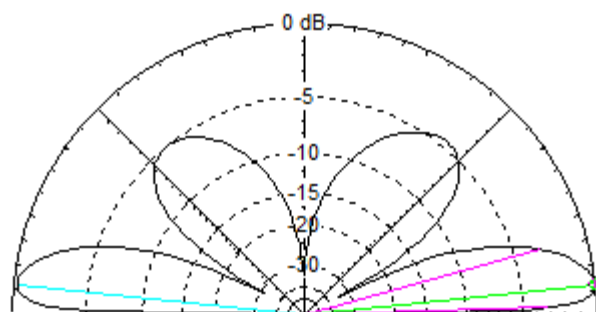


接收天線性能



Total Field

EZNEC+



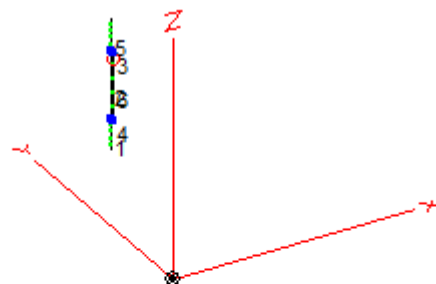
52 MHz

Elevation Plot	Cursor Elev	6.0 deg.
Azimuth Angle	Gain	6.0 dBi
Outer Ring		0.0 dBmax
		0.0 dBmax3D

3D Max Gain	6.0 dBi
Slice Max Gain	6.0 dBi @ Elev Angle = 6.0 deg.
Beamwidth	13.9 deg.; -3dB @ 1.7, 15.6 deg.
Sidelobe Gain	5.8 dBi @ Elev Angle = 174.0 deg.
Front/Sidelobe	0.2 dB

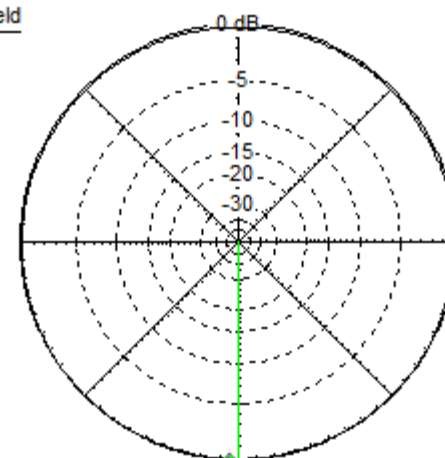
Elevation Plot	Cursor Elev	1.0 deg.
Azimuth Angle	Gain	1.1 dBi
Outer Ring		-4.9 dBmax
		-4.9 dBmax3D

3D Max Gain	6.0 dBi
Slice Max Gain	6.0 dBi @ Elev Angle = 6.0 deg.
Beamwidth	13.9 deg.; -3dB @ 1.7, 15.6 deg.
Sidelobe Gain	5.8 dBi @ Elev Angle = 174.0 deg.
Front/Sidelobe	0.2 dB



EZNEC Total Field

EZNEC+



52 MHz

Azimuth Plot	Cursor Az	268.0 deg.
Elevation Angle	Gain	6.0 dBi
Outer Ring		0.0 dBmax
		0.0 dBmax3D

3D Max Gain	6.0 dBi
Slice Max Gain	6.0 dBi @ Az Angle = 270.0 deg.
Front/Back	0.2 dB
Beamwidth	?
Sidelobe Gain	< -100 dBi
Front/Sidelobe	> 100 dB

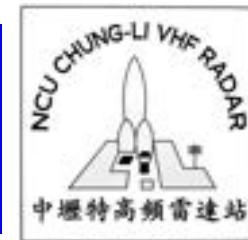
Azimuth Plot	Cursor Az	268.0 deg.
Elevation Angle	Gain	1.1 dBi
Outer Ring		0.0 dBmax
		-4.9 dBmax3D

3D Max Gain	6.0 dBi
Slice Max Gain	1.1 dBi @ Az Angle = 270.0 deg.
Front/Back	0.2 dB
Beamwidth	?
Sidelobe Gain	< -100 dBi
Front/Sidelobe	> 100 dB



測海雷達系統實驗

中央大學臨海工作站





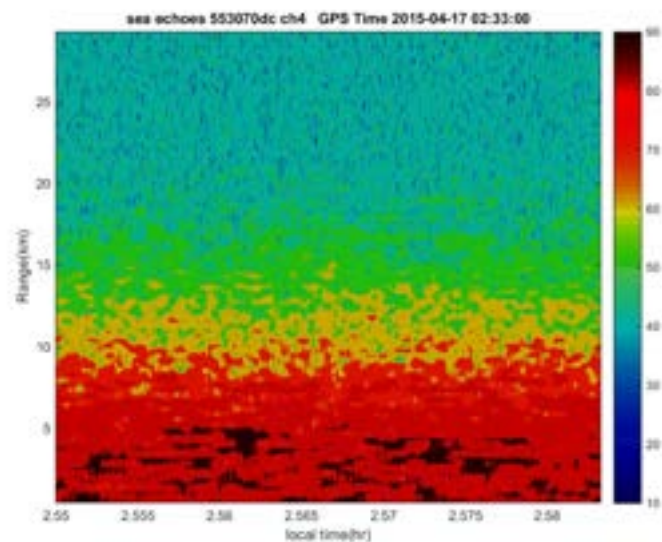
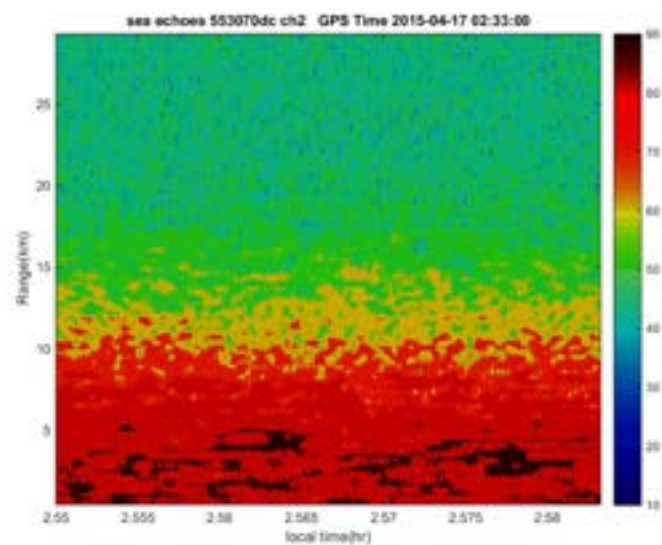
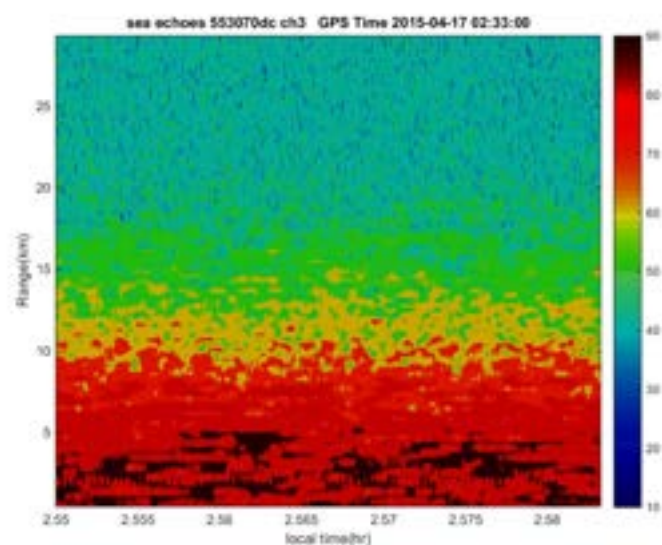
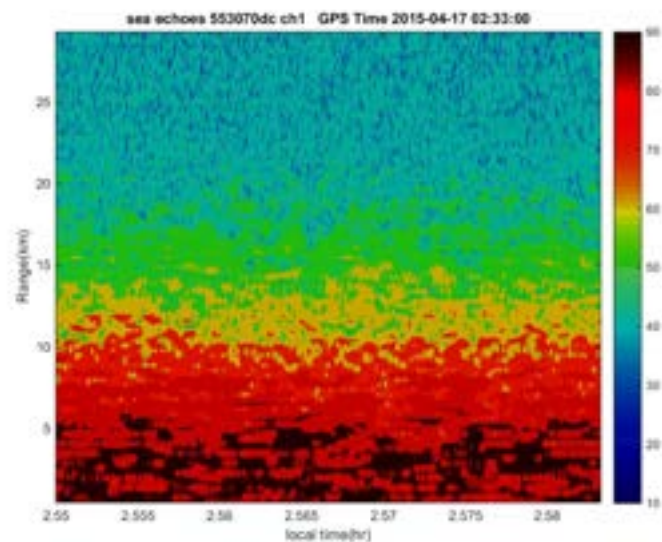
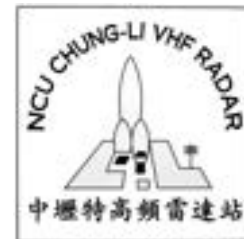
測海雷達系統 觀測參數



- **PRF 2000 Hz(IPP 0.5ms)**
- **同相積分(Number of Coherent Integration) 256次**
- **發射脈波長(pulse length) $2\mu\text{s}$**
- **空間解析度(range resolution) 300 m**
- **時間解析度(time resolution) 128ms**
- **發射功率 20 KW**
- **使用單根天線發射**
- **4根天線陣列接收**

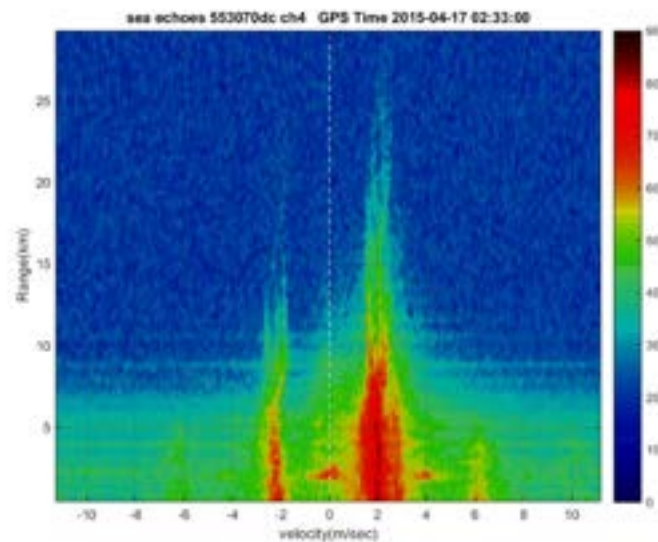
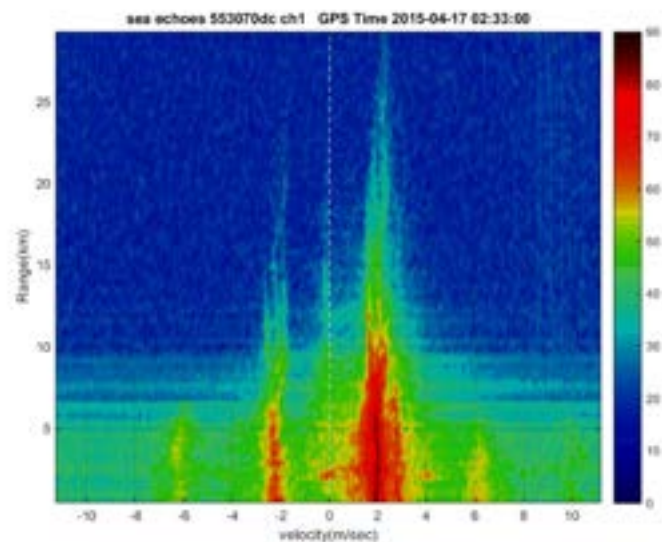
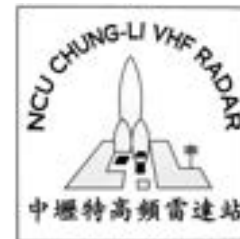


觀測結果



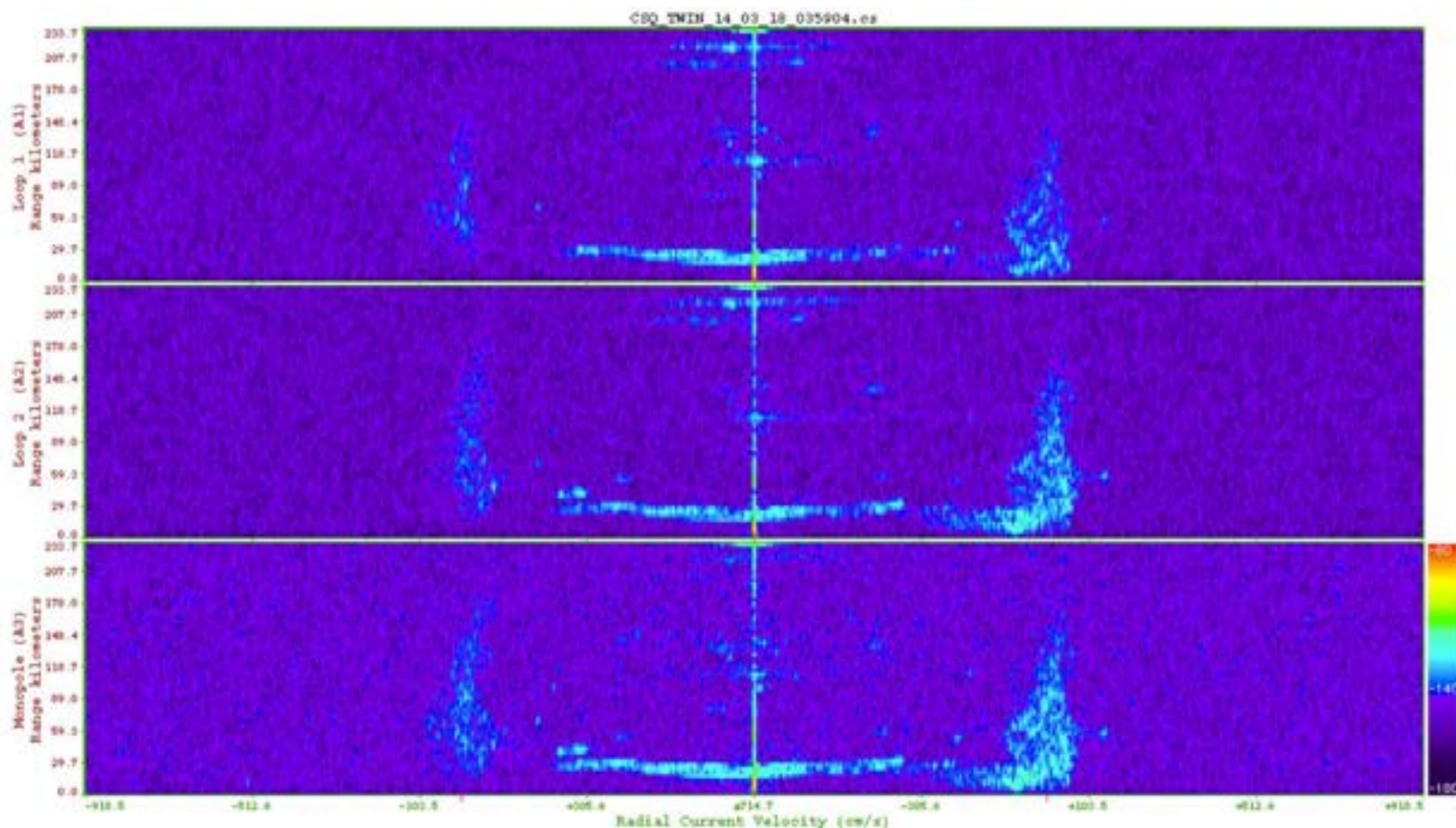
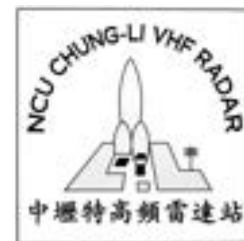


觀測結果(續)



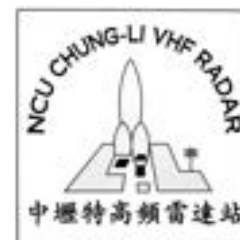


國研院海洋中心 CODAR資料

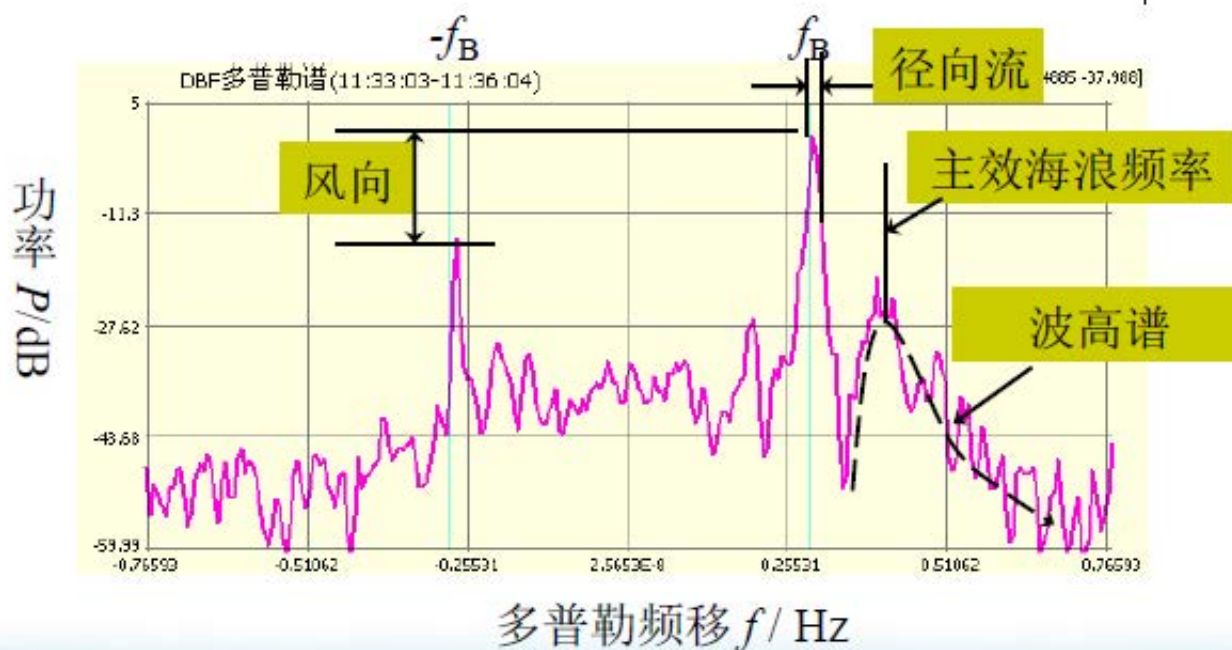




武漢大學吳雄斌資料



Sea State Parameters Correspondent to Spectra Characteristics





中央氣象局維修校驗剖風儀 主機系統

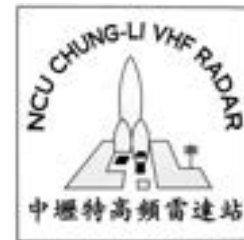


中心頻率 Central Frequency	449 MHz
放大器型式 Type of Amplifier	固態功率放大器 Solid state power amplifier
峰值總功率 Total peak power	2kW @ 17 % duty cycle
Wave form	pulse
脈波寬 Pulse width	1~10 μ s
Antenna array	planar phased array of 64 Yagi-Uda antenna elements.





中央氣象局維修校驗剖風儀 天線陣列





END



作業



- 中壢特高頻雷達站目前有哪些設備？這些設備相對應的研究領域？
- 雷達觀測電離層RTI資料顯示
- 雷達觀測低層大氣RTI資料顯示